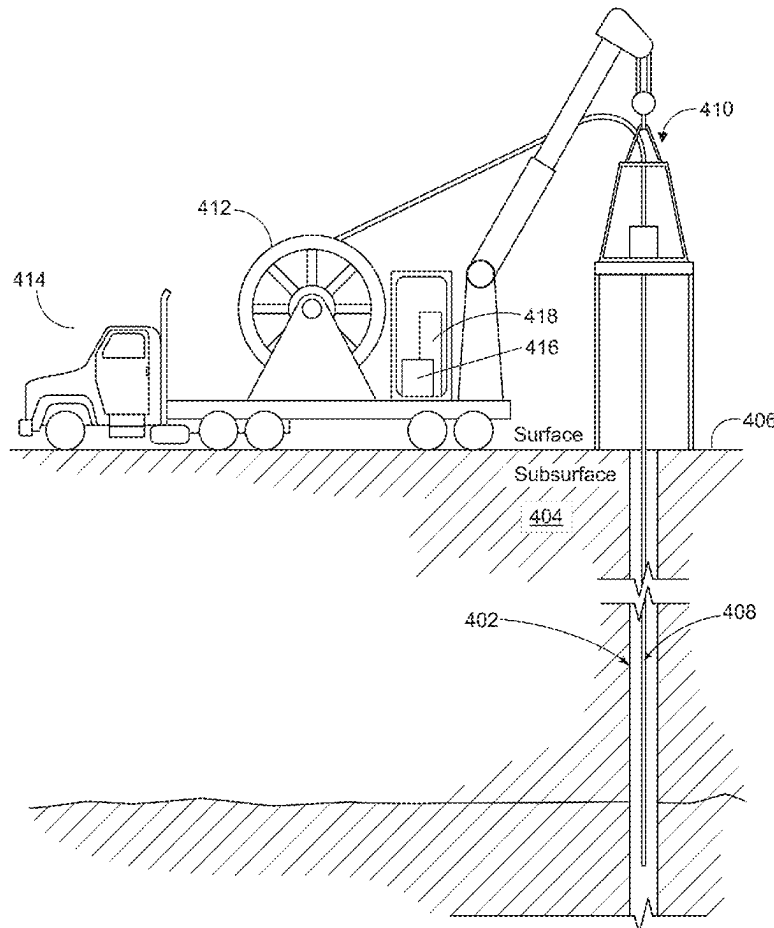




US 20250283382A1

(19) **United States**(12) **Patent Application Publication**  
**LIVESCU et al.**(10) **Pub. No.: US 2025/0283382 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SYSTEMS AND METHODS FOR  
GEOTHERMAL LOOP INSTALLATION  
USING COILED TUBING***F24T 10/00* (2018.01)*F24T 10/13* (2018.01)(71) Applicant: **Bedrock Energy, Inc.**, LOS  
ANGELES, CA (US)(52) **U.S. CL.**CPC ..... *E21B 19/24* (2013.01); *F16L 1/032*  
(2013.01); *F24T 10/13* (2018.05); *F24T*  
*2010/53* (2018.05)(72) Inventors: **Silviu LIVESCU**, Austin, TX (US);  
**Rio WHYTE**, Austin, TX (US); **Kyle**  
**CLARK**, Georgetown, TX (US); **Jesus**  
**Alberto CASTRO**, Kyle, TX (US)(57) **ABSTRACT**(21) Appl. No.: **19/189,179**(22) Filed: **Apr. 24, 2025****Related U.S. Application Data**(63) Continuation-in-part of application No. 18/629,546,  
filed on Apr. 8, 2024, now Pat. No. 12,298,043.(60) Provisional application No. 63/562,947, filed on Mar.  
8, 2024.**Publication Classification**(51) **Int. Cl.**  
*E21B 19/24* (2006.01)  
*F16L 1/032* (2006.01)

The various embodiments described herein include systems and methods for installing geothermal loops using coiled tubing. In one aspect, a system for installing geothermal loops includes a geothermal loop spooled on a geothermal loop reel that is mounted on a first axle. The system includes a coiled tubing string configured for inserting the geothermal loop into a borehole. The coiled tubing string is spooled on a coiled tubing reel that is mounted on a second axle. As the geothermal loop is inserted into the borehole, the geothermal loop is configured to unspool from the geothermal loop reel, the coiled tubing string is configured to unspool from the coiled tubing reel, and the geothermal loop and the coiled tubing string are configured to move at substantially similar speeds.



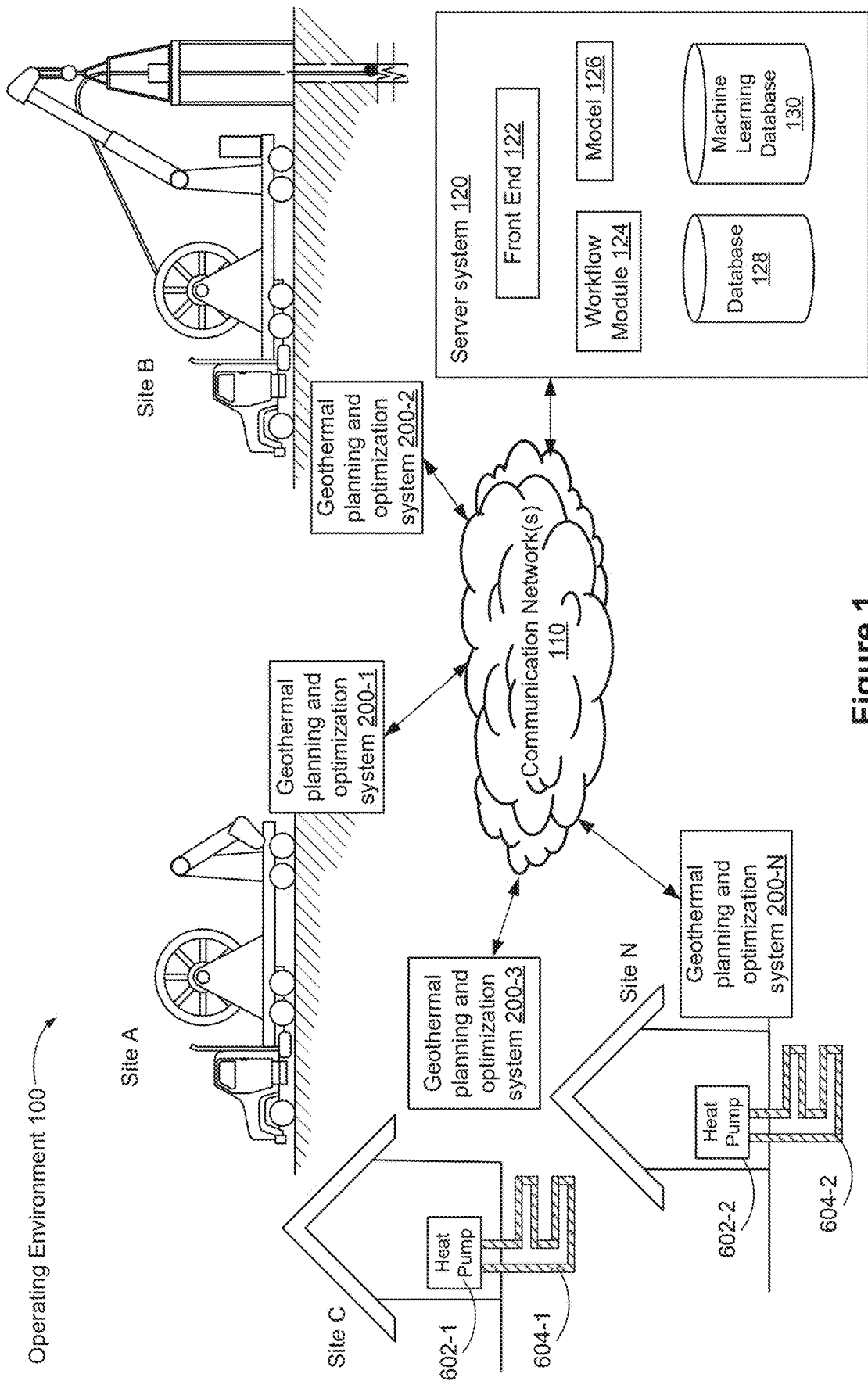


Figure 1

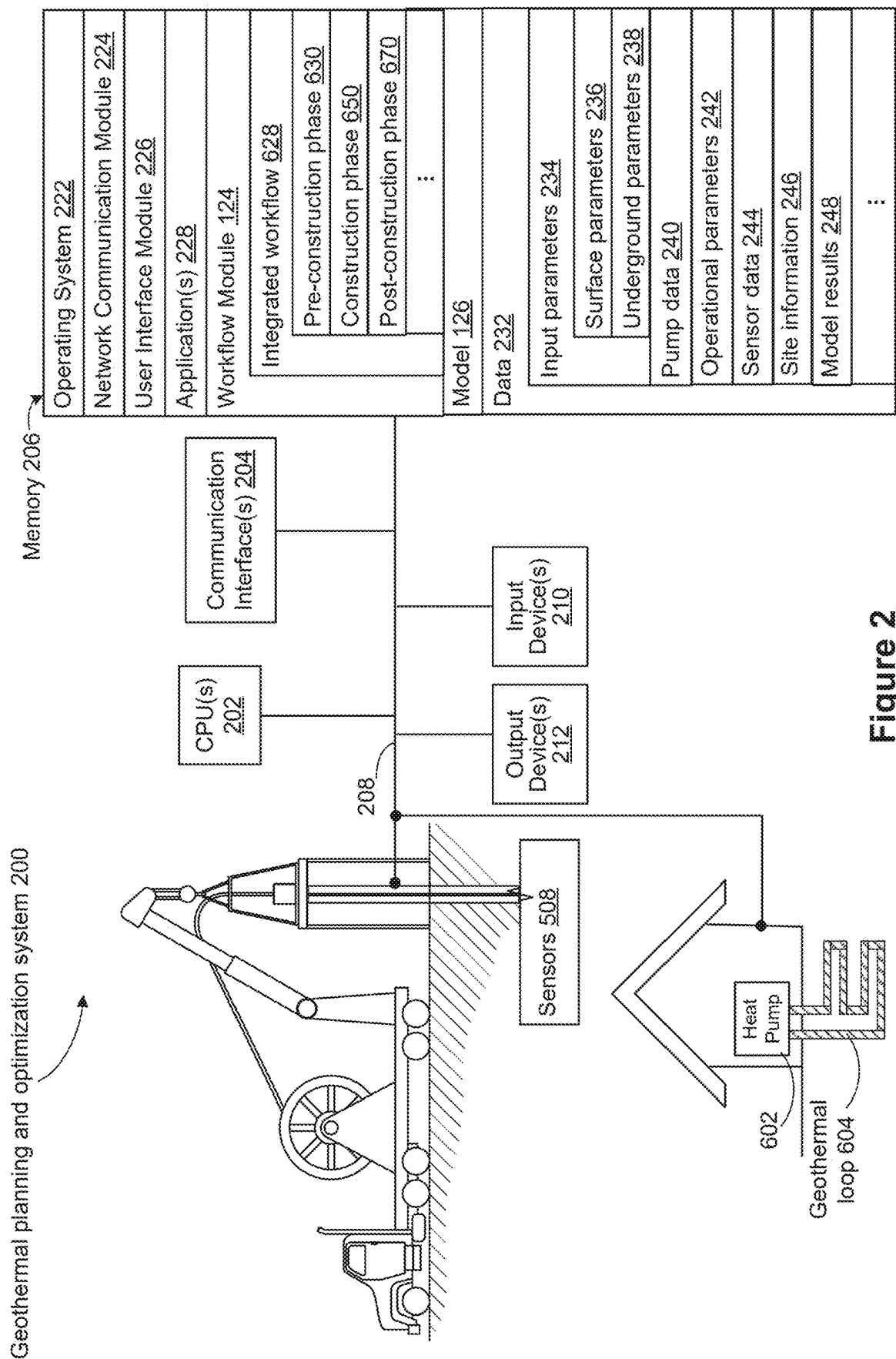


Figure 2

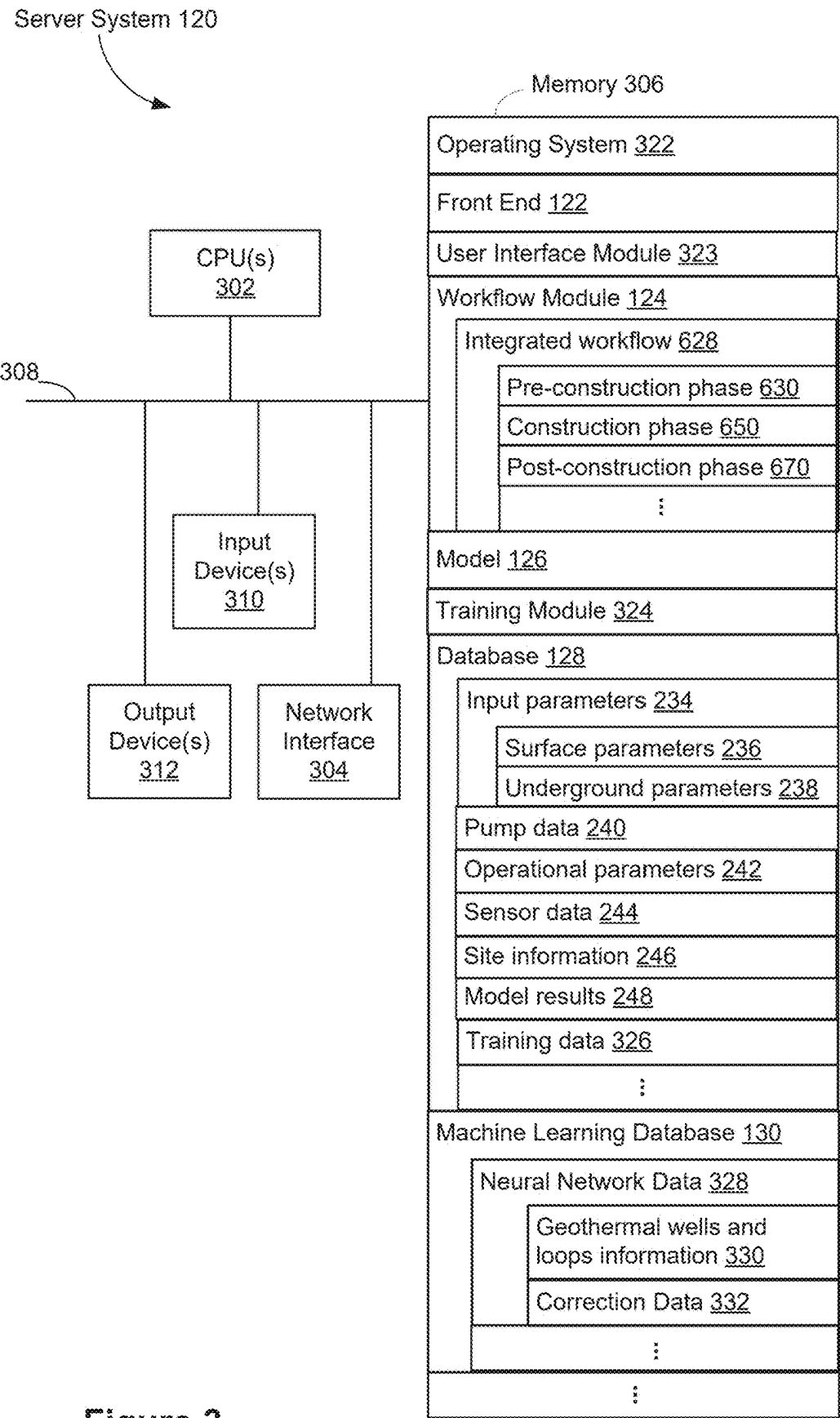


Figure 3

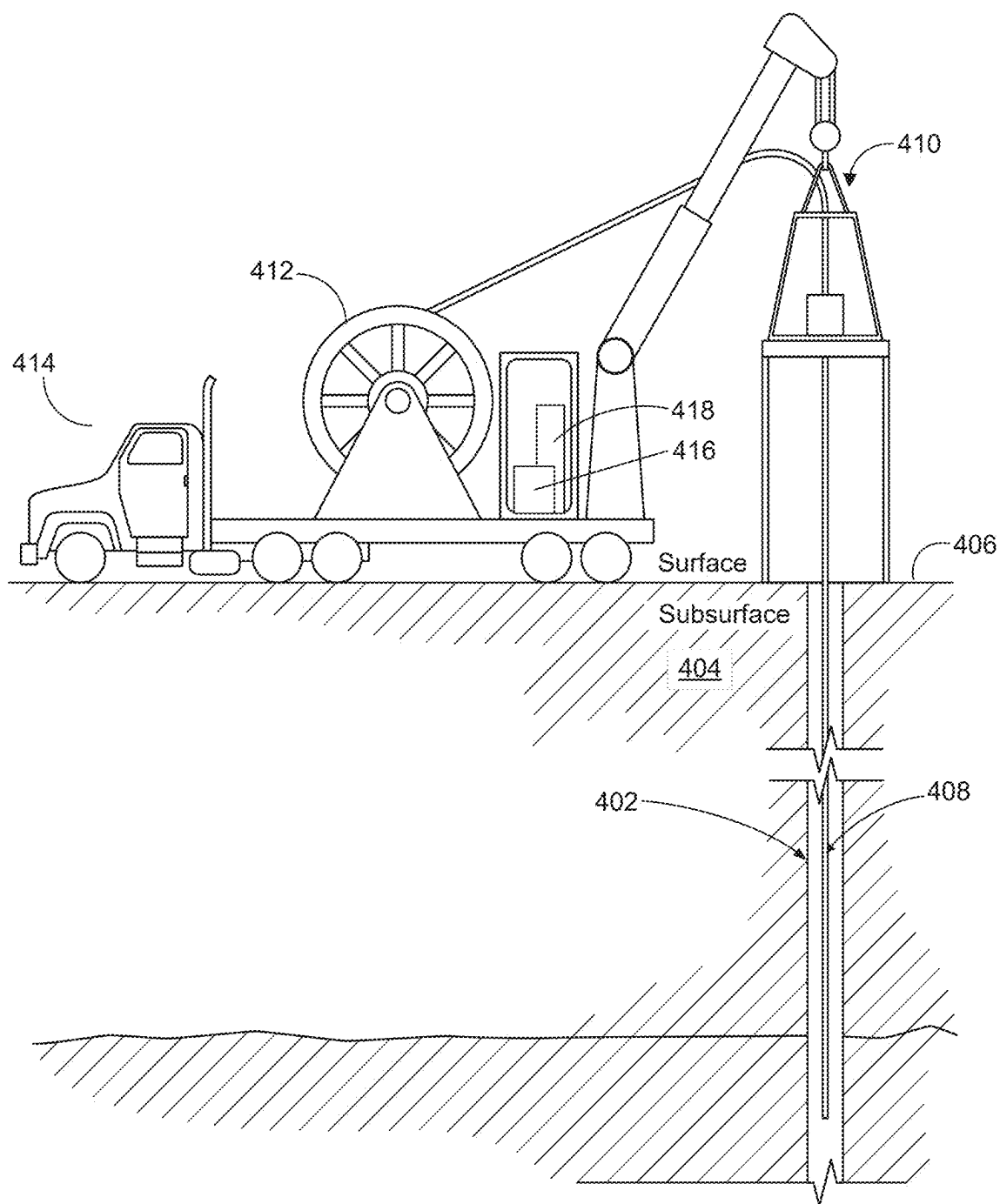


Figure 4A

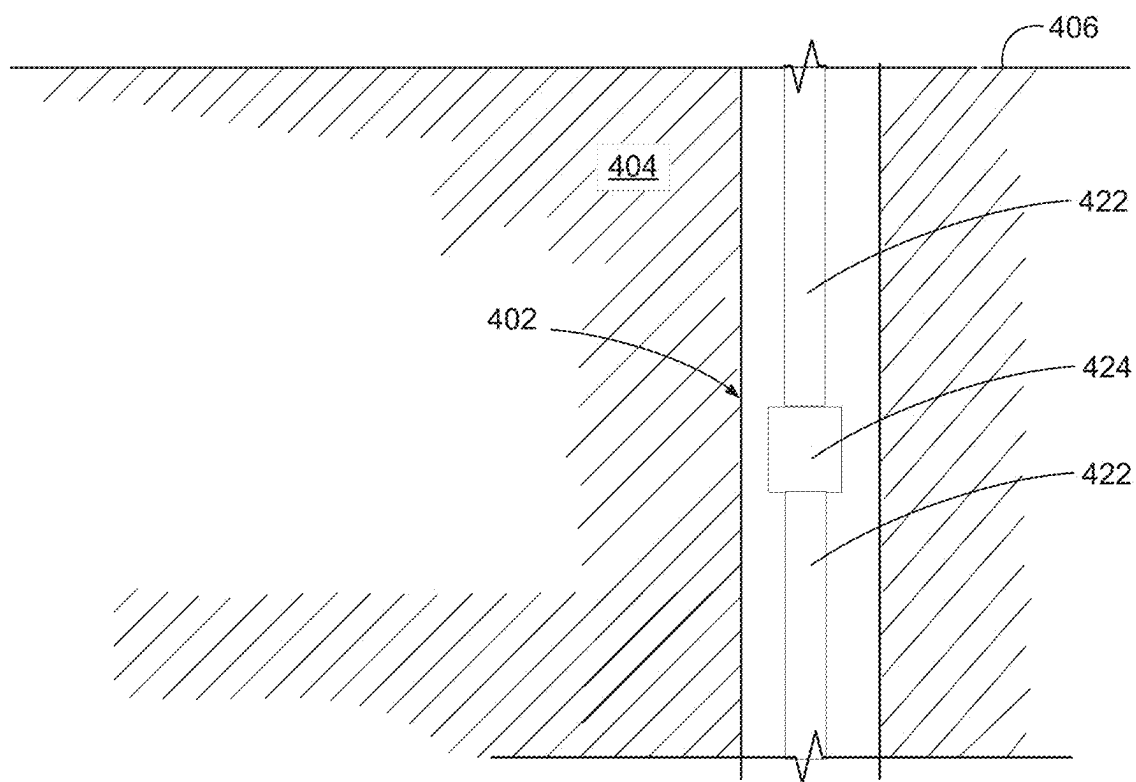


Figure 4B

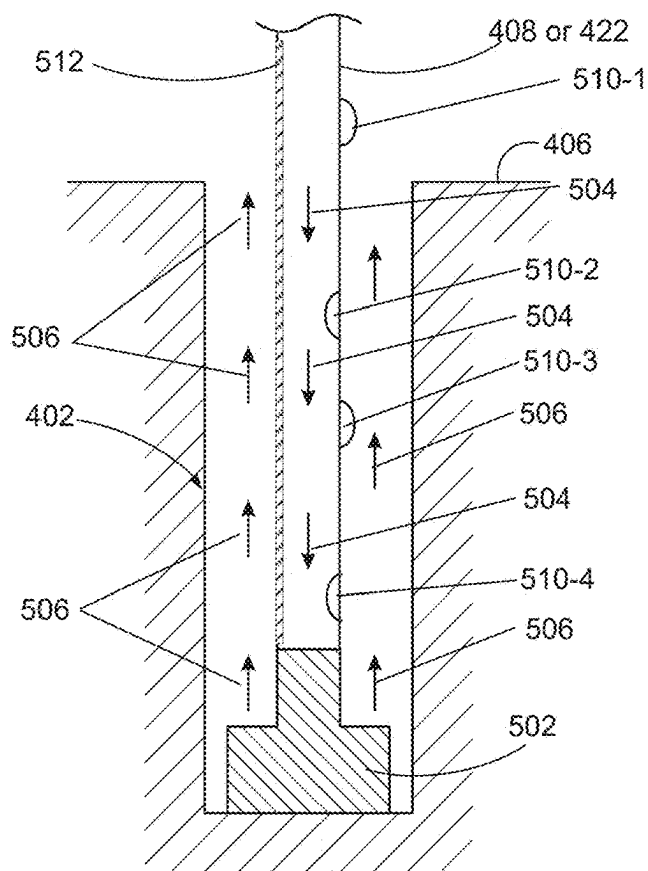


Figure 5A

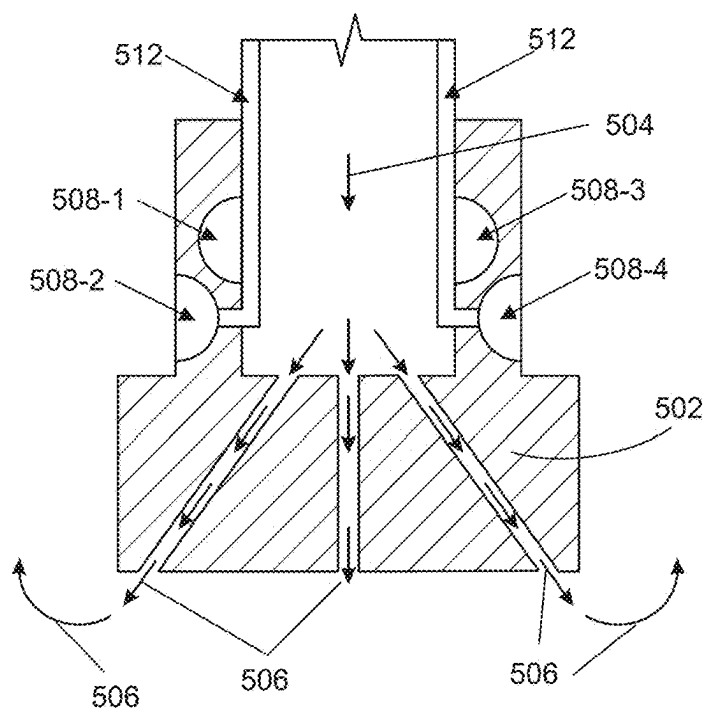
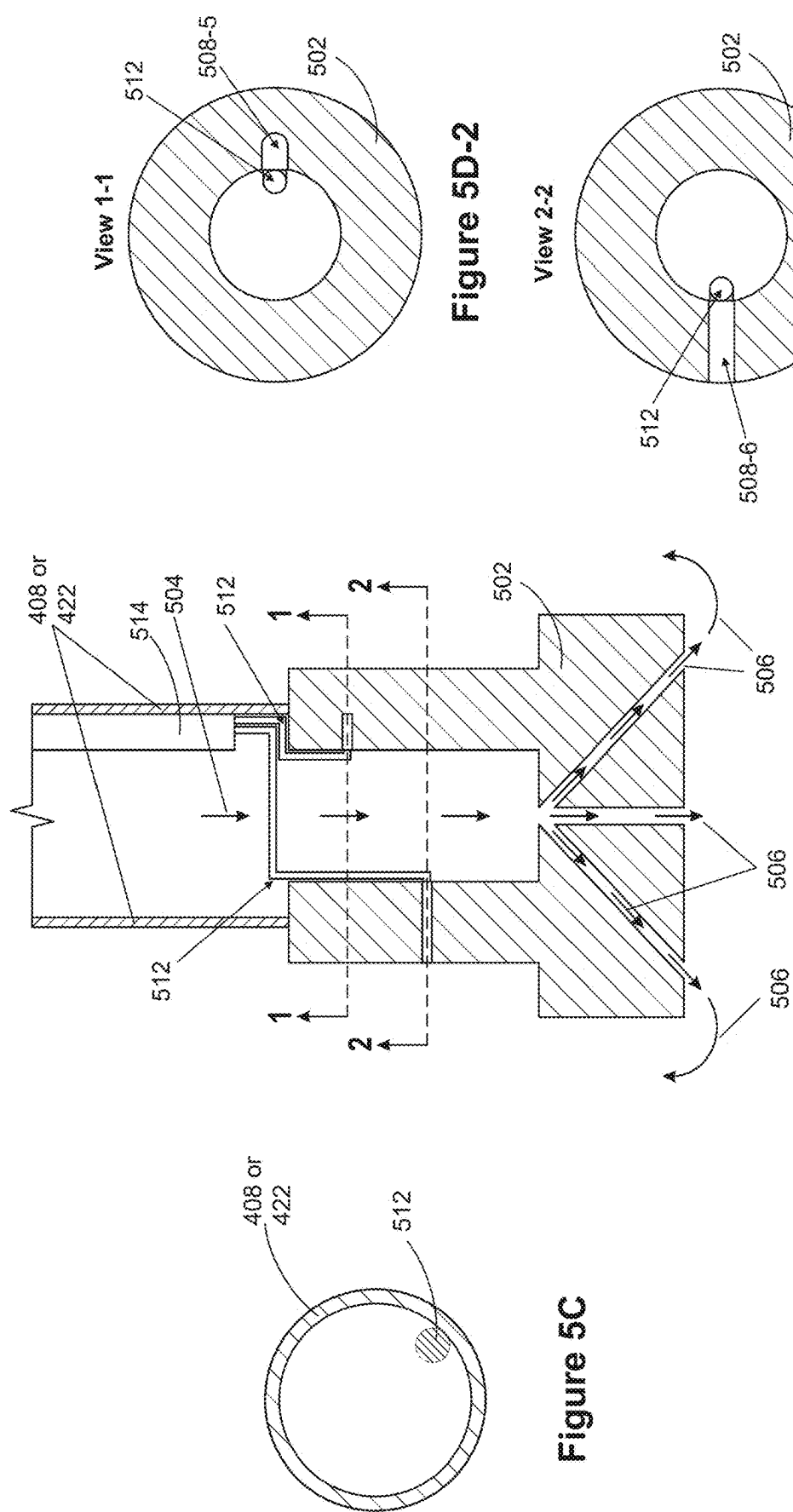


Figure 5B



View 1-1

Figure 5D-2

View 2-2

Figure 5D-3

Figure 5D-1

Figure 5C



|                           | Surface Data                                                                                          | Downhole data                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Mud Data</b>           | Pit volume<br>Mud temperature<br>Mud pressure<br>Mud weight<br>Pump strokes                           | N/A                                                                           |
| <b>Well Data</b>          | Temperature<br>Pressure<br>Gas measurements                                                           | Temperature<br>Pressure                                                       |
| <b>Directional Data</b>   |                                                                                                       | Inclination<br>Azimuth                                                        |
| <b>Drilling Mechanics</b> | RPM<br>Weight on bit<br>Torque<br>Bending moment<br>Rotary torque<br>Hook load<br>Rate of penetration | RPM<br>Weight on bit<br>Torque on bit<br>Bending moment<br>Downhole vibration |
| <b>Geological data</b>    | Cutting analysis                                                                                      | Density<br>Porosity<br>Resistivity<br>Gamma                                   |

**Figure 5E**

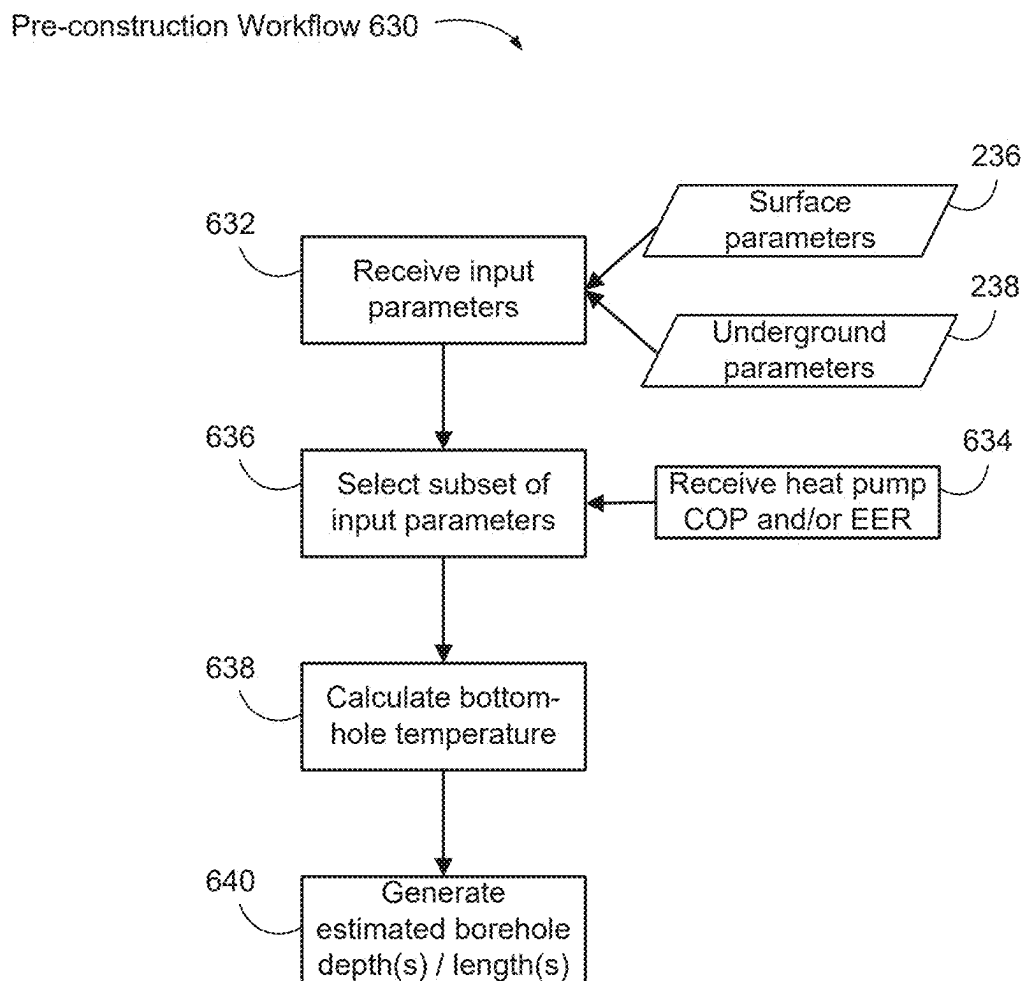


Figure 6A

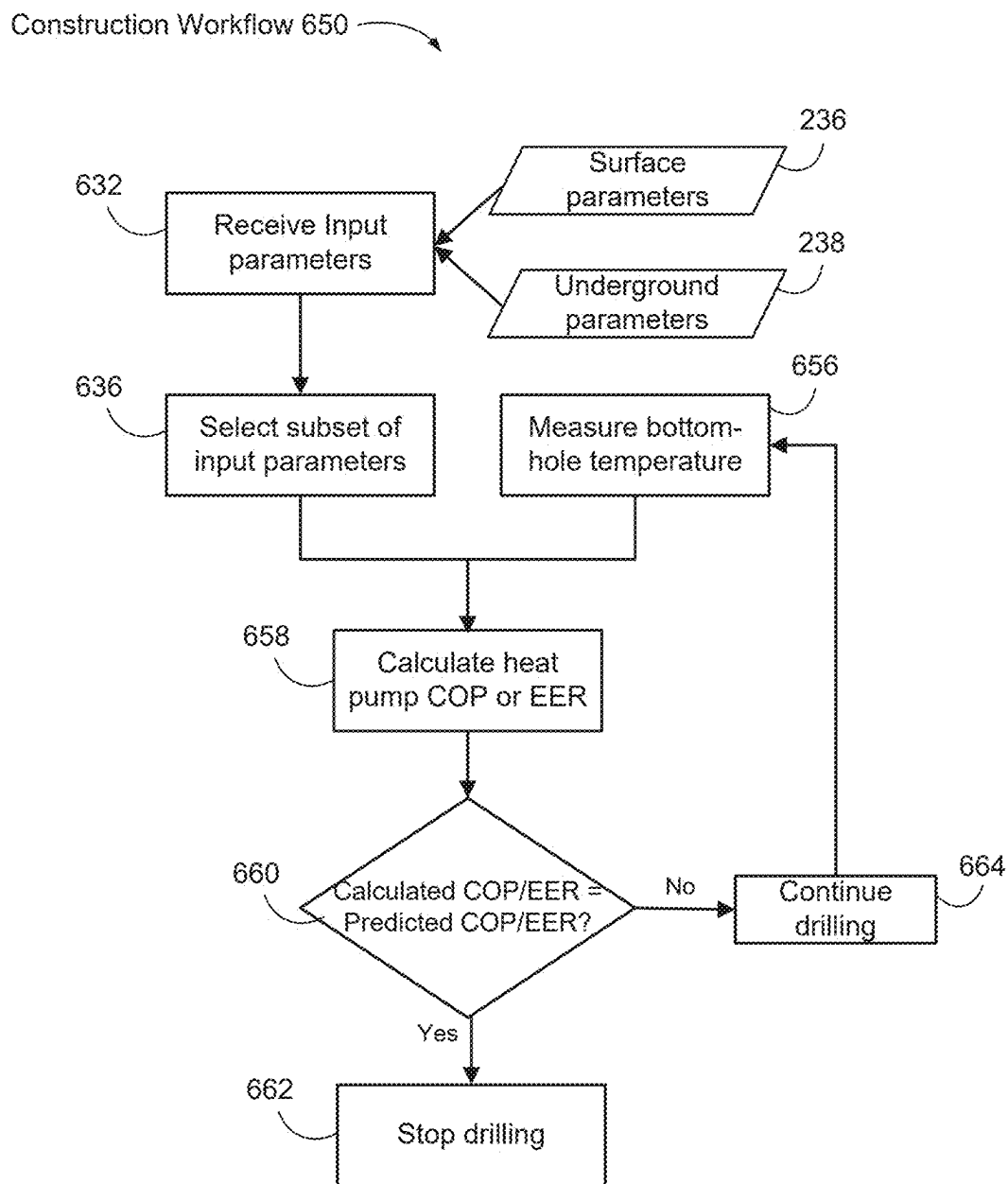


Figure 6B

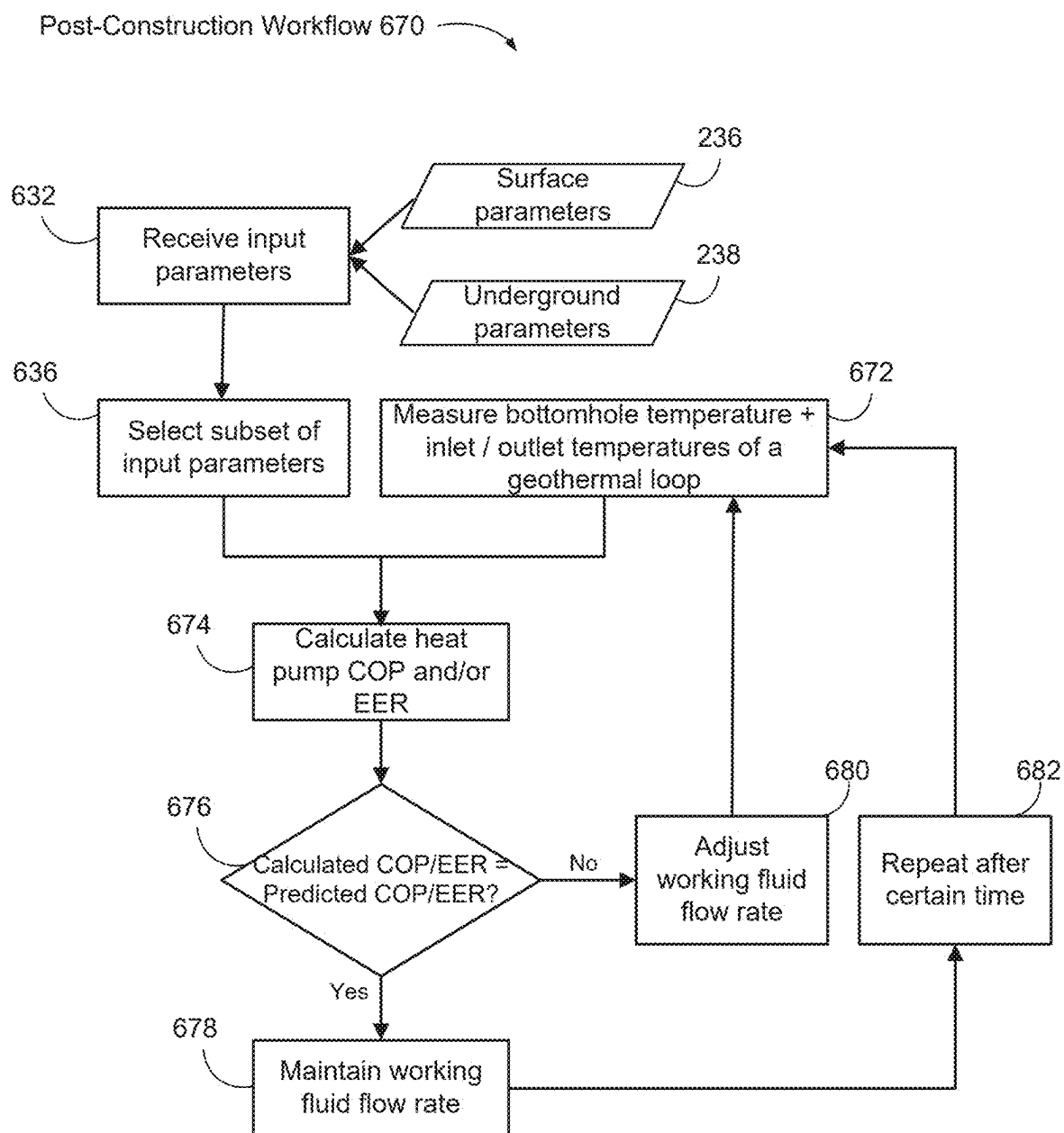


Figure 6C

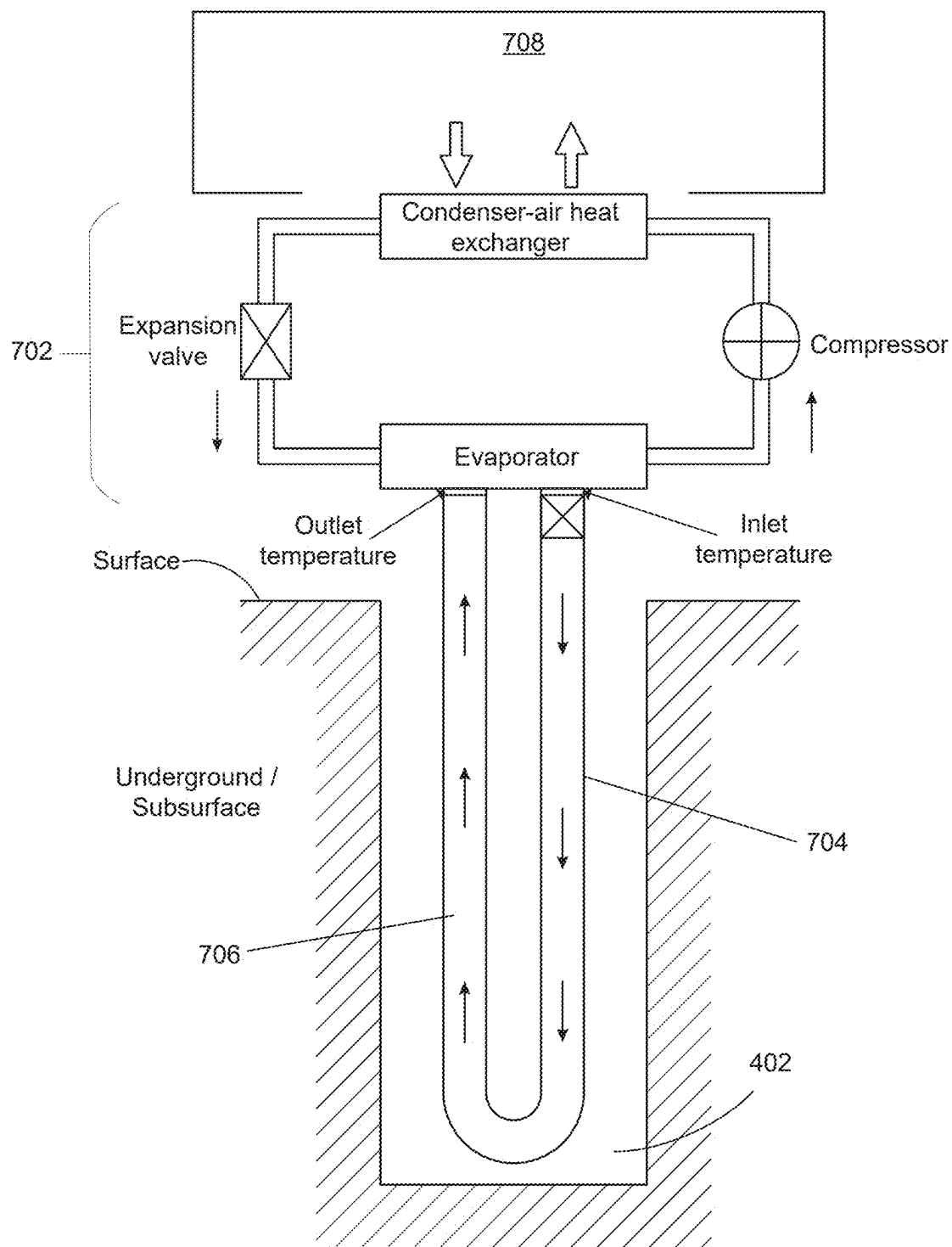


Figure 7

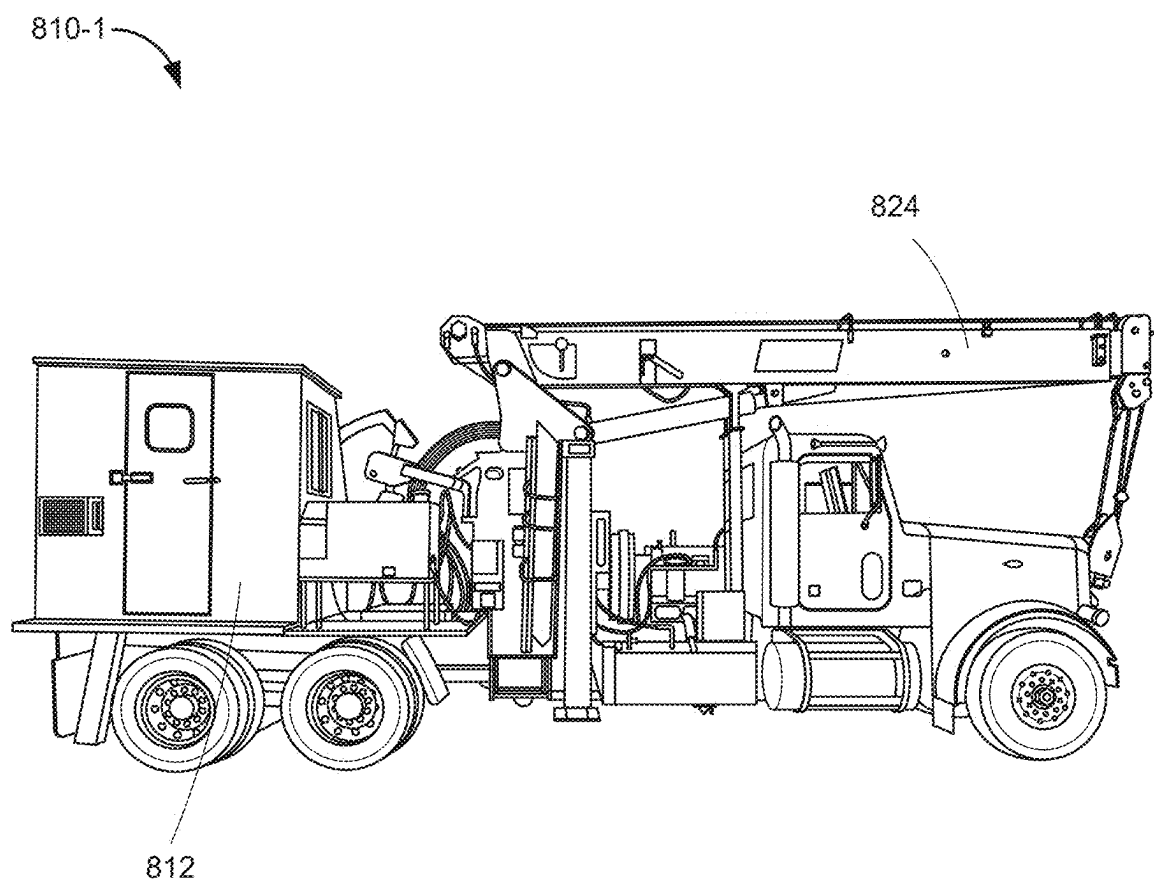
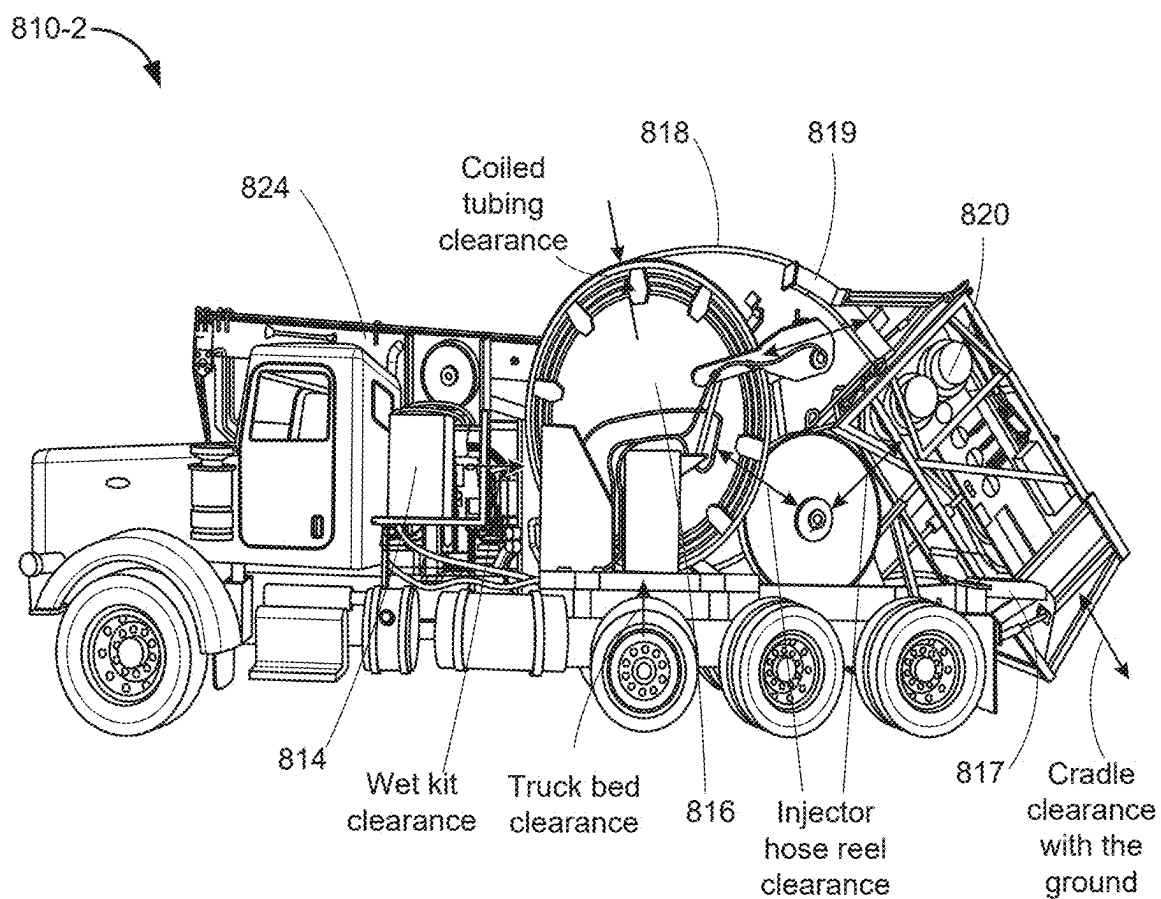


Figure 8A



**Figure 8B**

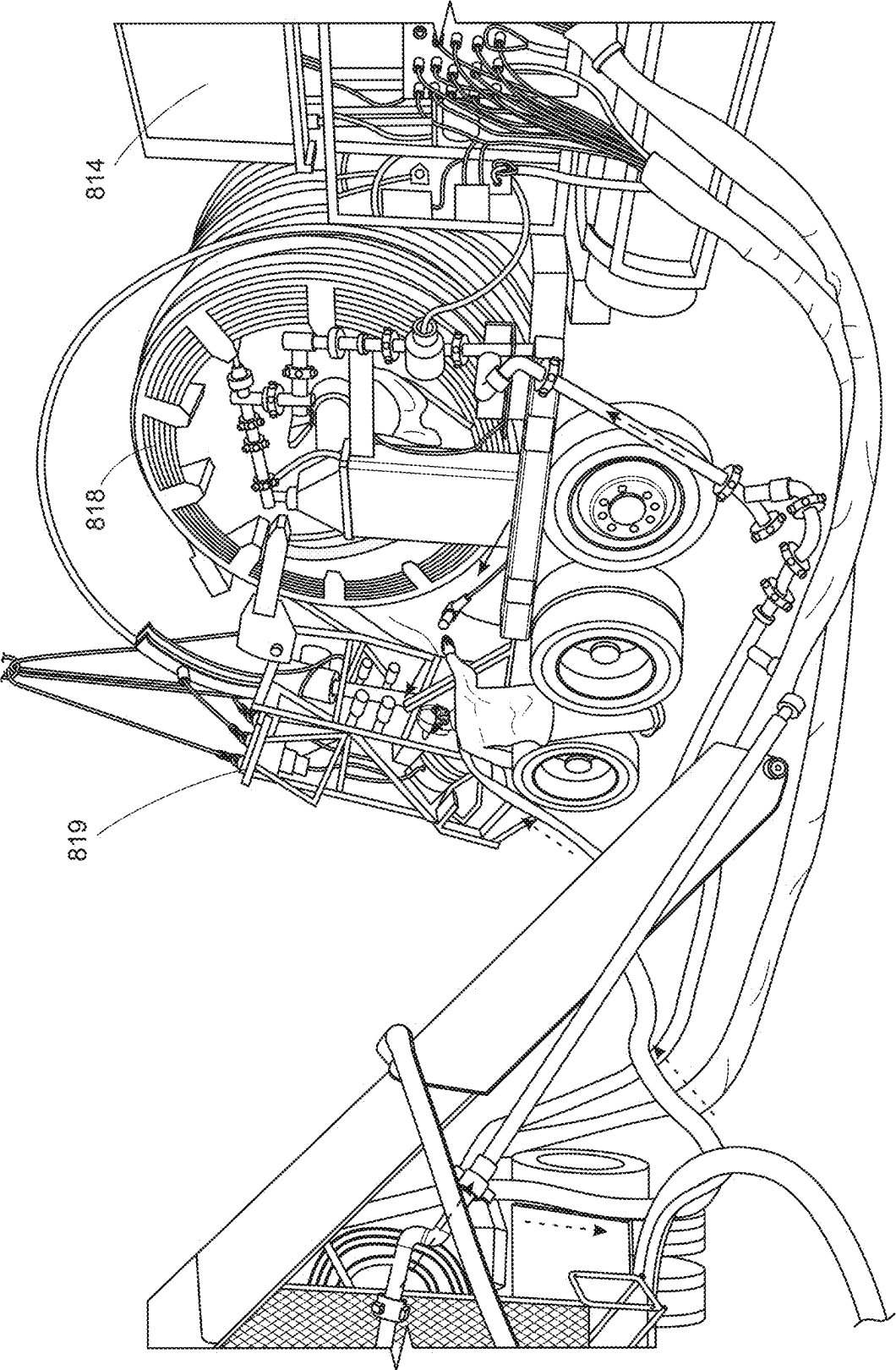


Figure 8C



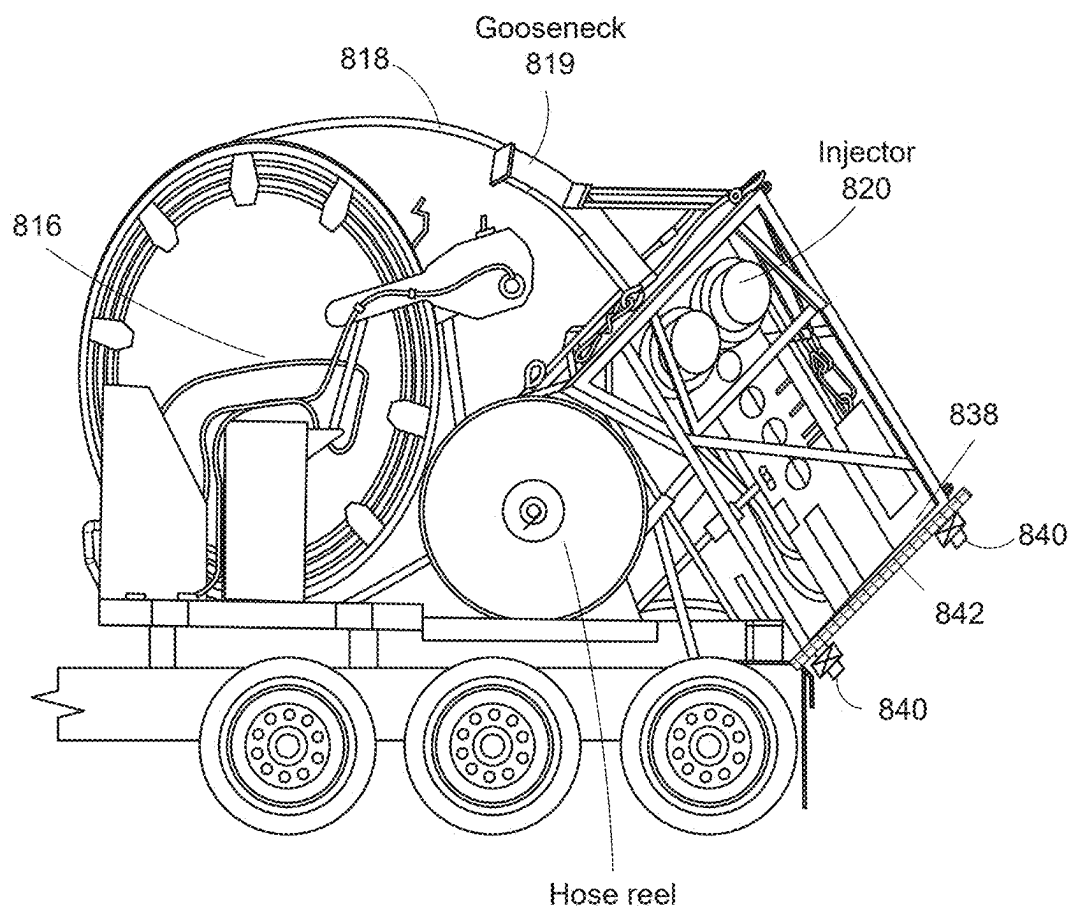


Figure 8D

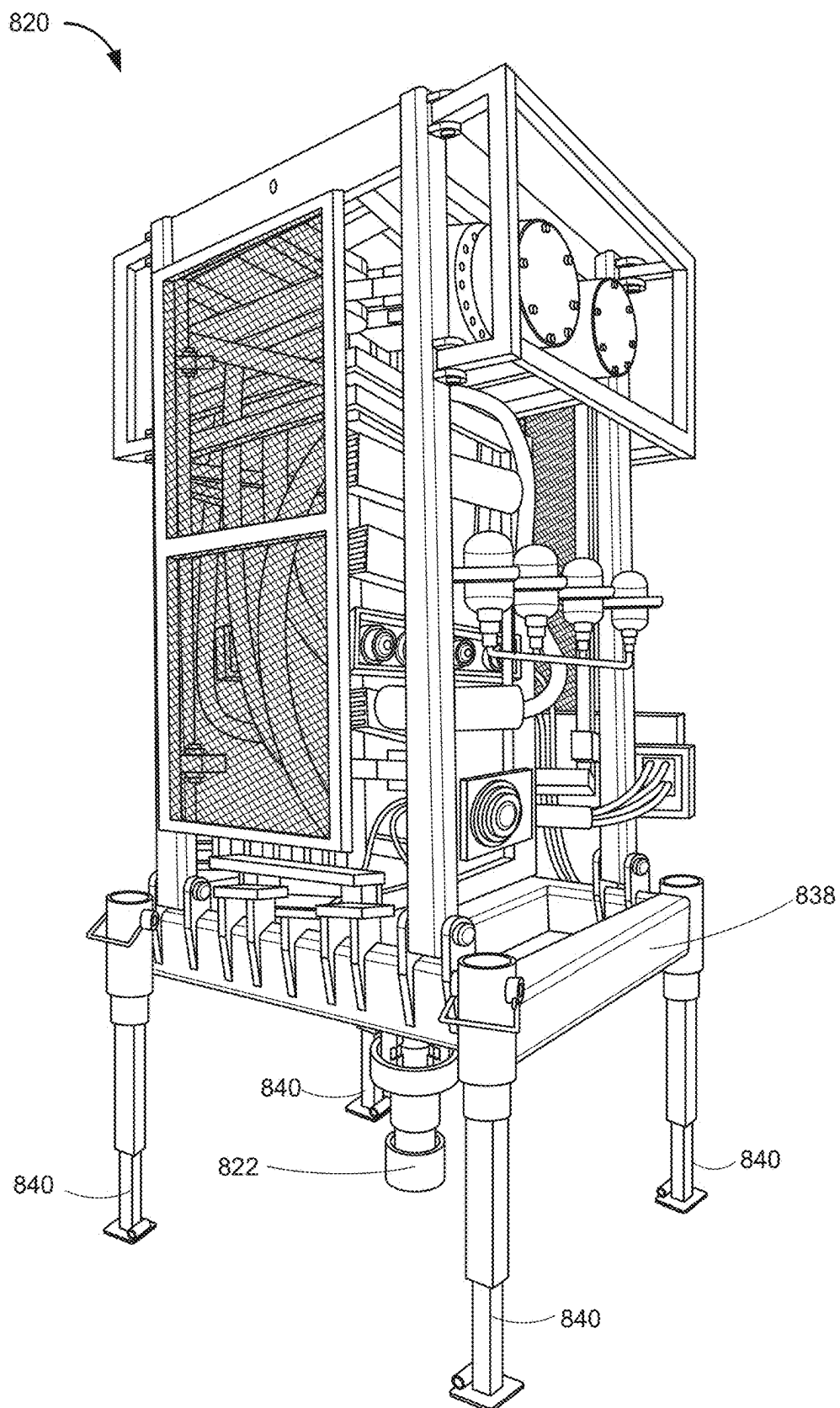
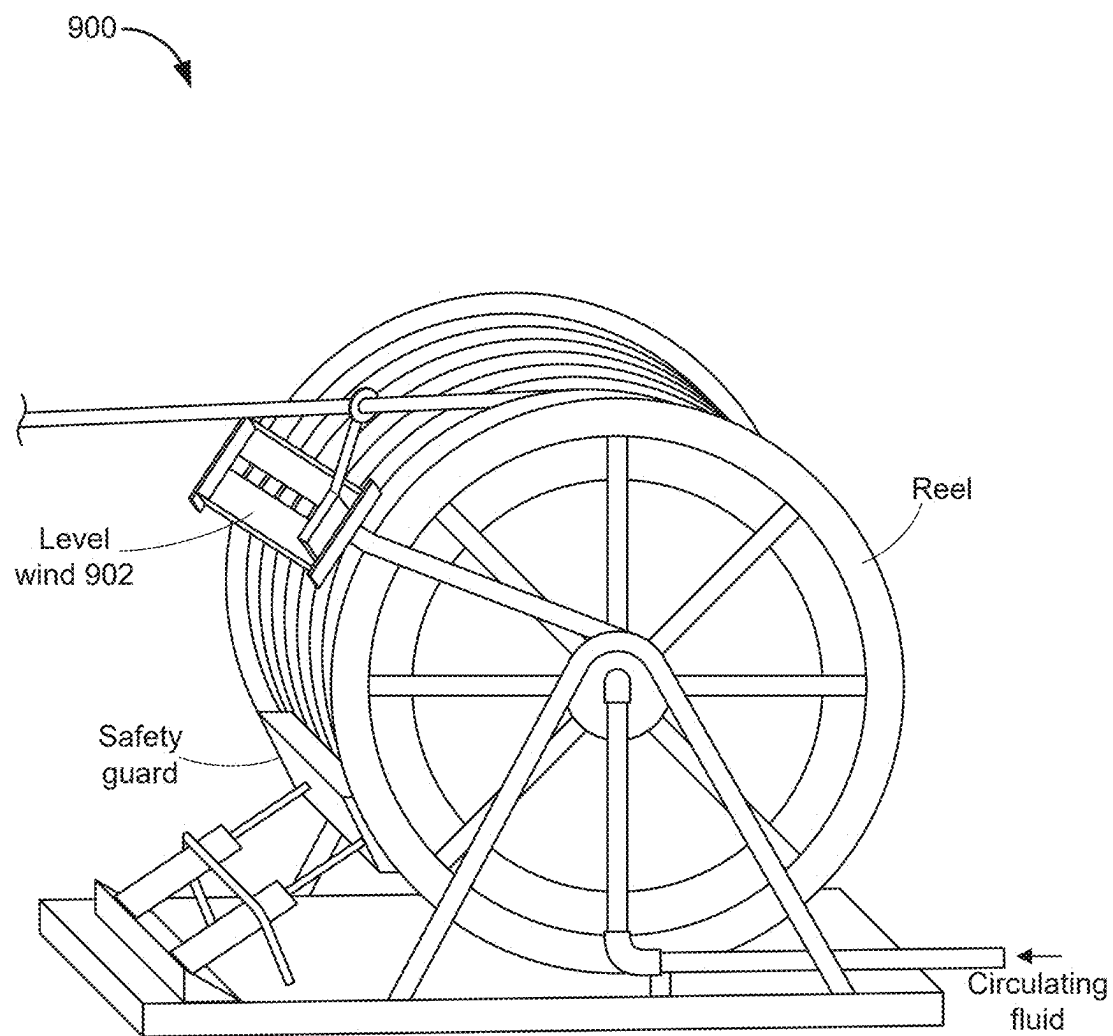
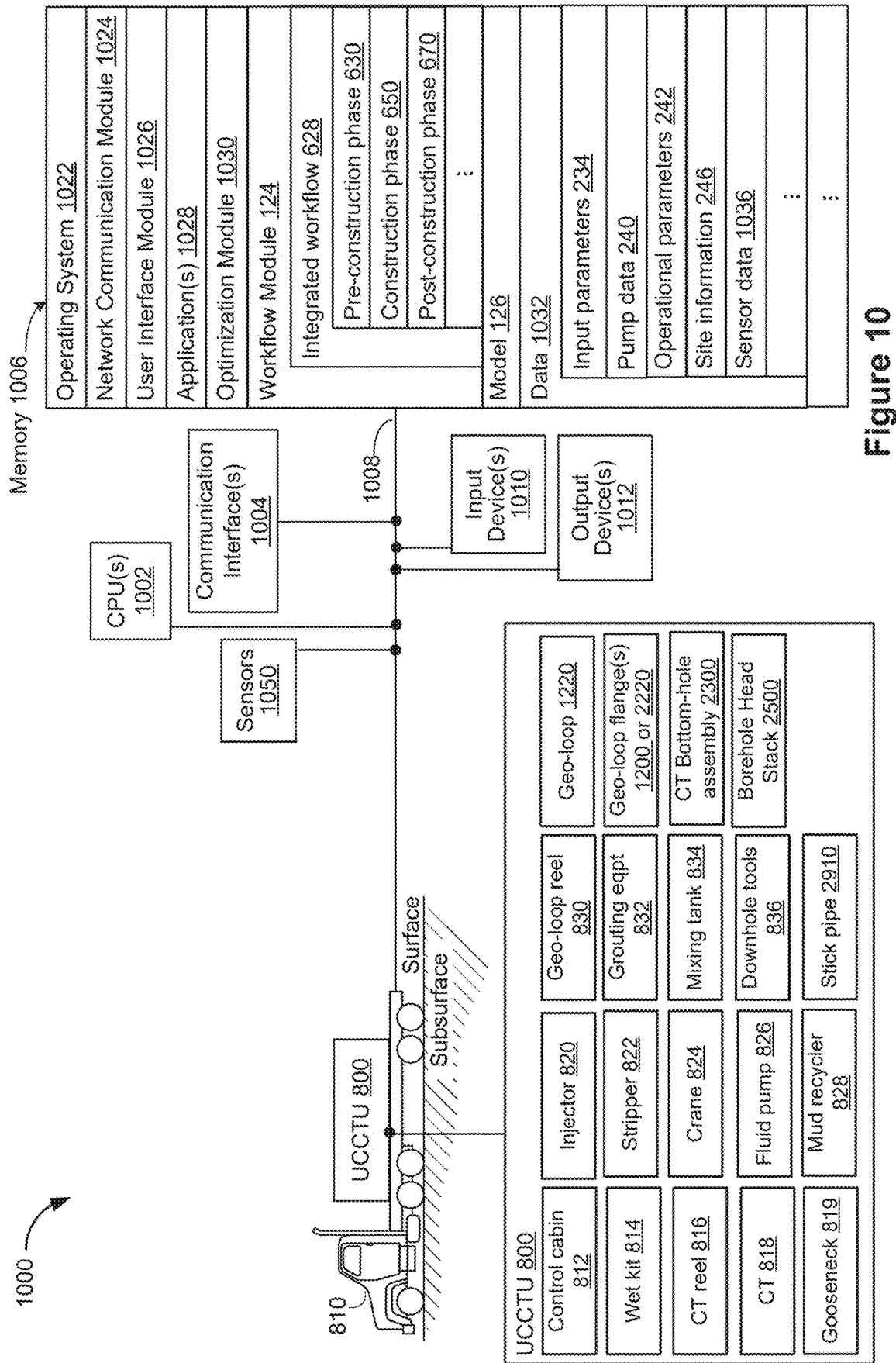


Figure 8E



**Figure 9 (PRIOR ART)**



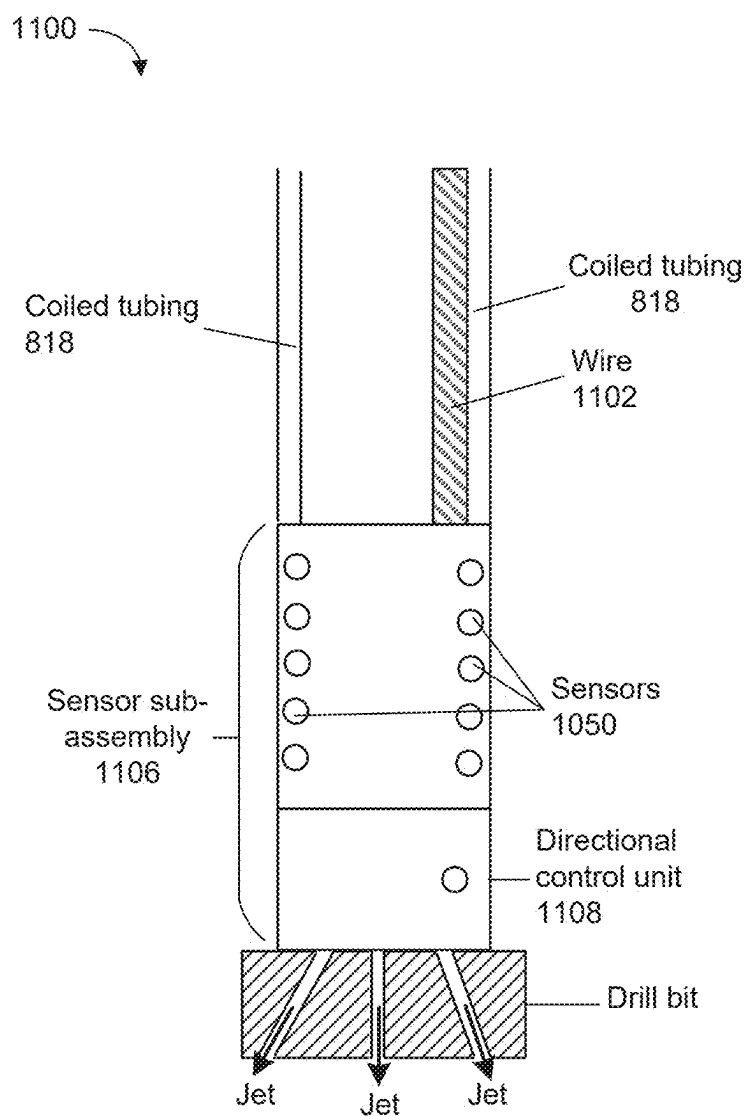


Figure 11

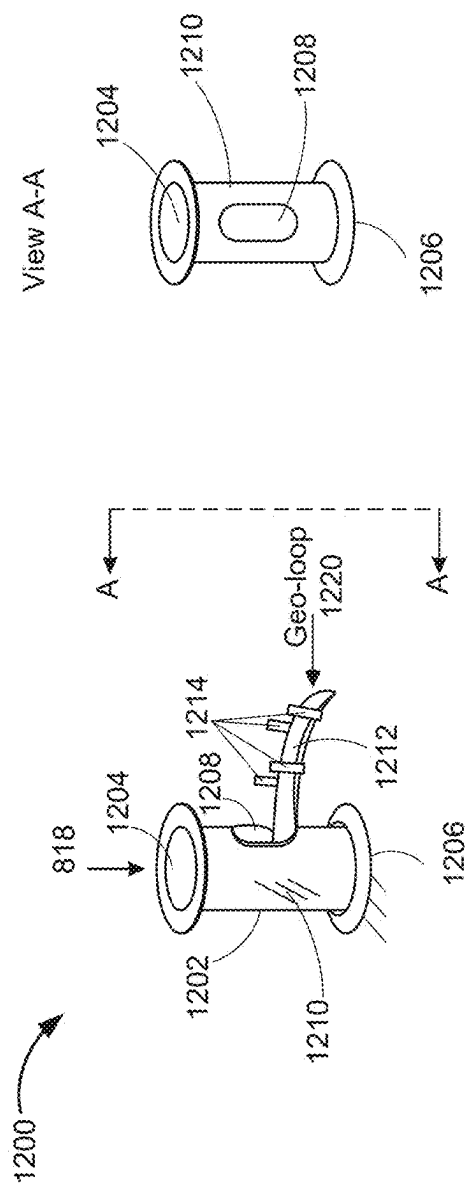


Figure 12A-1

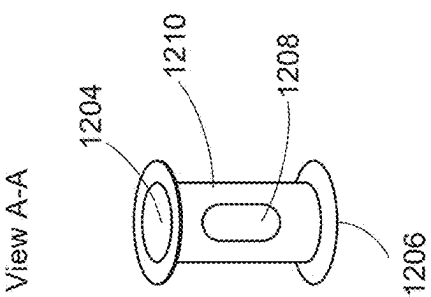


Figure 12A-2

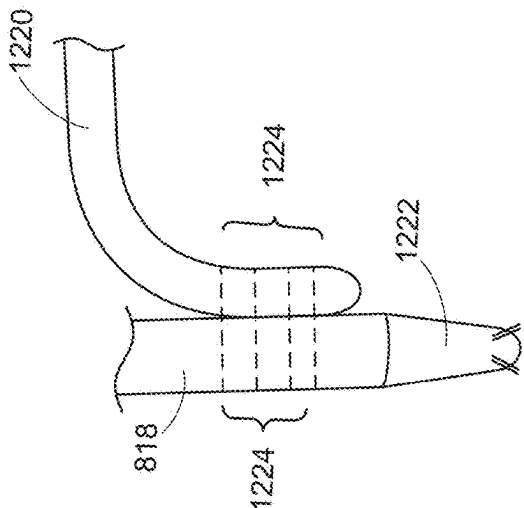


Figure 12B

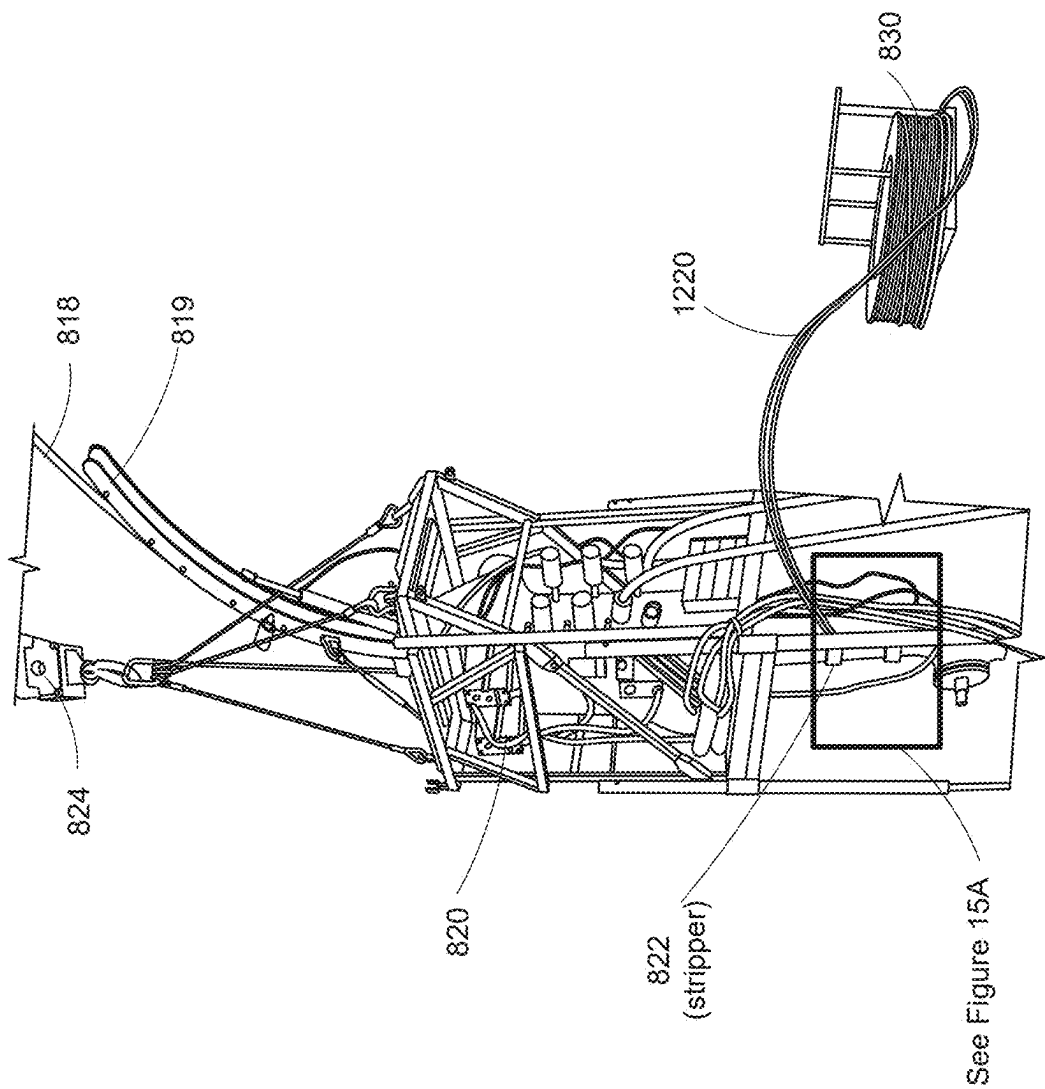


Figure 13

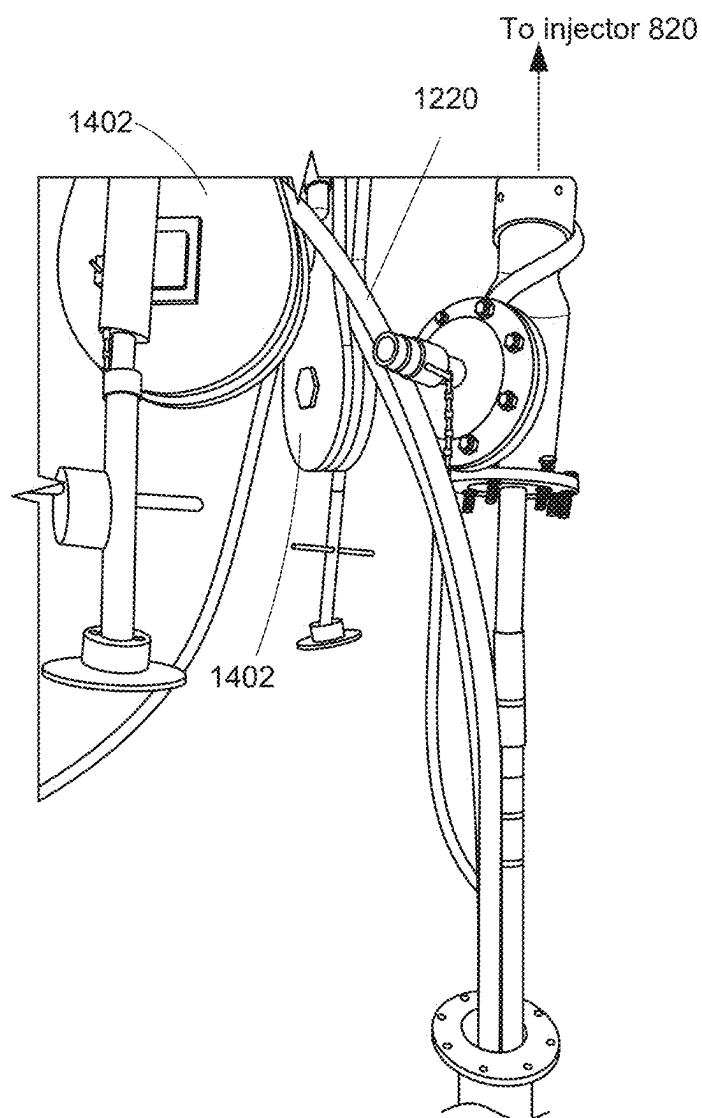


Figure 14



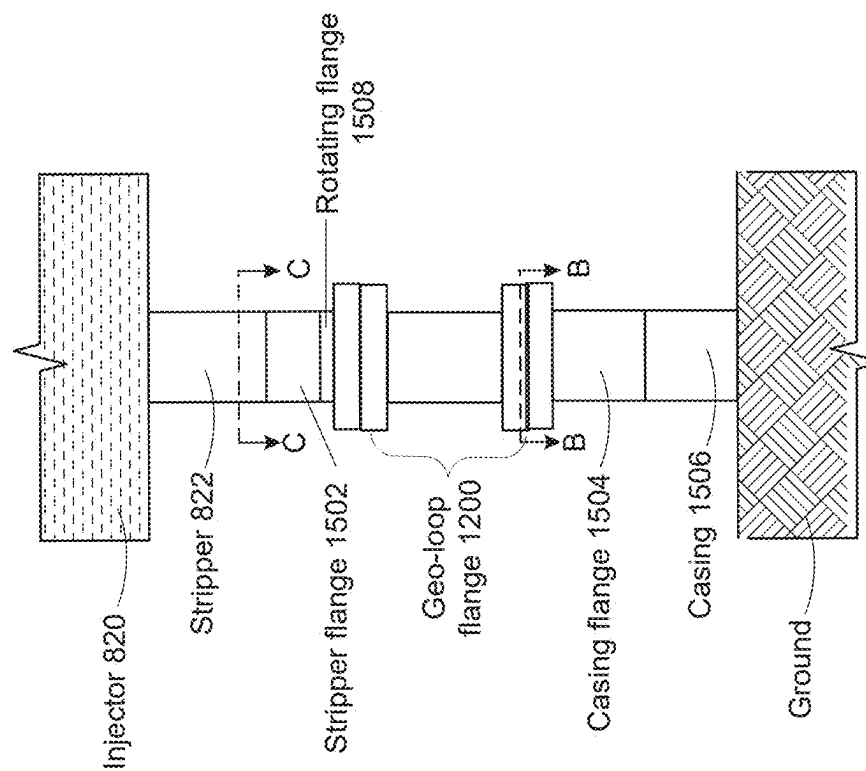


Figure 15A-1

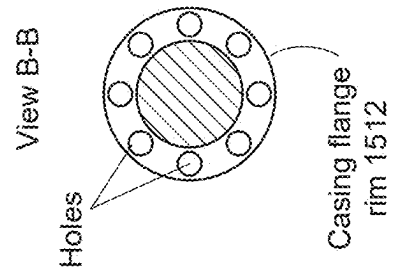


Figure 15A-2

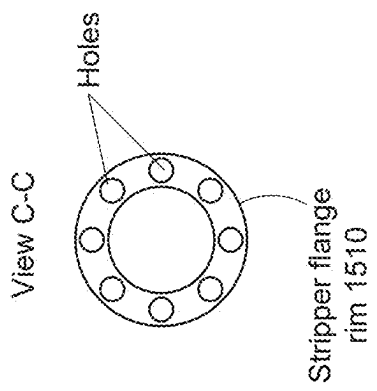


Figure 15A-3

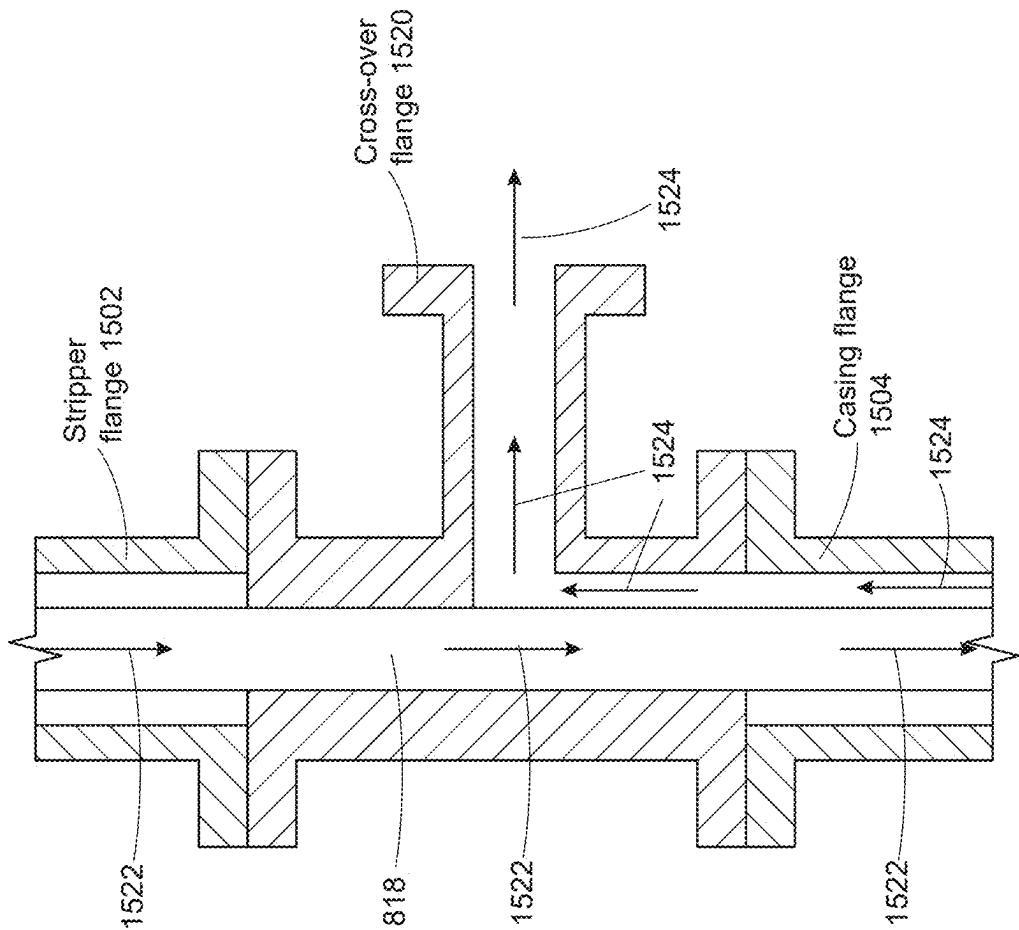


Figure 15B

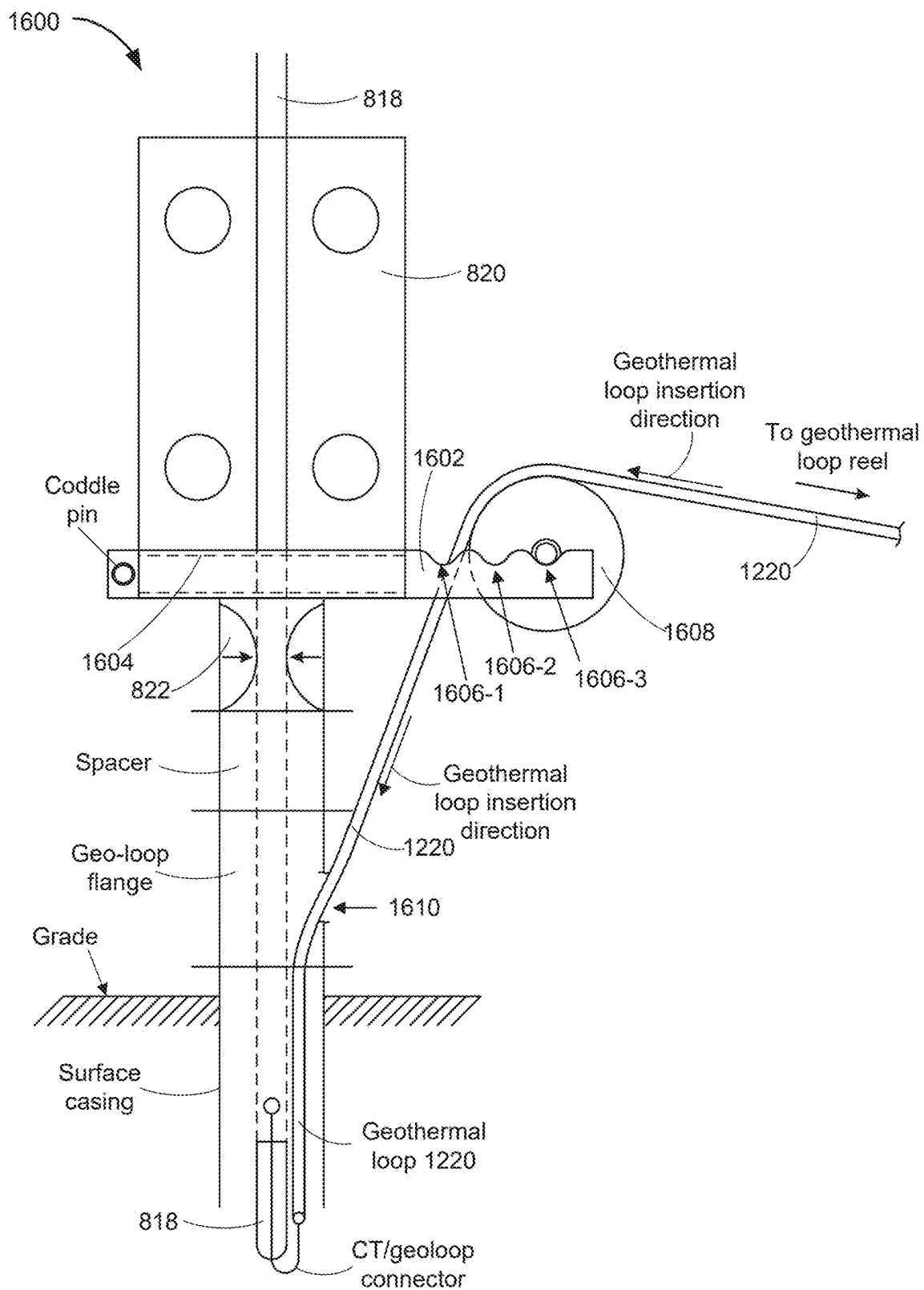


Figure 16

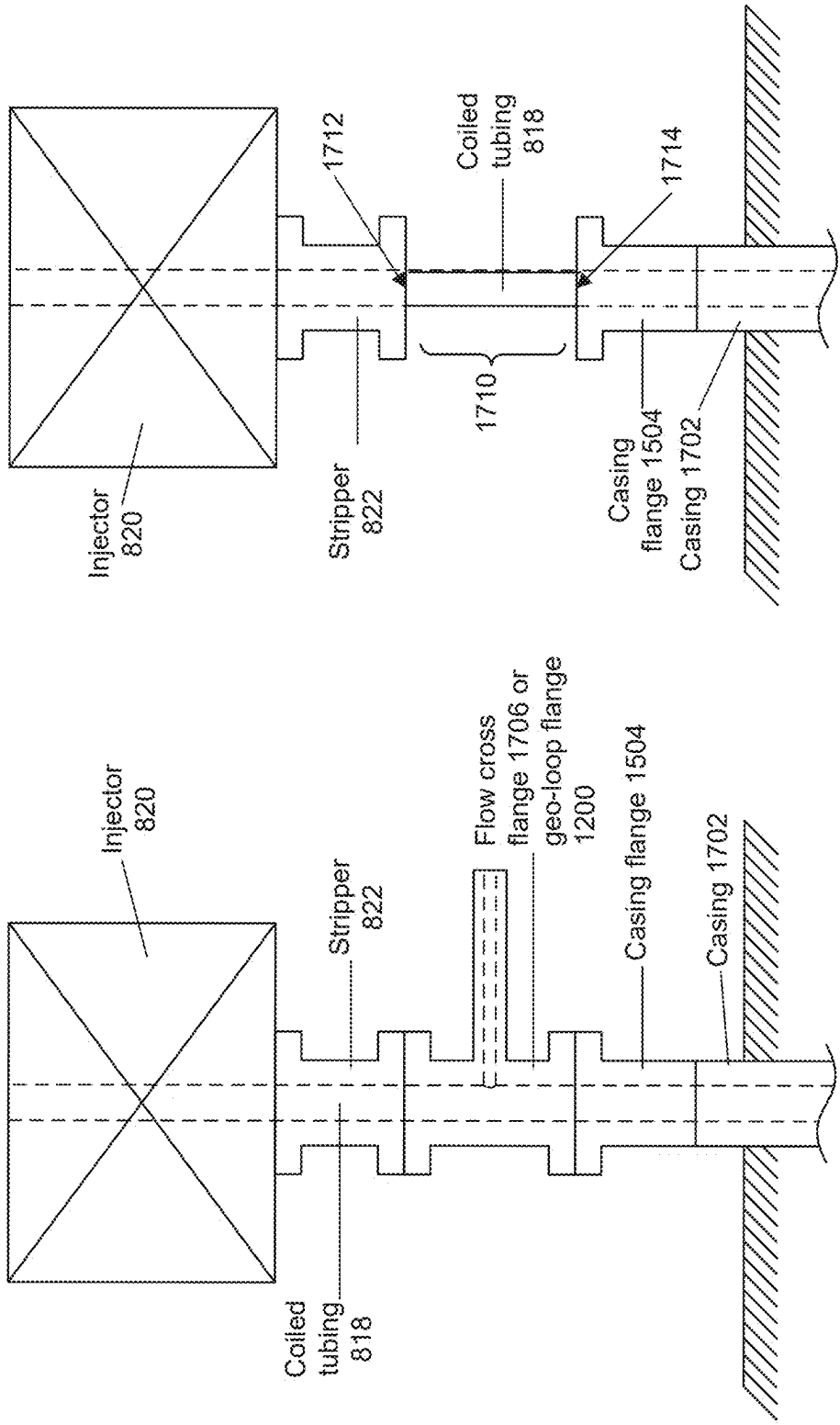


Figure 17A

Figure 17B (PRIOR ART)

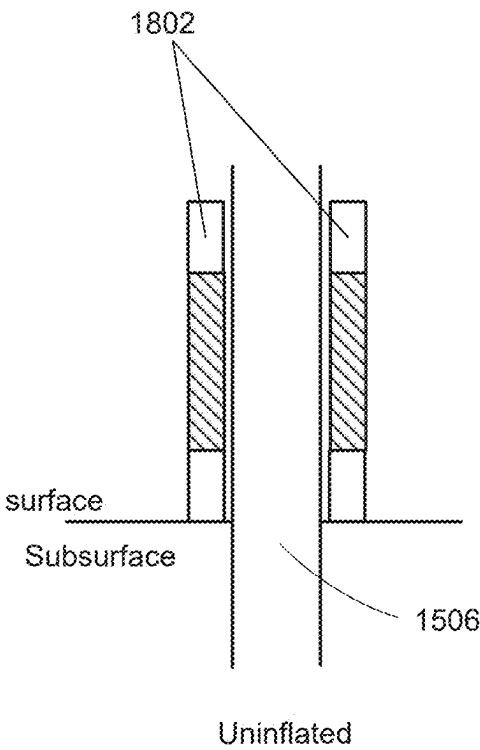


Figure 18A

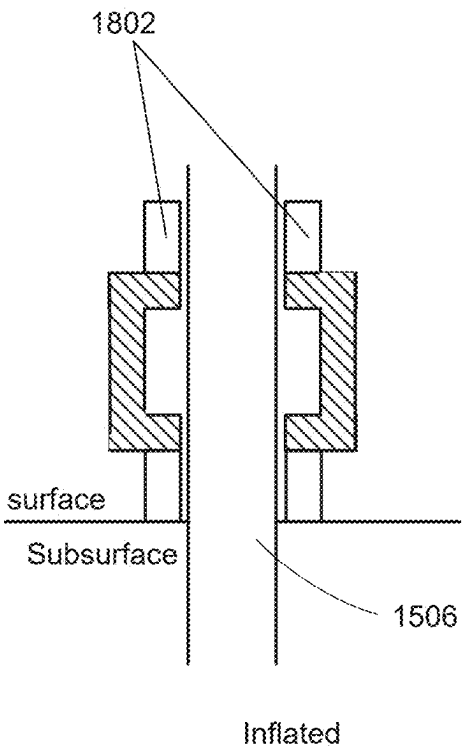


Figure 18B

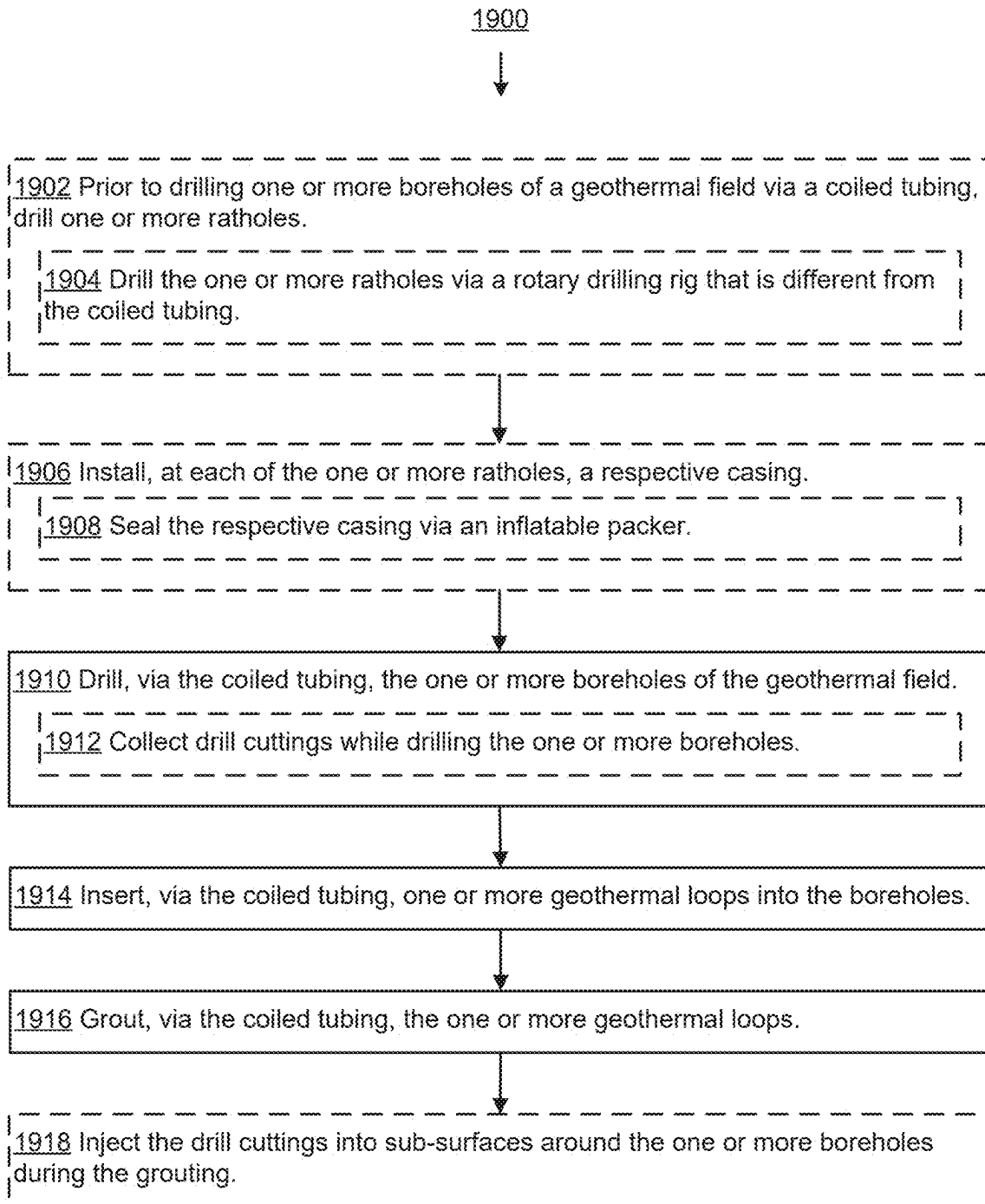


Figure 19

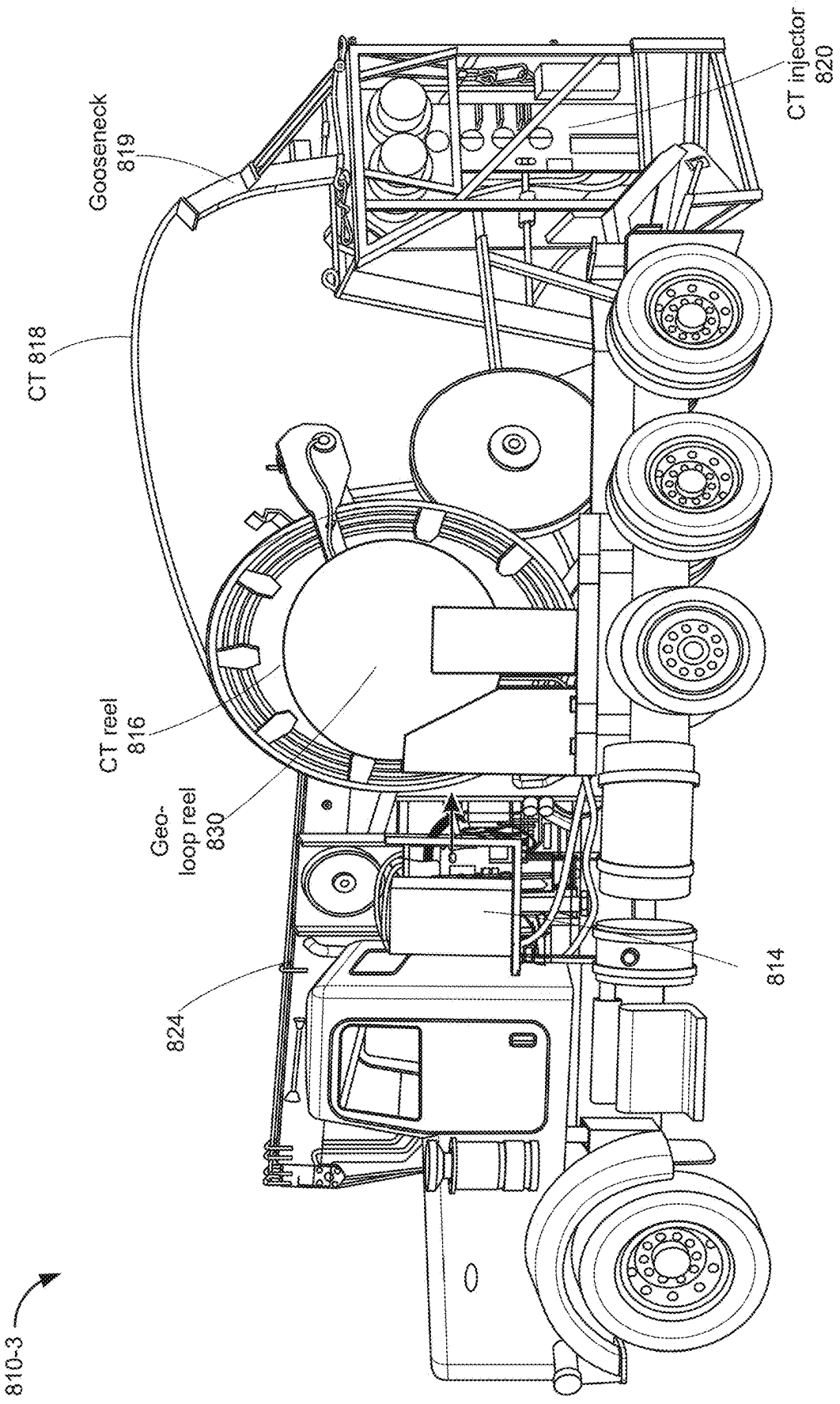


Figure 20

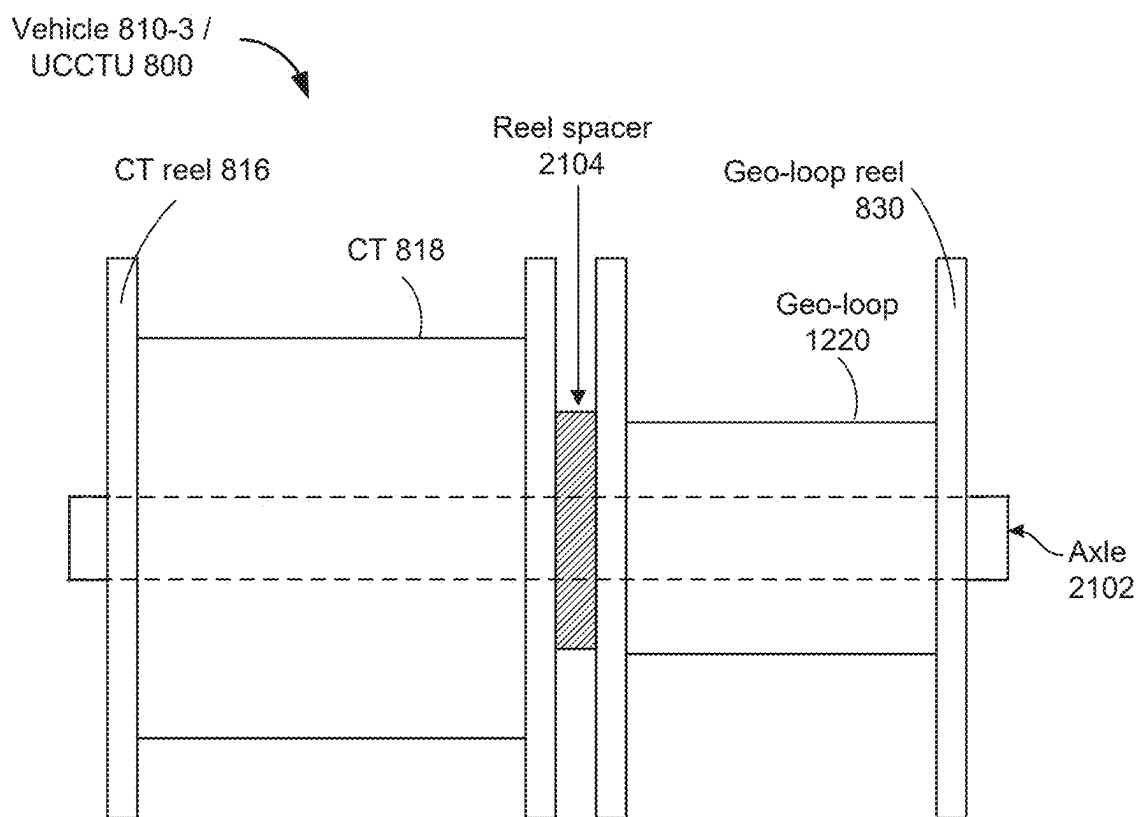


Figure 21A



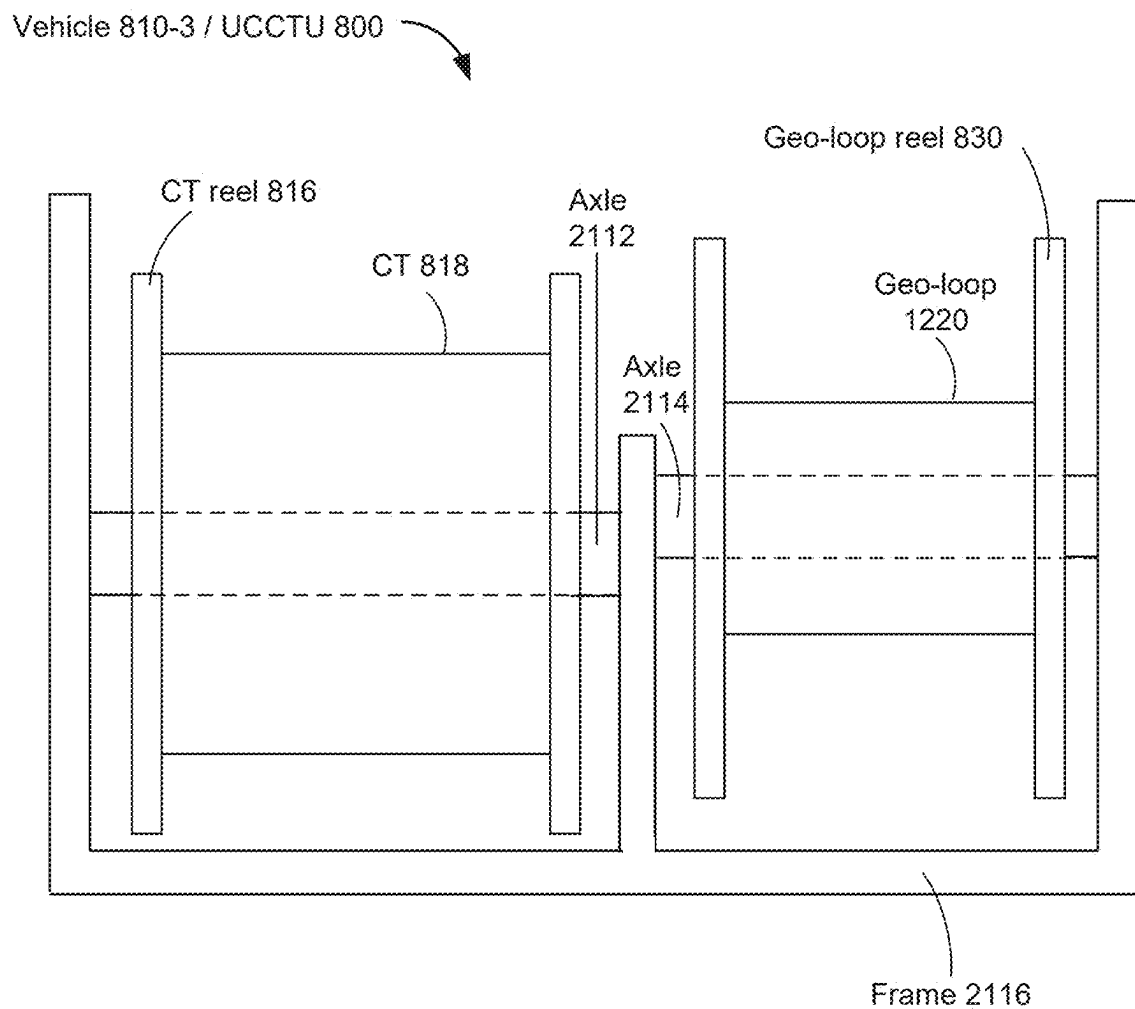


Figure 21B

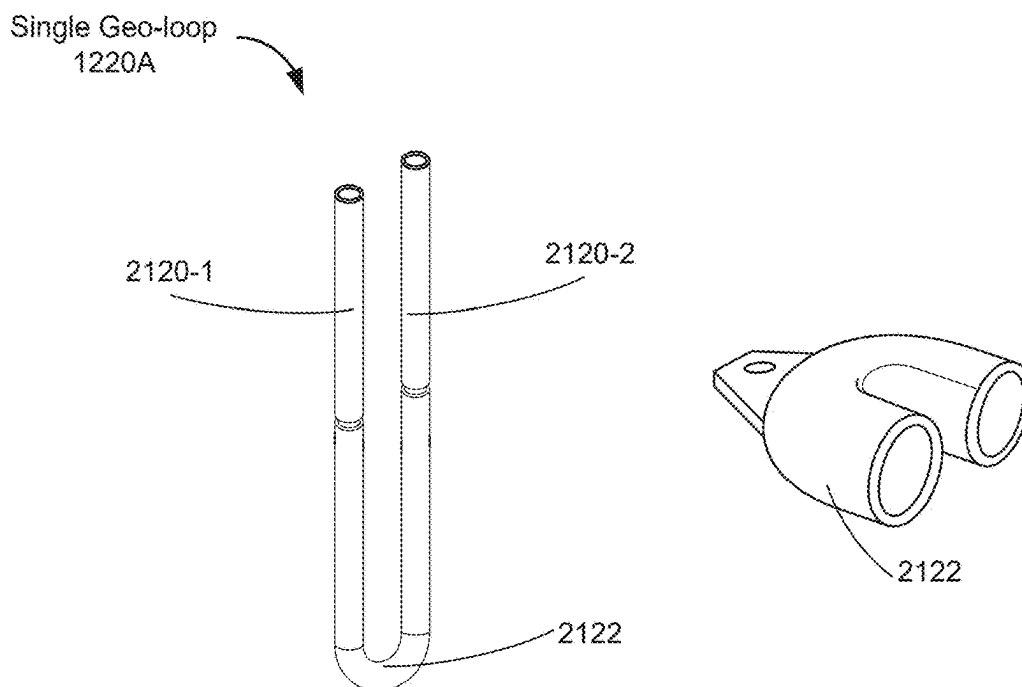


Figure 21C

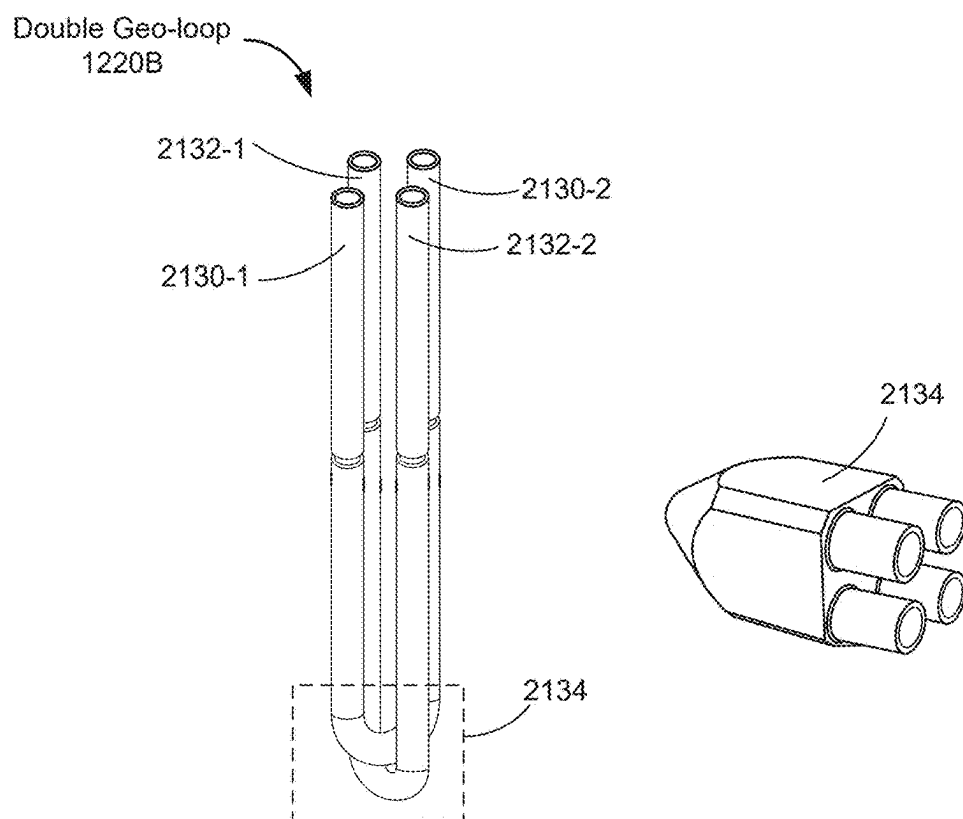


Figure 21D

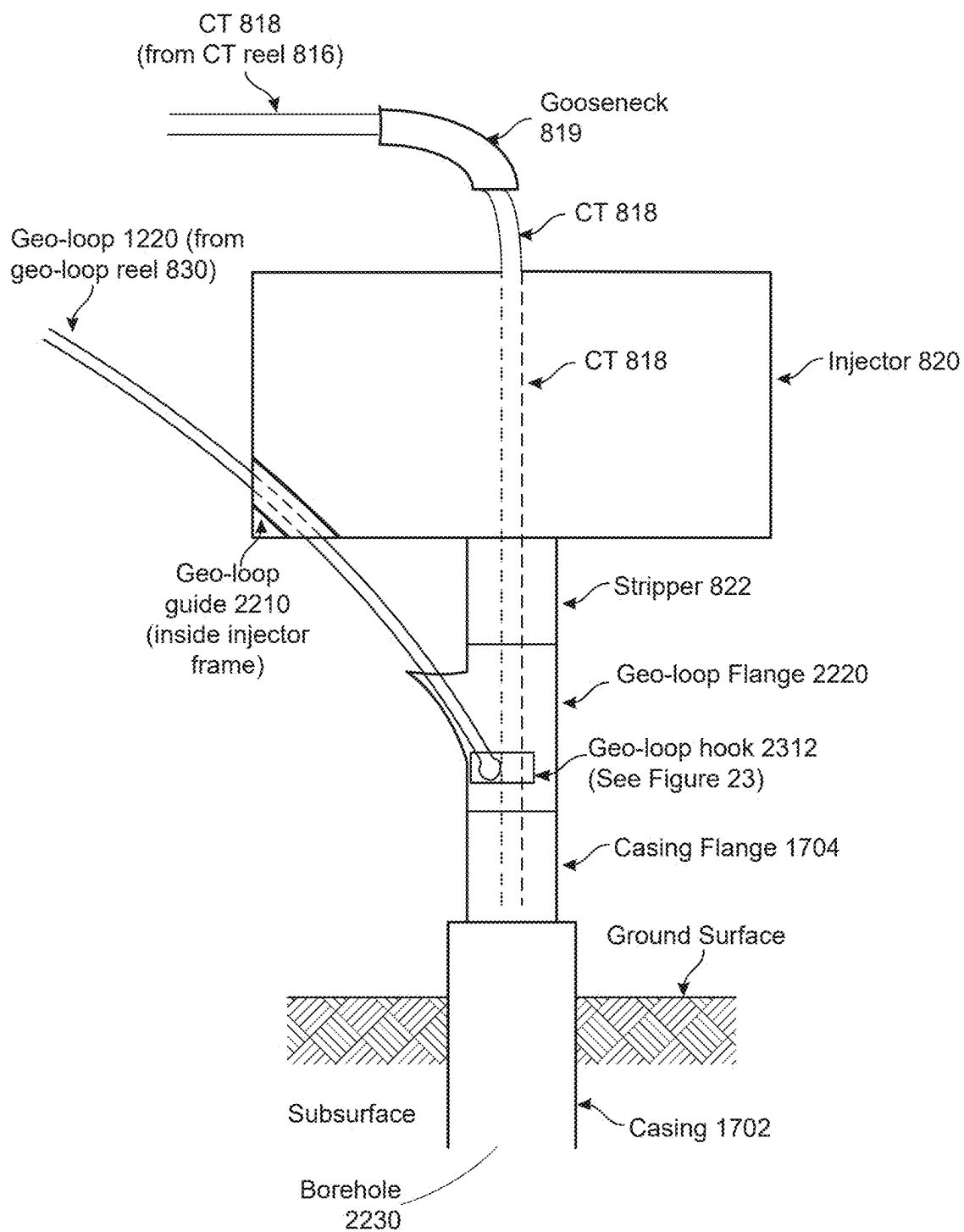


Figure 22

CT Bottom-hole  
assembly 2300

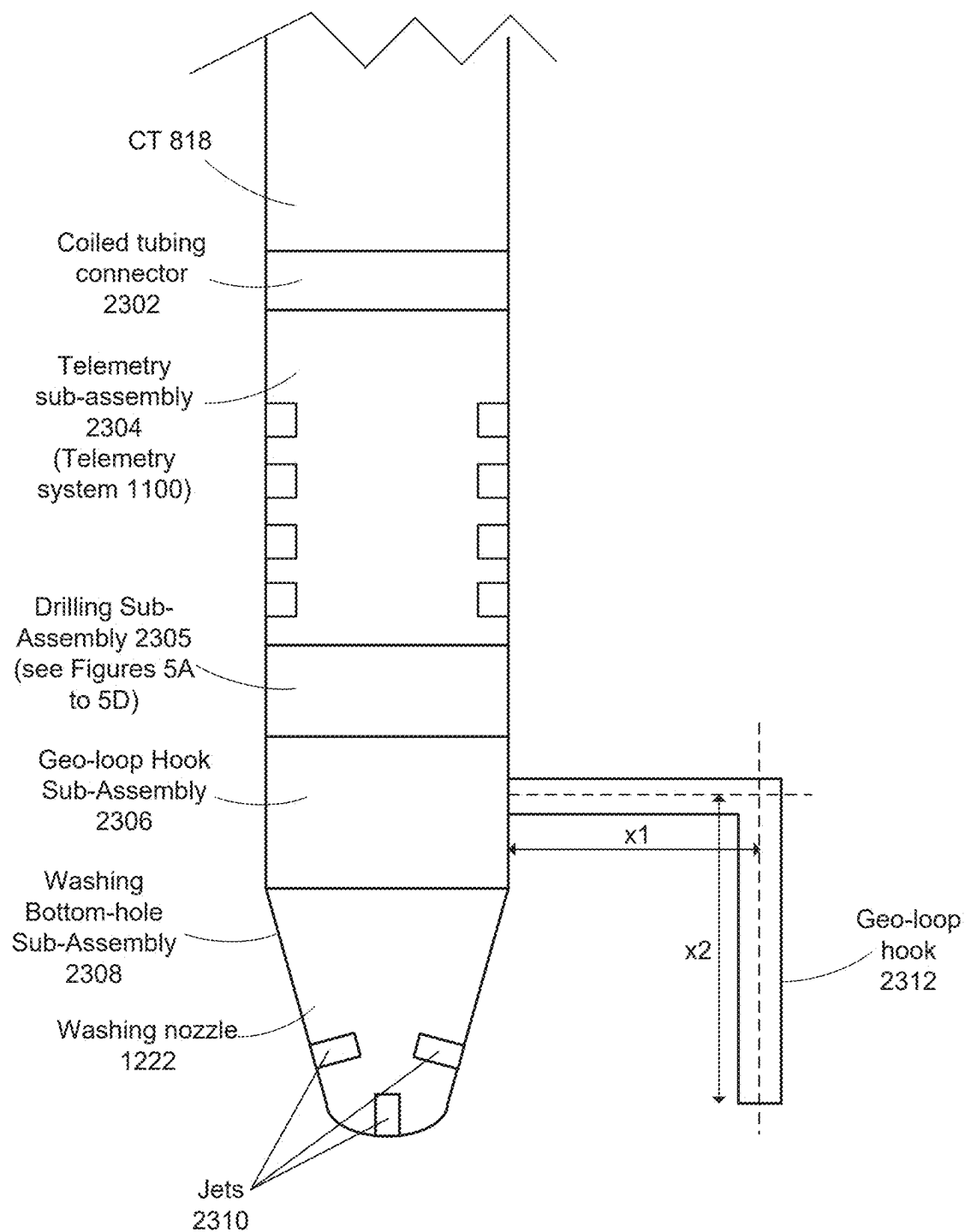


Figure 23

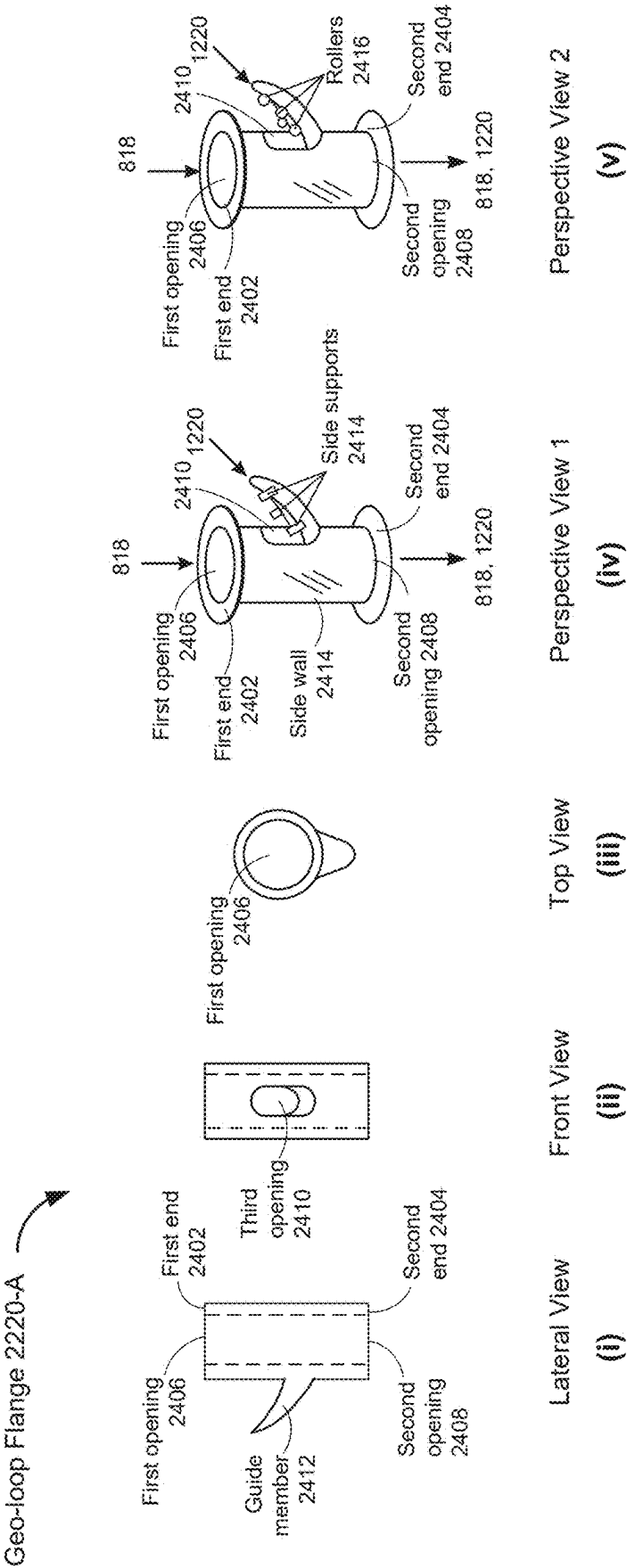


Figure 24A

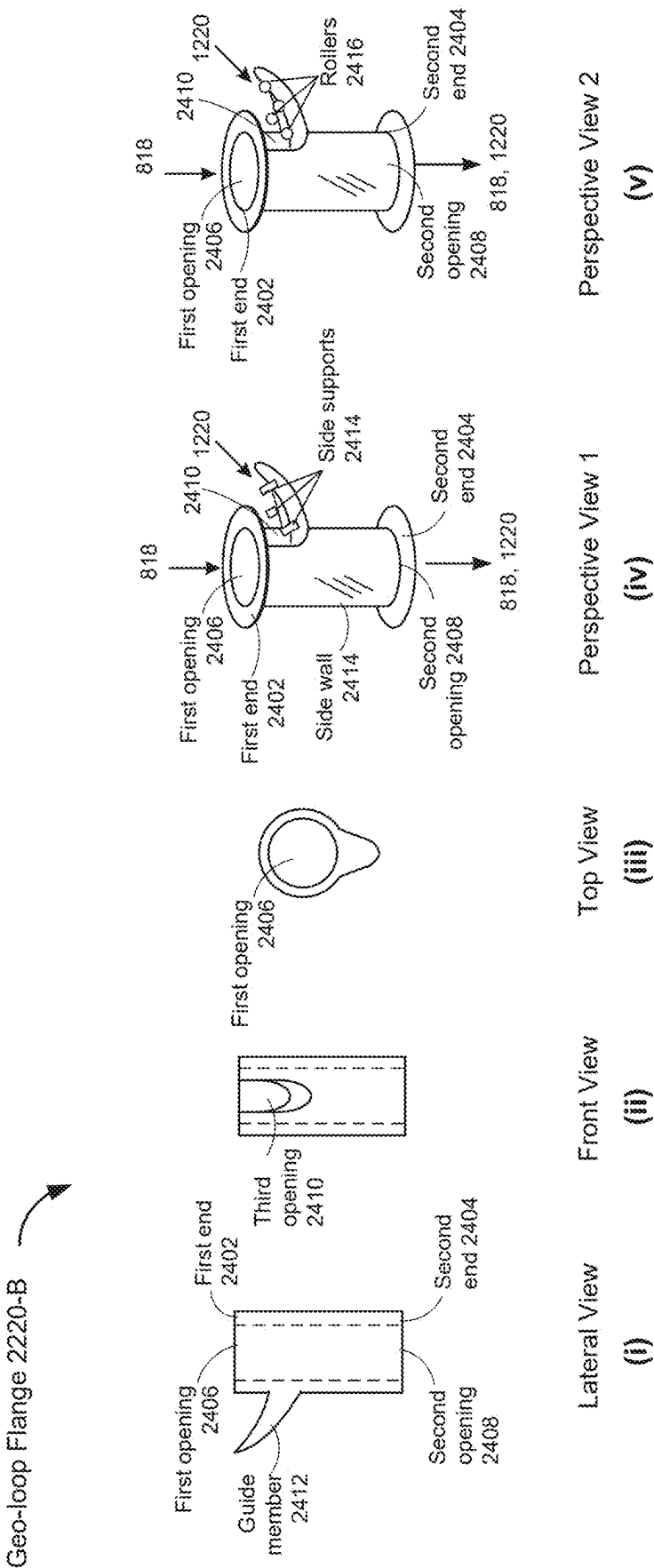


Figure 24B

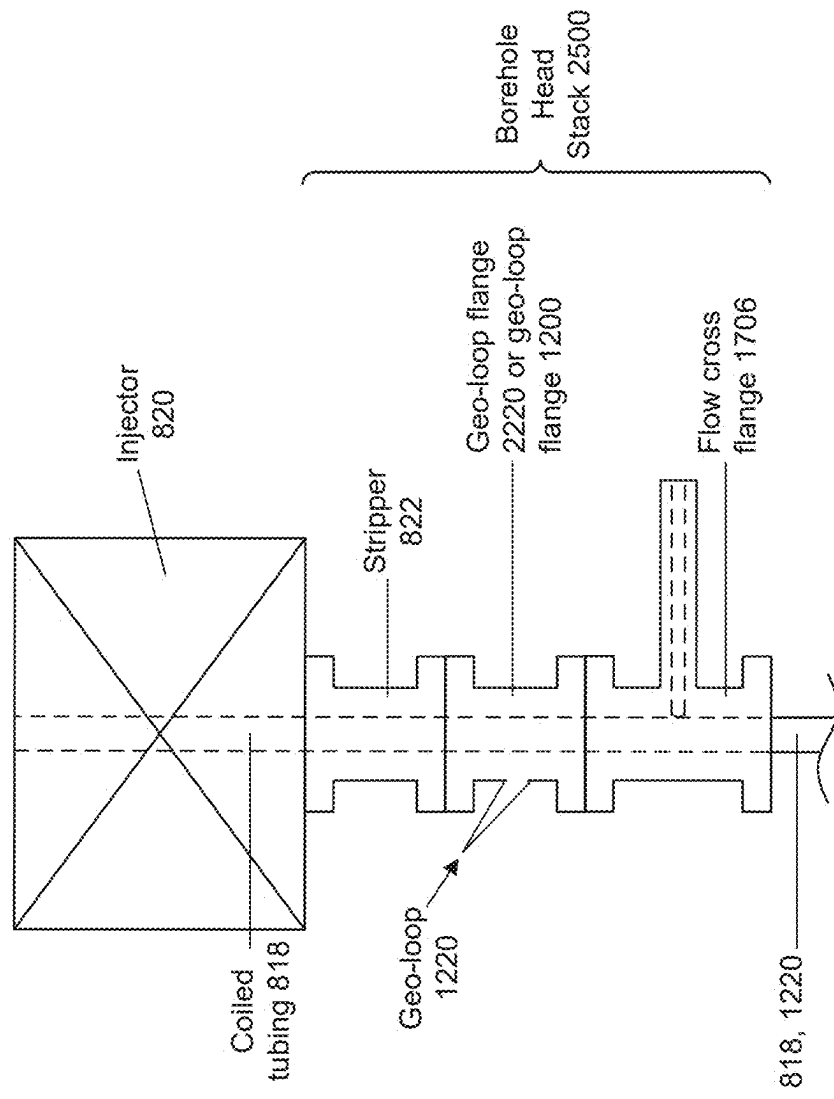


Figure 25

820

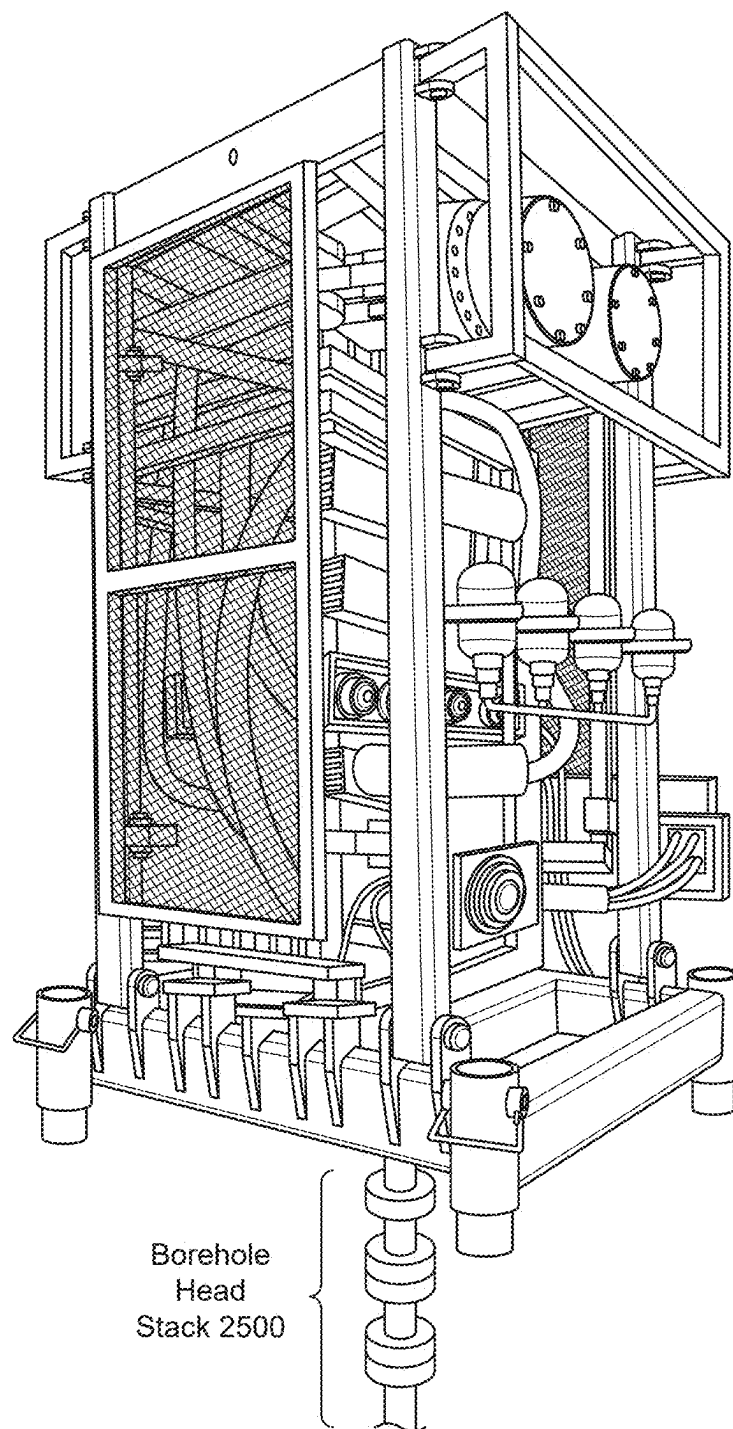


Figure 26



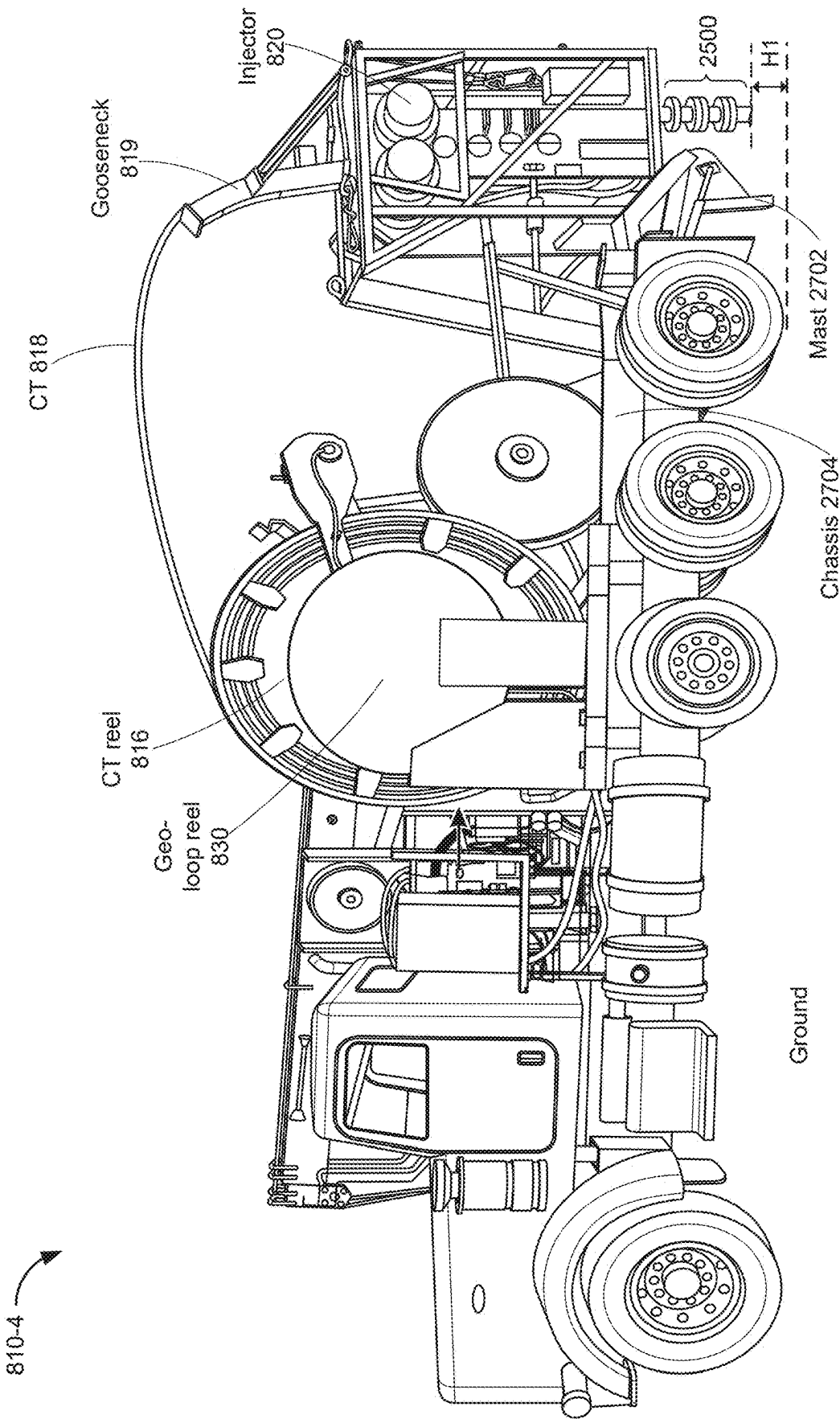


Figure 27

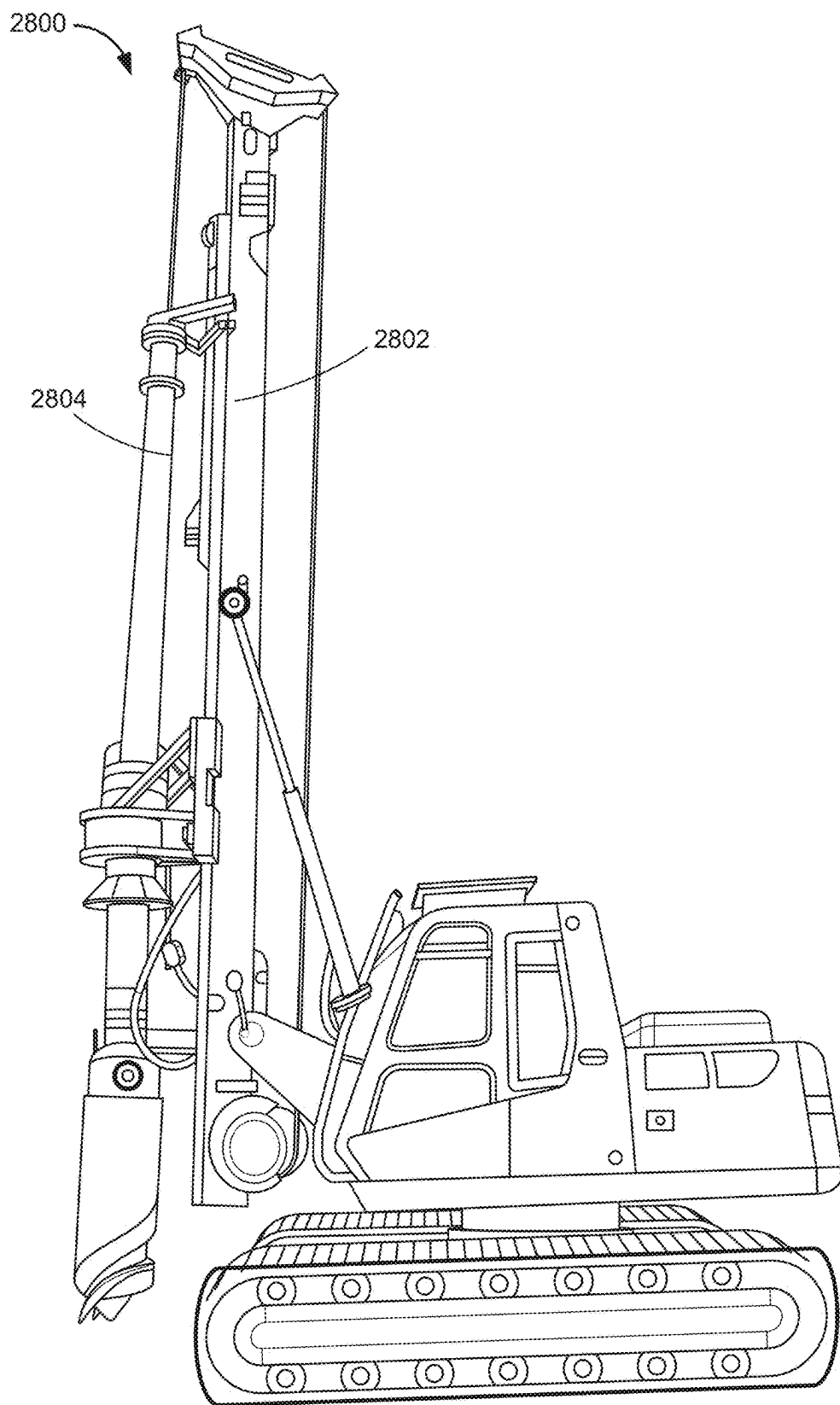


Figure 28 (Prior Art)

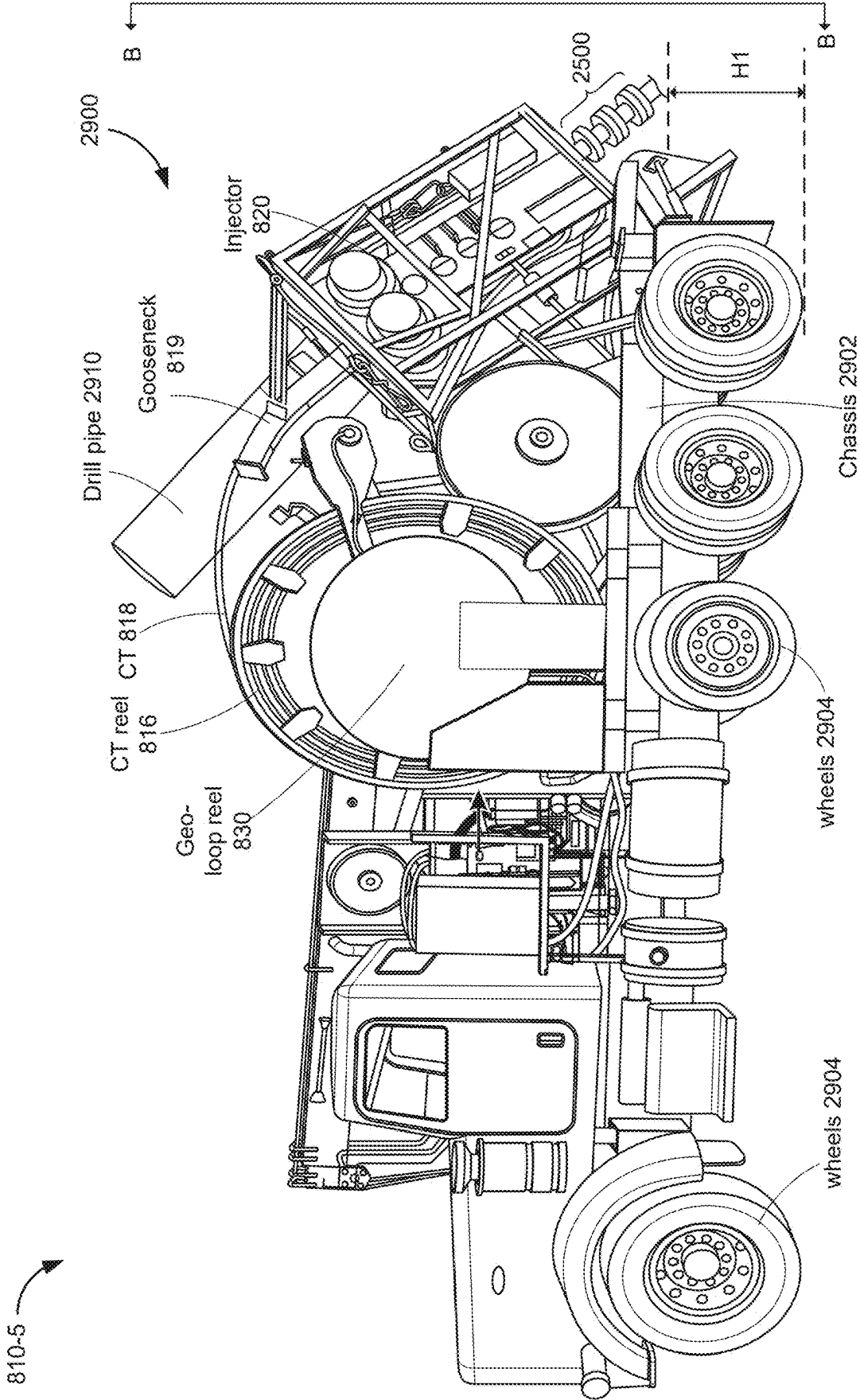


Figure 29

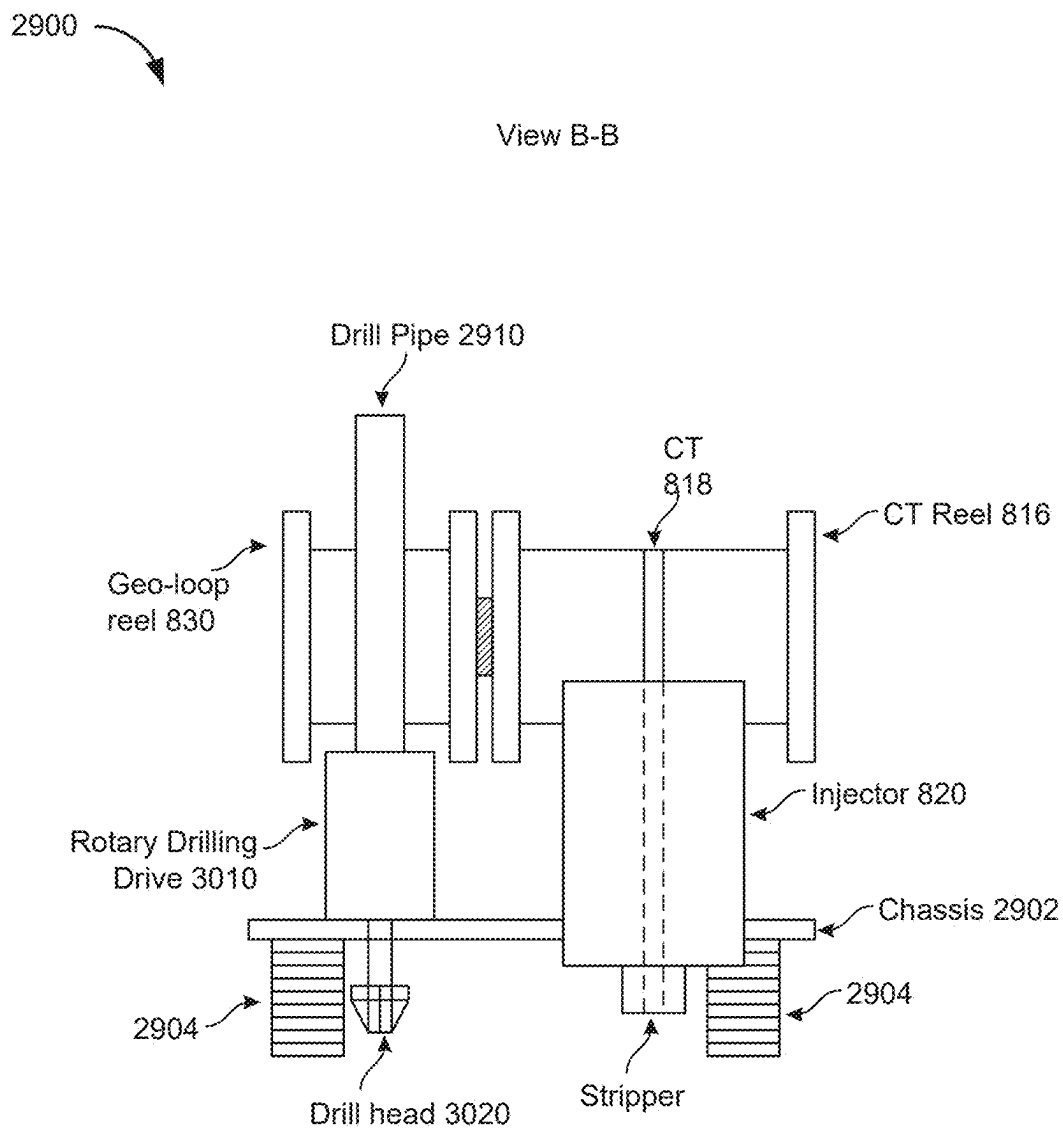


Figure 30

## SYSTEMS AND METHODS FOR GEOTHERMAL LOOP INSTALLATION USING COILED TUBING

### RELATED APPLICATIONS

**[0001]** This application is a continuation-in-part of U.S. patent application Ser. No. 18/629,546, filed Apr. 8, 2024, titled “Ultra-Compact Coiled Tubing (UCCT) Operations for Geothermal Field Construction,” which claims priority to U.S. Provisional Patent Application No. 63/562,947, filed Mar. 8, 2024, titled “Ultra-Compact Coiled Tubing (UCCT) Operations for Geothermal Field Construction,” each of which is incorporated by reference herein in its entirety.

**[0002]** This application is related to the following applications, all of which are incorporated by reference herein in their entireties:

**[0003]** (i) U.S. patent application Ser. No. 17/835,905, filed Jun. 8, 2022, titled “Geothermal Well Construction for Heating and Cooling Operations,” now U.S. Pat. No. 11,520,313, issued on Dec. 6, 2022; and

**[0004]** (ii) U.S. patent application Ser. No. 17/976,445, filed Oct. 28, 2022, titled “Coiled Tubing Drilling for Geothermal Heating and Cooling Applications.”

### TECHNICAL FIELD

**[0005]** The disclosed implementations relate generally to geothermal energy systems, and more specifically to systems, methods, and devices for geothermal field construction.

### BACKGROUND

**[0006]** Geothermal energy is a renewable resource that harnesses the Earth’s heat. Just a few feet below the surface, the Earth maintains a near-constant temperature, in contrast to the summer and winter extremes of the ambient air above ground.

**[0007]** Installing a geothermal heating and cooling system (GHCS) (also referred herein as a geothermal system) generally involves drilling one or more geothermal wells (e.g., geothermal boreholes) horizontally or vertically, depending on the characteristics of the site. A looped pipe is constructed from the geothermal wells, and heat is transferred between the building (e.g., a residential or commercial building) and the earth using fluid circulated through the looped pipes.

### SUMMARY

**[0008]** Current GHCS installations have several drawbacks. First, these installations generally do not take into account surface or subsurface conditions of the site in which the geothermal system would be installed. Instead, standard-sized geothermal wells (e.g., geothermal boreholes) are usually constructed, with depths, lengths, and spacings that are pre-determined based on prior rule-of-thumb experience of the drillers. Second, current residential geothermal systems tend to have fairly shallow well depths (e.g., around 20 feet deep for horizontal geothermal loops) with a large amount of pipes buried horizontally. For example, a typical 2,000-square-foot home uses around 1,500 to 1,800 feet of pipes. Larger buildings, such as industrial, commercial or multi-story residential, would require impractical lengths of horizontal pipes.

**[0009]** Drilling geothermal wells without proper planning and engineering design can lead to non-optimal performance of the geothermal systems, both short-term and in the long run. This directly translates into higher operating costs and unreliable heating and cooling systems for the building owners. Furthermore, the current installation model for residential geothermal systems, which utilizes shallow geothermal wells and large horizontal areas, is not scalable for buildings that are bigger in size (e.g., commercial buildings), have higher energy loads, and/or are located in areas of higher density. For example, a commercial building located in a city may require multiple deep geothermal boreholes, constructed vertically, to be drilled for a geothermal system to properly work.

**[0010]** Current drilling technologies for geothermal and hydrocarbon well construction are divided between (Category A) small drilling rigs for shallow water or geothermal wells (e.g., 100 to 400 feet long vertical (e.g., deep) wells, and (Category B) large drilling rigs for deep oil and gas and conventional geothermal wells (e.g., 10,000 to 30,000 feet long vertical wells). No off-the-shelf drilling rig exists commercially for mid-range-length wells that are between 500 and 10,000 feet deep. Therefore, no off-the-shelf drilling rig has been developed for this range, leaving a technology gap between Categories A and B. While shallow water well drilling rigs (Category A) are small and portable enough to be transported and operated in high-density urban areas, they do not go very deep and are not very fast. The deep oil and gas rigs (Category B) are fast and powerful, but also large and heavy, and are meant to operate in remote areas far from urban/suburban development. This leaves a gap for fast and strong, yet compact and portable drilling rigs that can operate in urban/suburban/rural areas.

**[0011]** In addition to the technology gap for mid-range length geothermal wells, several constraints exist when it comes to constructing geothermal fields in urban (e.g., high-density), suburban, and, or commercial areas. These include space limitations, weight limitations (e.g., the drill rig needs to be transported to and from the construction site via paved concrete roads), and noise generated during well construction. There is also a need for improved systems, methods, and devices for drilling mid-range depth geothermal wells (e.g., about 500-10,000 feet deep, 500-5,000 feet deep, or 500-2,500 feet deep) in urban (e.g., high-density), suburban, and, or commercial areas.

**[0012]** Accordingly, there is a need for improved systems, methods, and devices for geothermal well construction, particularly for mid-range depth geothermal wells.

**[0013]** The present disclosure describes improved systems, devices, and methods for geothermal field construction. As disclosed, an ultra-compact coiled tubing unit (UCCTU) includes coiled tubing is used in a drilling phase, a geothermal loop insertion phase, and a grouting phase of a geothermal field construction. The UCCTU disclosed herein fills the technology void for mid-range-length wells that are between ~500 feet and ~3,000 feet deep. The UCCTU is designed and built considering inner-city weight, width, length, and height limitations, and can be deployed for geothermal drilling in dense urban areas, for decarbonizing buildings and reducing their dependance on the electrical grid.

**[0014]** The present disclosure advantageously improves upon current geothermal well construction processes in several ways. First, the current construction process requires

geothermal loops to be manually inserted into boreholes. This is a difficult and physically demanding task that becomes even more challenging when the well depths are ~800 feet and upwards. Utilizing coiled tubing for geothermal loop insertion makes the process easier, faster, and more efficient. Coiled tubing can be driven into the ground using the injector. The driving force provided to the CT by the injector is large enough to push geothermal loops to depths unheard of by the current shallow geothermal drilling industry. Second, the industry currently uses Tremie lines for the grouting phase. Because grout is a viscous paste when pumped down, pumping grout through one-inch-diameter tremie lines at depths greater than ~500 to 600 feet is operationally impossible. The present disclosure improves the grouting practice by using washing nozzles attached to the coiled tubing to squeeze grout to fill spaces between a borehole and a geothermal loop.

**[0015]** The systems, methods, and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.

**[0016]** In accordance with some embodiments, a system for constructing a geothermal field comprises a coiled tubing that is configured for (i) drilling (e.g., constructing) one or more boreholes of the geothermal field; (ii) inserting one or more geothermal loops into the one or more boreholes; and (iii) grouting the one or more geothermal loops.

**[0017]** In accordance with some embodiments, an apparatus for facilitating insertion of a geothermal loop into a borehole includes a coiled tubing guide member having a first end, a second end, and an opening on a side wall of the first member. The coiled tubing guide member is configured for guiding a coiled tubing through the first end and the second end. The apparatus includes a geothermal loop guide member physically coupled to the coiled tubing guide member at the opening. The geothermal loop guide member is configured for guiding the geothermal loop through the opening and through the second end.

**[0018]** In accordance with some embodiments, a method for constructing geothermal fields includes (i) drilling, via a coiled tubing, one or more boreholes of a geothermal field; (ii) inserting, via the coiled tubing, one or more geothermal loops into the constructed boreholes; (iii) and grouting, via the coiled tubing, the one or more geothermal loops.

**[0019]** In accordance with some embodiments, a system for installing geothermal loops comprises a coiled tubing unit (CTU). The CTU includes a geothermal loop spooled on a geothermal loop reel that is mounted on a first axle. The CTU includes a coiled tubing string configured for inserting the geothermal loop into a borehole. The coiled tubing string is spooled on a coiled tubing reel that is mounted on a second axle. As the geothermal loop is inserted into the borehole, the geothermal loop is configured to unspool from the geothermal loop reel; the coiled tubing string is configured to unspool from the coiled tubing reel; and the geothermal loop and the coiled tubing string are configured to move at substantially similar speeds.

**[0020]** In accordance with some embodiments, a computing device includes one or more processors, memory, and one or more programs stored in the memory. The programs are configured for execution by the one or more processors. The one or more programs include instructions for performing any of the methods described herein.

**[0021]** In accordance with some embodiments, a non-transitory computer-readable storage medium stores one or more programs configured for execution by a computing device having one or more processors and memory. The one or more programs include instructions for performing any of the methods described herein.

**[0022]** Thus, methods, systems, and devices are disclosed that enable coiled tubing technology to be used drilling, geothermal loop insertion, and grouting phases of a geothermal field construction process.

**[0023]** Note that the various embodiments described above can be combined with any other embodiments described herein. The features and advantages described in the specification are not all inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes and may not have been selected to delineate or circumscribe the inventive subject matter.

#### BRIEF DESCRIPTION OF THE DRAWINGS

**[0024]** FIG. 1 illustrates an operating environment in accordance with some embodiments.

**[0025]** FIG. 2 illustrates a block diagram of a geothermal planning and optimization system according to some embodiments.

**[0026]** FIG. 3 is a block diagram illustrating a server system, in accordance with some embodiments.

**[0027]** FIGS. 4A and 4B illustrate drilling of a geothermal borehole in accordance with some embodiments.

**[0028]** FIGS. 5A, 5B, 5C, 5D-1, 5D-2, and 5D-3 illustrate a drill bit for generating geothermal boreholes in accordance with some embodiments.

**[0029]** FIG. 5E illustrates exemplary types of surface data and downhole data that are collected by sensors in accordance with some embodiments.

**[0030]** FIGS. 6A to 6C illustrate an integrated workflow for planning, constructing, and/or optimizing a GHCS operation, in accordance with some embodiments.

**[0031]** FIG. 7 illustrates a heat pump and a geothermal loop in accordance with some embodiments.

**[0032]** FIGS. 8A to 8E illustrate components of an ultra-compact coiled tubing unit (UCCTU), in accordance with some embodiments.

**[0033]** FIG. 9 illustrates a coiled tubing reel in prior art systems.

**[0034]** FIG. 10 illustrates a block diagram of a system for constructing geothermal loops, according to some embodiments.

**[0035]** FIG. 11 illustrates a downhole telemetry system in accordance with some embodiments.

**[0036]** FIGS. 12A-1 and 12A-2 illustrate a geo-loop flange, in accordance with some embodiments.

**[0037]** FIG. 12B illustrates physically coupling a coiled tubing and a geothermal loop, in accordance with some embodiments.

**[0038]** FIG. 13 illustrates geothermal loop insertion via a geo-loop flange and a coiled tubing, in accordance with some embodiments.

**[0039]** FIG. 14 illustrates rollers for stabilizing a geothermal loop 1220, in accordance with some embodiments.

[0040] FIGS. 15A-1, 15A-2, and 15A-3 illustrate a stripper flange, a geo-loop flange, and a casing flange, in accordance with some embodiments.

[0041] FIG. 15B illustrates a cross-over flange, in accordance with some embodiments.

[0042] FIG. 16 illustrates a geothermal loop insertion method, in accordance with some embodiments.

[0043] FIG. 17A illustrates a wellhead connection to an injector, in accordance with some embodiments of the present disclosure. FIG. 17B illustrates a wellhead connection to an injector in existing systems.

[0044] FIGS. 18A and 18B illustrate cross-sectional views of an inflatable packer in uninflated and inflated states, respectively, in accordance with some embodiments.

[0045] FIG. 19 provides a flowchart of a method for constructing geothermal fields, in accordance with some embodiments.

[0046] FIG. 20 illustrates a view of a vehicle that includes a coiled tubing unit, in accordance with some embodiments.

[0047] FIGS. 21A and 21B are example views of a coiled tubing reel and a geothermal loop reel, in accordance with some embodiments.

[0048] FIG. 21C shows a cross-sectional view of a single geothermal loop, in accordance with some embodiments.

[0049] FIG. 21D shows a cross-sectional view of a double geothermal loop, in accordance with some embodiments.

[0050] FIG. 22 illustrates respective paths of a geothermal loop and a coiled tubing string during the geothermal loop insertion phase, in accordance with some embodiments.

[0051] FIG. 23 illustrates a coiled tubing bottom-hole assembly, in accordance with some embodiments.

[0052] FIGS. 24A and 24B illustrate various embodiments of a geo-loop flange and their respective views, in accordance with some embodiments.

[0053] FIG. 25 illustrates a borehole head stack, in accordance with some embodiments.

[0054] FIG. 26 illustrates a borehole head stack, in accordance with some embodiments.

[0055] FIG. 27 illustrates a view of a vehicle for transporting a coiled tubing injector with a connected borehole head stack, in accordance with some embodiments.

[0056] FIG. 28 illustrates a stick pipe derrick 2800 in existing systems, in accordance with some embodiments.

[0057] FIG. 29 illustrates a view of a vehicle that includes a hybrid coiled tubing-stick pipe rig, in accordance with some embodiments.

[0058] FIG. 30 illustrates a view of the hybrid coiled tubing-stick pipe rig, in accordance with some embodiments.

[0059] Reference will now be made to implementations, examples of which are illustrated in the accompanying drawings. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. However, it will be apparent to one of ordinary skill in the art that the present invention may be practiced without requiring these specific details.

#### DESCRIPTION OF IMPLEMENTATIONS

[0060] Some methods, devices, and systems disclosed in the present specification improve upon geothermal heating and cooling system (GHCS) operations by providing an integrated workflow that combines data modeling, hardware

components, and a workflow for planning, executing, and/or optimizing a GHCS installation operation.

[0061] In accordance with some aspects of the present disclosure, the GHCS operation includes drilling one or more geothermal wells (e.g., boreholes). GHCS drilling employ various technologies and methods to drill and construct geothermal boreholes in urban, suburban, and rural areas. Methods of drilling geothermal boreholes include joint pipe drilling and CT drilling. Performing an optimal geothermal drilling operation involves modeling design before the operation, using downhole parameters such as a borehole size and length, mechanical friction forces, tensile forces, pumping rates, soil and rock properties, pressure and/or temperature. Some aspects of the present disclosure describe using surface and subsurface (e.g., underground) sensors for acquiring data such as pressure, temperature, depth correlation and azimuth, rock hardness, pumping rates, etc., for adjusting the pre-operation design in real time while drilling. Some aspects of the present disclosure describe using the same model after the GHCS drilling and construction operation is finished, to monitor the geothermal loop inlet and outlet temperature and enthalpy, the indoor temperature, and/or the coefficient of performance of the heat pump in real time, and to adjust the pump rate to maintain an optimal heating and cooling performance.

[0062] In this disclosure, a “geothermal field” is also referred to as a “geo-field.” Both of these terms have the same meaning.

[0063] In this disclosure, a “geothermal loop” is also referred to as a “geo-loop.” Both of these terms have the same meaning.

[0064] FIG. 1 illustrates an operating environment 100 in accordance with some embodiments. In some embodiments, the operating environment 100 includes one or more geothermal planning and optimization systems 200 (e.g., a computer system a computing device, or an electronic device). In the example of FIG. 1, each of the systems 200-1, 200-2, 200-3, and 200-N is located at a respective (e.g., distinct) site (e.g., Site A, Site B, Site C, Site N), corresponding to a respective (e.g., distinct) city, state, or country.

[0065] In some embodiments, each of the geothermal planning and optimization systems 200 is configured to be operable at various phases (e.g., all phases, all stages, or a subset thereof) of a GHCS operation. The phases can include a pre-construction phase, a construction phase, and a post-construction phase. For example, in FIG. 1, the geothermal planning and optimization system 200-1 located in Site A corresponds to the pre-construction phase. The system 200-2 located in Site B corresponds to the construction phase. The systems 200-3 and 200-N, in Site C and Site N respectively, correspond to the post-construction phase.

[0066] In some embodiments, the geothermal planning and optimization system 200 is communicatively coupled through communication network(s) 110 to a server system 120.

[0067] In some embodiments, the server system 120 includes a front end server 122 that facilitates communication between the server system 120 and the geothermal planning and optimization system 200. The front end server 122 is configured to receive information from the geothermal planning and optimization system 200. For example, during the planning (e.g., pre-construction) phase for a geothermal borehole construction project at Site A, the front

end server **122** can receive (e.g., in real-time), from the geothermal planning and optimization system **200-1**, information such as building size, site information, and/or data regarding rock type found at Site A. As another example, during the construction of a geothermal borehole at Site B, the front end server **122** can receive (e.g., in real-time) from the geothermal planning and optimization system **200-2** information such as sensor data collected by one or more sensors of a drilling rig that is performing the borehole construction at Site B. As another example, the front end server **122** can receive (e.g., in real-time) from the geothermal planning and optimization system **200-3** information such as a fluid flow rate, a heat pump coefficient of performance (COP) for heating, and/or a heat pump energy efficiency ratio (EER) for cooling, corresponding to a geothermal loop **604-1** at Site C.

**[0068]** In some embodiments, the front end server **122** is configured to send information to the one or more geothermal planning and optimization systems **200**. For example, the front end server **122** can send operational parameters (e.g., drilling depth, diameter, and/or direction) to the geothermal planning and optimization system **200-1**, to facilitate drilling of the geothermal borehole at Site A. As another example, in response to receiving the sensor data from the geothermal planning and optimization system **200-2**, the front end server **122** can send updated operational parameters to the geothermal planning and optimization system **200-2**, to control (e.g., optimize) the dimensions of the geothermal borehole that is being constructed at Site B. As another example, in response to receiving data associated with the geothermal loop **604-1**, the front end server **122** can send to the geothermal planning and optimization system **200-3** information such as a desired flow rate for the working fluid, so as to optimize the operation of the geothermal loop **604-1**.

**[0069]** In some embodiments, the server system **120** includes a workflow module **124** for providing an integrated workflow **628** corresponding a geothermal heating and cooling system operation. Details of the workflow module **124** are described in FIGS. **2** and **3**, and **6A** to **6C**, as well as U.S. patent application Ser. No. 17/835,905 and U.S. patent application Ser. No. 17/976,445, both of which are incorporated by reference herein in their entireties.

**[0070]** In some embodiments, the server system **120** includes a model **126** (e.g., a physics-based model, a mathematical-based model, a data-driven model, a machine learning algorithm) for modeling the heating and cooling loads of the building, the heat pump, and the GHCS well (e.g., geothermal borehole(s)) sizes and lengths. Details of the model **126** are discussed in FIGS. **2** and **3**.

**[0071]** In some embodiments, the server system **120** includes a database **128**, which is described with respect to FIG. **3**.

**[0072]** In some embodiments, the server system **120** includes a machine learning database **130** that stores machine learning information. In some embodiments, the machine learning database **130** is a distributed database. In some embodiments, the machine learning database **130** includes a deep neural network database. In some embodiments, the machine learning database **130** includes supervised training and/or reinforcement training databases.

**[0073]** FIG. **2** illustrates a block diagram of a geothermal planning and optimization system **200** according to some embodiments.

**[0074]** The system **200** typically includes one or more processors (e.g., processing units, or CPUs) **202**, one or more network or other communication interfaces **204**, memory **206**, and one or more communication buses **208** for interconnecting these components. In some embodiments, the communication buses **208** include circuitry (sometimes called a chipset) that interconnects and controls communications between system components.

**[0075]** In some embodiments, the system **200** is communicatively connected to one or more sensors **508** that are positioned on a drill bit **502** of a CT **408** and/or a joint pipe **422**, the details of which are described in FIGS. **4**, **5**, and **6**.

**[0076]** In some embodiments, the system **200** is communicatively connected to one or more heat pumps **602** that are operably coupled to one or more respective geothermal loops **604**. Details of the heat pump **602** and the geothermal loop are described in FIG. **7**.

**[0077]** The system **200** includes one or more input devices **210** that facilitate user input, such as a keyboard, a mouse, a voice-command input unit or microphone, a touch screen display, a touch-sensitive input pad, a gesture capturing camera, or other input buttons or controls. In some embodiments, the system **200** includes one or more cameras or scanners for capturing data. The system **200** also includes one or more output devices **212** that enable presentation of user interfaces and display content, including one or more speakers and/or one or more visual displays.

**[0078]** In some embodiments, the memory **206** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and, optionally, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. In some embodiments, the memory **206** includes one or more storage devices remotely located from one or more processing units **202**. The memory **206**, or alternatively the non-volatile memory device(s) within the memory **206**, includes a non-transitory computer-readable storage medium. In some embodiments, the memory **206** or the computer-readable storage medium of the memory **206** stores the following programs, modules, and data structures, or a subset or superset thereof:

**[0079]** an operating system **222**, including procedures for handling various basic system services and for performing hardware dependent tasks;

**[0080]** a communications module **224**, which is used for connecting the system **200** to other systems, computers and devices via the one or more communication interfaces **204** (wired or wireless), such as the communication network(s) **110**, Internet, other wide area networks, local area networks, metropolitan area networks, and so on;

**[0081]** a user interface module **226**, which enables presentation of information (e.g., a graphical user interface for application(s) **228**, widgets, websites and web pages thereof, and/or audio and/or video content, text, etc.) at the system **200** via one or more output devices **212** (e.g., displays, speakers, etc.);

**[0082]** one or more user applications **228**, which are executed by the system **200** (e.g., web or non-web based applications for controlling another system, electronic device, or sensors, or for reviewing data captured by such devices);



[0083] a workflow module **124**, which provides an integrated workflow **628** corresponding to a GHCS operation;

[0084] a model **126**, which models the heating and cooling loads of the building, the heat pump, and the GHCS well (e.g., geothermal borehole(s)) sizes and lengths); and

[0085] data **232**, including:

[0086] input parameters **234**. In some embodiments, the input parameters **234** include surface parameters **236**. The surface parameters **236** include: building size, building type (e.g., wall material and insulation; roof type; number, size, and orientation of windows; ventilation system; etc.), location, outdoor temperature, heat pump size, heat pump type, and/or inlet/outlet temperatures of the geothermal loop. In some embodiments, the input parameters **234** include underground parameters **238**. The underground parameters **238** include: fluid type (e.g., density and/or viscosity, and thermal conductivity), geothermal loop roughness, inlet temperatures of the geothermal loop(s), geothermal loop length and/or depth, size (e.g., diameter), and trajectory (e.g., vertical, horizontal, inclined, and/or deviated), geothermal loop material/heat transfer coefficient, soil and/or rock type and properties, aquifer depth and height, grout/cement/casing heat transfer coefficients, and geothermal working fluid flow rate;

[0087] pump data **240**, including manufacturer heat pump COP (for heating) and/or EER (for cooling) and/or calculated heat pump COP and/or EER (e.g., during borehole construction or during operation of a geothermal loop);

[0088] operational parameters **242**, including drilling depth(s), drilling diameter(s), and/or drilling direction(s);

[0089] sensor data **244**. In some embodiments, the sensor data **244** is collected automatically and/or in real time. In some embodiments, the sensor data **244** is collected directly (e.g., by sensors **508** or sensors **510**) or by indirect measurements. In some embodiments, the sensor data **244** includes surface data and/or down-hole data, as illustrated in FIG. 5E;

[0090] site information **246**, including geographical location(s) of geothermal borehole(s), rock conditions, heat pump(s), geothermal loop(s) and their inlet and outlet temperatures, working fluid type(s), and/or working fluid flow rates; and

[0091] model results **248** generated by the model **126**.

[0092] In some embodiments, the integrated workflow **628** includes a combination of one or more workflows, each corresponding to a respective phase of a geothermal heating and cooling operation (e.g., a geothermal project). The integrated workflow **628** can include a combination of: a pre-construction phase workflow **630**, a construction phase workflow **650**, and a post-construction phase workflow **670**. In some embodiments, each of the workflows **630**, **650**, and **670** can be executed as a standalone workflow. Further details of the workflows **630**, **650**, and **670** are discussed in FIGS. 6A to 6C, as well as in U.S. patent application Ser. No. 17/835,905 and U.S. patent application Ser. No. 17/976,445, both of which are incorporated by reference in their entireties.

[0093] In some embodiments, the model **126** comprises a physics-based model, a mathematical-based model, a data-

driven model, and/or a machine learning algorithm. In some embodiments during a pre-construction phase, the model **126** uses data **232**, such as input parameter(s) **234**, a heat pump COP for heating, and/or a heat pump EER for cooling, to predict (e.g., estimate) the length and/or depth of the borehole(s) to be drilled and/or estimate one or more locations for the boreholes. The model **126** generates operational parameters **242** for constructing a geothermal borehole. In some embodiments, during a construction phase, the model **126** uses sensor data **244** (e.g., collected in real time using sensors **508** and/or sensors **510**) to further refine and/or optimize the operational parameters **242**. In some embodiments, during a post-construction phase, the model **126** uses inputs such as a surface air temperature, a building size, a heat pump COP for heating, and/or a heat pump EER for cooling, for determining heat exchange parameters at a geothermal loop and determining an optimized flow rate of the fluid in the geothermal loop.

[0094] In some embodiments, the model **126** comprises a subsurface fluid transport and heat transfer model. The model **126** allows a user to automatically engineer an entire subsurface system (e.g., engineer the number, size(s), length(s) of geothermal wells (e.g., boreholes), and the distance between wells) specifically designed for each building.

[0095] In some embodiments, the model **126** includes a surface sub-model and an underground sub-model. The surface sub-model solves energy balance equations for the building and heat pump(s). For example, the surface sub-model uses a subset of the surface parameters **236** and generates, as outputs, the heat pump efficiency and the geothermal working fluid flow rate. The underground sub-model solves mass, momentum, and energy conservation equations in the geothermal loop(s) and radial diffusivity equation(s) in the underground rocks around the geothermal borehole(s). The underground sub-model uses a subset of the underground parameters **238** and generates (e.g., outputs) an outlet temperature of the geothermal loop. In some embodiments, the model **126** couples the surface sub-model and the underground sub-model and solves them together. In this instance, the model **126** uses a subset of the surface parameters **236** and the underground parameters **238** as inputs, and calculates an optimum heat pump efficiency by adjusting the geothermal working fluid flow rate. The unknowns in the model **126** include: underground heat transfer coefficients in the rocks, aquifer, and grout/cement (e.g., the grout/cement thickness may vary along the borehole length). Thus, inlet/outlet temperatures of the geothermal loops and geothermal working fluid flow rates (e.g., site information **246**) are measured over time and data sets (e.g., training data training data **326**) are built to predict the underground heat transfer coefficients using non-linear least square solvers such as the gradient-based Levenberg-Marquardt algorithm. This, in turn, allows for adjusting the geothermal working fluid flow rates for continuously optimal heat pump efficiencies (e.g., during the day, at night or every day of the year).

[0096] Although FIG. 2 shows a system **200**, FIG. 2 is intended more as a functional description of the various features that may be present rather than as a structural schematic of the implementations described herein. In practice, and as recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated.

[0097] Each of the above identified executable modules, applications, or sets of procedures may be stored in one or

more of the memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various implementations. In some embodiments, the memory 206 stores a subset of the modules and data structures identified above. Furthermore, the memory 206 may store additional modules or data structures not described above (e.g., module(s) for machine learning and/or training models). In some embodiments, a subset of the programs, modules, and/or data stored in the memory 206 can be stored on and executed by server system 120 or by system 1000.

[0098] FIG. 3 is a block diagram illustrating a server system 120, in accordance with some embodiments.

[0099] The server system 120 includes one or more processors 302 (e.g., processing units of CPU(s)), one or more network interfaces 304, memory 306, and one or more communication buses 308 for interconnecting these components (sometimes called a chipset), in accordance with some embodiments.

[0100] In some embodiments, the server system 120 includes one or more input devices 310 that facilitate user input, such as a keyboard, a mouse, a voice-command input unit or microphone, a touch screen display, a touch-sensitive input pad, a gesture capturing camera, or other input buttons or controls. In some embodiments, the server system 120 uses a microphone and voice recognition or a camera and gesture recognition to supplement or replace the keyboard. In some embodiments, the server system 120 includes one or more output devices 312 that enable presentation of user interfaces and display content, such as one or more speakers and/or one or more visual displays.

[0101] The memory 306 includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and, in some embodiments, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. In some embodiments, the memory 306 includes one or more storage devices remotely located from the one or more processors 302. The memory 306, or alternatively the non-volatile memory within the memory 306, includes a non-transitory computer-readable storage medium. In some embodiments, the memory 306, or the non-transitory computer-readable storage medium of the memory 306, stores the following programs, modules, and data structures, or a subset or superset thereof:

[0102] an operating system 322, including procedures for handling various basic system services and for performing hardware dependent tasks;

[0103] a front end 122, which communicatively couples the server system 120 to other devices and/or systems (e.g., the geothermal planning and optimization systems 200-1, 200-2, 200-3, . . . , 200-N) via the network interface(s) 304 (wired or wireless) and one or more networks, such as communication network(s) 110, the Internet, other wide area networks, local area networks, metropolitan area networks, and so on;

[0104] a user interface module 323, which enables presentation of information (e.g., a graphical user inter-

face for presenting application(s), widgets, websites and web pages thereof, games, audio and/or video content, text, etc.) either at the server system or at a geothermal planning and optimization system 200 (e.g., a computing device);

[0105] a workflow module 124, which provides an integrated workflow 628 corresponding a GHCS operation. In some embodiments, the integrated workflow 628 includes one or more of: a pre-construction phase workflow 630, a construction phase workflow 650, and a post-construction phase workflow 670;

[0106] a model 126;

[0107] a training module 324, which generates and/or trains geothermal planning and optimization models (e.g., model 126) for predicting underground heat transfer coefficient(s). In some embodiments, the training module 324 uses training data 326 to generate and/or train the models. In some embodiments, the training data 326 (e.g., training data sets) is generated from geothermal loop inlet/outlet temperature measurements over time (e.g., one month, six months, one year, five years) and/or geothermal working fluid flow rate measurements over time. In some embodiments, the training data 326 is generated from nearby buildings or from buildings with similar thermal energy loads. In some embodiments, the training module 324 uses non-linear least square solvers such as the gradient-based Levenberg-Marquardt algorithm, for predicting underground heat transfer coefficient(s) for a GHCS site location. This, in turn, allows for adjusting the geothermal working fluid flow rates for continuously optimal heat pump efficiencies (i.e., during the day or at night, every day of the year);

[0108] a database 128, which stores data used, collected, and/or created by the workflow module 124, the model 126, and/or the training module 324. The database 128 may store input parameters 234, including surface parameters 236 and/or underground parameters 238, which provide the data used in the model 126, the integrated workflow 628, and/or the workflows 630, 650, and 670. In some embodiments, the database 128 stores pump data 240 of heat pumps that are installed at sites utilizing the system 200. In some embodiments, the database 128 stores operational parameters 242 for constructing geothermal borehole(s), including drilling depth(s), drilling diameter(s), and/or drilling direction(s). In some embodiments, the database 128 stores sensor data 244 that are collected by surface and/or subsurface sensors (e.g., sensors 508 and/or sensors 510) located at various geothermal sites. In some embodiments, the database 128 stores site information 246 of sites equipped with the system 200, including geographical location(s) of geothermal borehole(s), rock conditions, heat pump(s), geothermal loop(s) and their inlet and outlet temperatures, working fluid type(s), and/or working fluid flow rates. In some embodiments, the database 128 stores model results 248 generated by the model 126. In some embodiments, the database 128 stores training data 326. In some embodiments, the database 128 stores the data by site locations. In some embodiments, the database 128 stores the data by a respective identifier of the system(s) 200;

[0109] In some embodiments, the memory 306 includes a machine learning database 130 for storing machine learning

information. In some embodiments, the machine learning database **130** includes the following datasets or a subset or superset thereof:

**[0110]** neural network data **328**, including information corresponding to the operation of one or more neural network(s), including, and not limited to:

**[0111]** geothermal wells and loops information **330**, including information (e.g., feature vectors) corresponding to different locations and/or configurations of geothermal wells (e.g., geothermal boreholes) and geothermal loops; and

**[0112]** correction data **332** corresponding to the geothermal wells and loops information **330**.

**[0113]** In some embodiments, the server system **120** includes a device registration module for registering devices (e.g., the computer device and system) for use with the server system **120**.

**[0114]** In some embodiments, the server system **120** includes a notification module (not shown) for generating alerts and/or notifications for users of the geothermal planning and optimization system(s) **200**. For example, in some embodiments the model **126** (or the workflow module **124**) is stored locally on the system **200** of the user, the server system **120** may generate notifications to alert the user to download the latest version(s) or update(s) to the model.

**[0115]** Each of the above identified elements may be stored in one or more of the memory devices described herein, and corresponds to a set of instructions for performing the functions described above. The above identified modules or programs need not be implemented as separate software programs, procedures, modules or data structures, and thus various subsets of these modules may be combined or otherwise re-arranged in various implementations. In some embodiments, the memory **306** stores a subset of the modules and data structures identified above. In some embodiments, the memory **306** stores additional modules and data structures not described above. In some embodiments, a subset of the programs, modules, and/or data stored in the memory **306** can be stored on and/or executed by the geothermal planning and optimization system **200**. In some embodiments, a subset of the programs, modules, and/or data stored in the memory **306** can be stored on and/or executed by system **1000**.

**[0116]** FIGS. **4A** and **4B** illustrate drilling of a geothermal borehole (e.g., a geothermal well) in accordance with some embodiments. In FIG. **4A**, an exemplary geothermal borehole **402** has been (e.g., is being) drilled through the earth **404** from the surface **406**. While the geothermal borehole **402** in FIG. **4A** is illustrated as a substantially vertical borehole, it might, in practice, have portions that are inclined or horizontally-oriented.

**[0117]** In some embodiments, as illustrated in FIG. **4A**, a drill rig comprising CT **408** is used for constructing the geothermal borehole **402**. The CT **408** is tubing that is sufficiently flexible that long lengths can be coiled onto a spool and stored on a CT reel **412** that is mounted on a truck **414**, so that it can be injected into a geothermal borehole **402** using a CT injector **410**. In some embodiments, the truck **414** is provided with a radio frequency (RF) power source or generator **416** and motorized equipment **418** of a type known in the art to rotate the reel **412**.

**[0118]** In accordance with some embodiments of the present disclosure, the CT reel **412** and/or the CT injector **410** are purposefully re-designed and re-sized for drilling geo-

thermal boreholes with depths of about 1,000 to 2,000 feet. Current drilling technologies for geothermal and hydrocarbon well construction are divided between two categories: (A) small drilling rigs for shallow water or geothermal wells, usually drilling 100- to 400-foot long vertical wells, and (B) large drilling rigs for deep oil & gas and conventional geothermal wells, usually for 10,000 to 30,000 feet long wells. In oil and gas drilling and conventional geothermal drilling, no significant demand has historically existed for mid-range lengths, e.g., between 500 and 10,000 feet. Therefore, no off-the-shelf drilling rig has been developed for this range, leaving a technology gap between (A) and (B). While shallow water well drilling rigs (Category A) are small and portable enough to be transported and operated in high-density urban areas, they do not go very deep and are not very fast. The deep oil and gas rigs (Category B) are fast and powerful, but also large and heavy, and are meant to operate in remote areas far from urban/suburban development. This leaves a gap for fast and strong, yet compact and portable drilling rigs that can operate in urban/suburban/rural areas.

**[0119]** In some embodiments, as illustrated in FIG. **4B**, a drill rig comprising one or more joint pipes **422** (e.g., joint drill pipes) are used for constructing the geothermal borehole **402**. The joint pipes **422** are straight pipes having a fixed length (e.g., 10 feet long) and that are connected together via one or more tool joints **424** while drilling. For example, rigs for drilling geothermal wells for single-family residential applications (Category A) have joint pipes between 10 and 20 feet long.

**[0120]** Current drilling rigs (Categories A and B) use joint pipes. Drilling down, and then exiting the borehole on the way back up, must slow to a standstill when two joint pipes must be connected together. This repetitive process limits the average drilling speed to about 10-100 feet/hour for the entire well drilling operation. In contrast, with continuous CT drilling, the average drilling speeds can reach 150-200 feet/hour. These faster drilling speeds allow a significant reduction in time and operational costs: for instance, a 400-foot long geothermal well can be drilled in only 2 hours, instead of 20+ hours required by the shallow water well drilling rigs (Category A).

**[0121]** FIGS. **5A**, **5B**, **5C**, **5D-1**, **5D-2**, and **5D-3** illustrate a drill bit **502** for generating geothermal boreholes according to some embodiments.

**[0122]** FIG. **5A** illustrates a cross-sectional view that shows a drill bit **502** mounted on one end of a CT **408** and/or a joint pipe **422**, to facilitate drilling of the geothermal boreholes. FIG. **5B** illustrates a cross-sectional view of the drill bit **502** according to some embodiments. The drill bit **502** is rotated by a motor (not shown) that is hydraulically activated by pumping drilling fluid (e.g., water and/or bentonite) through the CT **408** or the joint pipes **422**. The arrows **504** in FIGS. **5A** and **5B** indicate the direction of the drilling fluid as it is pumped through the CT **408** and/or a joint pipe **422**. The arrows **506** in FIGS. **5A** and **5B** indicate the direction of drilling mud (e.g., water and/or bentonite and debris) as it exits the drill bit **502** and motor.

**[0123]** In some embodiments, the drill bit **502** includes sensors **508** (e.g., underground or downhole sensors) that are designed to measure underground parameters in-situ, as (e.g., during, while) the geothermal boreholes are being drilled/constructed. The underground parameters include temperature, pressure, humidity, lithology, azimuth, stresses,

and/or depth correlation. In some embodiments, the sensors **508** collect downhole data having data types that are illustrated in FIG. 5E.

[0124] In some embodiments, the sensors **508** are positioned on both the interior surface and the exterior surface of the drill bit **502**. For example, FIG. 5B illustrates four downhole sensors **508-1** to **508-4**. The sensors **508-1** and **508-3** are positioned on the interior surface of the drill bit **502**. The sensors **508-2** and **508-4** are positioned on the exterior surface of the drill bit **502**.

[0125] In some embodiments, at least two of the sensors **508** have the same sensor type. As an example, the sensor **508-1** and the sensor **508-2** can both be temperature sensors. As another example, the sensor **508-1** and the sensor **508-3** can both be pressure sensors.

[0126] In some embodiments, the sensors **508** include two sensors having the same sensor type. One of the two sensors is positioned on the interior surface and the other of the two sensors is positioned on the exterior surface. For example, in FIG. 5B, both the sensor **508-1** and the sensor **508-2** can be temperature sensors. The positioning of temperature sensors on both the interior and exterior surface of the drill bit **502** enables one to determine the temperature of the drilling fluid (e.g., drilling mud) before it enters the drill bit **502** after it exits the drill bit **502**. By knowing these temperature values, one can calculate (e.g., determine) parameters such as frictional forces and heat generated by the frictional forces (e.g., by applying relations such as mass-momentum conservation energy in pipes).

[0127] In some embodiments, the CT **408** and/or joint pipes **422** also include one or more sensors **510** (e.g., surface sensors) that can be positioned along (e.g., between) the surface **406** and a depth of the geothermal borehole **402**. In some embodiments, the surface sensors **510** can be positioned on the surface **406**, on the CT injector **410**, on the reel **412**, and/or anywhere on the truck **414**. The surface sensors **510**, such as the sensors **510-1** to **510-4** as shown in FIG. 5A, measure surface data in-situ, as the geothermal boreholes are being drilled/constructed. In some embodiments, the sensors **510** measure (e.g., collect) data such as fluid rate, pressure, temperature, and/or a length of the CT or joint pipe running into the borehole. In some embodiments, the sensors **510** collect surface data having data types that are illustrated in FIG. 5E.

[0128] In some embodiments, the sensors **510** are positioned on both the interior surface and the exterior surface along the CT **402**/joint pipes **422**.

[0129] In some embodiments, at least two of the sensors **510** have the same sensor type.

[0130] In some embodiments, the sensors **510** include two sensors having the same sensor type. One of the two sensors is positioned on the interior surface along the CT **402**/joint pipes **422**, and the other of the two sensors is positioned on the exterior surface along the CT **402**/joint pipes **422**.

[0131] FIG. 5E illustrates exemplary types of surface data and downhole data that are collected by the sensors **508** and the sensors **510** in accordance with some embodiments.

[0132] With continued reference to FIG. 5, in some embodiments, the underground data collected by the sensors **508** and/or the sensors **510** can be transferred to the surface (e.g., to the geothermal planning and optimization system **200**) automatically and in real-time using telemetry wires **512**. The telemetry wires **512** also transfer power from the surface to the sensors **508** and/or the sensors **510**. In some

embodiments, the telemetry wires **512** are associated with telemetry technologies such as electrical conduits or mud pulse telemetry, both of which are used for oil and gas CT operations. The electrical conduits (e.g., wires) have the dual advantage of being capable of much higher data transmission rates and being capable to also power to the underground sensors **508** (and to the surface sensors **510**) from the surface. In the case of mud pulse telemetry, the underground sensors **508** would need to be powered by batteries, making the drill bits much bigger.

[0133] FIG. 5C is a cross-sectional view of a CT **408**/joint pipe **422**. In this example, the telemetry wire **512** is located on an internal wall (e.g., interior surface) of the CT **408**/joint pipe **422**.

[0134] FIG. 5D-1 is a cross-sectional view of the CT **408**/joint pipe **422** and the drill bit **502**, showing positions of the telemetry wire **512**, a telemetry tube **514**, and the flow directions **504** and **506** of the drilling fluid/drilling mud. The view in direction **1-1**, as illustrated in FIG. 5D-2, shows the telemetry wire **512** positioned on the interior surface of the drill bit **502**. The view in direction **1-1** shown in FIG. 5D-2 also illustrates coupling (e.g., an electrical coupling, a communicative coupling) between the telemetry wire **512** and a sensor **508-5**. In this example, the sensor **508-1** is an internal sensor (e.g., it is positioned on an interior surface of the drill bit **502**). The view in direction **2-2**, as depicted in FIG. 5D-3, illustrates coupling (e.g., an electrical coupling, a communicative coupling) between the telemetry wire **512** and a sensor **508-6-5**. In this example, the sensor **508-6** is an external sensor (e.g., it is positioned on an exterior surface of the drill bit **502**).

[0135] FIGS. 6A to 6C illustrate an integrated workflow (e.g., integrated workflow **628**) for planning, constructing, and/or optimizing a GHCS operation, in accordance with some embodiments. In some embodiments, the workflow **628** includes a combination of one or more of: pre-construction workflow **630**, a construction workflow **650**, and a post-construction workflow **670**. The workflows **630**, **650**, and **670** are executed by a processor (e.g., processor **202** or processor **302**), in accordance with some embodiments. In some embodiments, each of the workflows **630**, **650**, and **670** can be executed as a standalone workflow.

[0136] FIG. 6A illustrates a pre-construction workflow **630** that is executed by a processor (e.g., processor **202** or processor **302**) during a pre-construction phase of a GHCS operation, according to some embodiments.

[0137] The workflow **630** includes receiving (632) input parameters (e.g., specified by a user, the system **200**, or the server **120**). The input parameters can include surface parameters **236** and/or underground parameters **238**. In some embodiments, the input parameters include default (e.g., pre-defined) values that can be modified and/or overridden by a user.

[0138] The workflow **630** includes receiving (634) (e.g., from a user) a heat pump COP for heating and/or EER for cooling. In some embodiments, the heat pump COP/EER comprises a manufacturing COP/EER of the heat pump as specified by the pump manufacturer. In some embodiments (e.g., for large buildings), two or more heat pumps that work in tandem or independently may be used. The workflow **630** includes receiving respective heat pump COPs/EERs corresponding to each of the two or more heat pumps. In some embodiments, in addition to (or instead of) receiving the heat pump COP/EER, the workflow can measure heat trans-

fer (e.g., directly, by measuring a temperature difference in an object) or indirectly, by calculation) and use the measured heat transfer as a proxy for determining the COP/EER. For example, in some embodiments, the workflow **630** includes measuring (or receiving) one or more temperatures of a geothermal fluid, such as an inlet and/or an outlet temperature. In some embodiments, the workflow **630** includes measuring (or receiving) a heat transfer coefficient of the heat between a subsurface and a geothermal borehole.

[0139] The workflow **630** includes selecting (**636**) (e.g., by the processor), or receiving user selection of, a subset of input parameters. The workflow **630** includes calculating (**638**) a bottom-hole temperature of a geothermal borehole to be constructed, based on the selected subset of input parameters. The workflow **630** includes generating (**640**) (e.g., by the processor, by applying the model **126**) estimated borehole parameters, such as the borehole depth, length, and/or diameter.

[0140] In some embodiments, in accordance with receiving the heat pump COP and/or EER in step **634**, the processor generates a range of COP and/or EER values (e.g., by applying the model **126**), to obtain a range of borehole parameters, which in turn facilitates a builder to properly design and plan the drilling job, obtain the right drilling permits, and/or for plan the geothermal loops for an optimal long-term underground heat transfer process.

[0141] In some embodiments, execution of the workflow **630** provides (e.g., generates) operational parameters of one geothermal borehole. In a construction that involves multiple boreholes, the workflow **630** is executed repeatedly, each of the iterations generating a set of borehole parameters for a respective borehole.

[0142] FIG. 6B illustrates a construction workflow **650** that is executed by a processor (e.g., processor **202** or processor **302**) during the construction phase of a GHCS operation, according to some embodiments. In some embodiments, the workflow **650** includes the step **632** and the step **636** as described previously with respect to FIG. 8A.

[0143] In some embodiments, the workflow **650** includes, in step **656**, measuring (e.g., by one or more sensors) the bottom-hole temperature of a geothermal borehole during the construction of the borehole. For example, as illustrated in FIGS. 5B, 5D-1, 5D-2, and 5D-3, sensors **508** can be mounted on a drill bit **502** that is used for drilling a geothermal borehole. In some embodiments, the sensors **508** include temperature sensors for measuring (e.g., automatically and in real time) a bottom-hole temperature as the geothermal borehole is being drilled. In some embodiments, the workflow **650** includes measuring surface and downhole conditions (e.g., pressure, temperature, and/or flow rate) during the construction of the borehole, from surface sensors (e.g., sensors **510**) and underground sensors (e.g., sensors **508**).

[0144] In some embodiments, the workflow **650** includes, in step **658**, calculating (e.g., by the processor) the heat pump COP/EER based on the selected subset of input parameters and the measured bottom-hole temperature. For example, the processor updates the model **126** based on the measured bottom-hole temperature, and applies the updated model to determine a calculated (e.g., actual, modified) heat pump COP/EER.

[0145] In some embodiments, the workflow **650** includes comparing (**660**) the calculated COP/EER with a predicted COP/EER (e.g., predicted by the model **126** or a machine

learning database **130**). The workflow **650** includes, in step **662**, terminating the drilling process when the calculated COP/EER matches the predicted COP. The workflow **650** includes, in step **644**, continuing the drilling of the geothermal borehole (e.g., drilling deeper, varying a drilling angle and/or a borehole diameter) when the calculated COP/EER does not match the predicted COP/EER.

[0146] Stated another way, the downhole sensors (e.g., sensors **508**) encased in the drill bit **502** allow re-calibration of the model **126** in real time. That is, actual temperature data is acquired in real time during drilling, which changes the output of initial design model dynamically. This empowers the field personnel to optimize the operational parameters on-the-fly, e.g., drilling shorter, longer, or wider wells to ensure the optimal performance of each GHCS installation.

[0147] FIG. 6C illustrates a post-construction workflow **670** that is executed by a processor (e.g., processor **202** or processor **302**) during the post-construction phase of a GHCS operation, according to some embodiments. In some embodiments, the workflow **670** includes the step **632** and the step **636** as described previously with respect to FIG. 6A.

[0148] In some embodiments, the workflow **670** includes, in step **672**, measuring the bottom-hole temperature (e.g., of a borehole or a geothermal loop) and an inlet/outlet temperature of a geothermal loop (e.g., geothermal loop **604**). In some embodiments, after the geothermal borehole **402** has been constructed, an optical fiber is placed inside the constructed borehole, for collecting the temperature along a geothermal loop constructed based on the borehole. In some embodiments, the bottom-hole temperature is determined indirectly via calculations (e.g., solving a loop-pipe flow thermodynamic equation, or determining a temperature profile along the geothermal loop loop).

[0149] In some embodiments, the workflow **670** includes, in step **674**, calculating a heat pump COP for heating and/or a heat pump EER for cooling. The workflow **670** includes, in step **676**, comparing the calculated COP with a predicted COP/EER (e.g., predicted by the model **126**). The workflow **670** includes, in step **678**, maintaining a current flow rate of a working fluid in the geothermal loop when the calculated COP/EER matches the predicted COP/EER. The workflow **670** includes, in step **680**, adjusting the flow rate of the working fluid in the geothermal loop when the calculated COP/EER does not match the predicted COP/EER.

[0150] In some embodiments, the workflow **670** includes, in step **682**, repeating the measuring and calculating (e.g., steps **672**, **674**, and **676**) after a certain time (e.g., every month, every three months, every year, or every change of season), and maintaining or adjusting the flow rate of the working fluid accordingly.

[0151] In a geothermal system, heat exchange occurs (1) between the geothermal loop and the building and (2) between the geothermal loop and the near constant-temperature thermal reservoir from the underground rocks. The underground heat transfer process between the working fluid **606** and the underground rocks (e.g., the underground region surrounding the geothermal loop) can be calculated by solving a radial (i.e., one-dimensional) diffusion equation in the near-borehole region (e.g., 10 to 30 feet radially from the borehole). The initial temperature of the near-borehole region is different from the long-term pseudo-steady-state temperature of the same region. The temperature of the near-borehole region changes over time due to several

reasons. First, for heating, the working fluid picks up heat from the underground rocks and brings it to the surface, so the inlet temperature of the geothermal loop (see, e.g., FIG. 7) is lower than the outlet temperature of the geothermal loop. If the fluid is continuously circulated, heat is produced from the underground, so the temperature in the near-borehole region decreases over time until it reaches a pseudo-steady-state temperature, which is lower than the initial underground temperature. On the other hand, for cooling, the working fluid loses heat to the underground rocks, so the inlet temperature of the geothermal loop is higher than the outlet temperature of the geothermal loop. If the fluid is continuously circulated, heat is lost to the underground, so the temperature in the near-borehole region increases over time until it reaches a pseudo-steady-state temperature, which is higher than the initial underground temperature. Stated another way, because the pseudo-steady-state temperature of the near-borehole region is expected to vary from the initial temperature when the geothermal loop is first installed, the working fluid flow rate may need to be adjusted over time to ensure optimal long-term performance of the GHCS.

**[0152]** FIG. 7 illustrates a heat pump **702** and a geothermal loop **704** in accordance with some embodiments. In some embodiments, the geothermal loop **704** is constructed from a geothermal borehole **402**. The geothermal loop **704** comprises a pipe that is made of a material such as high-density polyethylene (HDPE), fiberglass or carbon steel. The geothermal loop **704** can be installed as a U-loop into a borehole **402** and connected to a heat pump **702** of a building **708**. The heat pump **702** uses the nearly constant temperature underground to heat or cool the interior of a building **708**. The geothermal loop **704** is filled with a fluid **706** (e.g., a working fluid, such as a refrigerant) that circulates inside the geothermal loop **704**. In summer, the surface air has a higher temperature compared to the underground temperature. The fluid **706** absorbs heat from the surface and releases it underground. In winter, the surface air has a lower temperature compared to the underground temperature. The fluid **706** absorbs heat from the underground and releases it to the surface. In some embodiments, the heat pump **702** includes a reversing valve, which lets the heat pump **702** switch directions to either heat or cool the building.

#### A. Ultra-Compact Coiled Tubing (UCCT) Operations for Geothermal Well Construction

**[0153]** Some aspects of the present disclosure are directed to coiled tubing (CT) drilling technology for constructing geothermal wells.

**[0154]** Current drilling technologies for geothermal and hydrocarbon well construction are divided between (Category A) small drilling rigs for shallow water or geothermal wells (e.g., 100 to 400 feet long vertical (e.g., deep) wells, and (Category B) large drilling rigs for deep oil & gas and conventional geothermal wells (e.g., 10,000 to 30,000 feet long vertical wells). No off-the-shelf drilling rig exists commercially for mid-range-length wells that are between 500 and 10,000 feet deep. Therefore, no off-the-shelf drilling rig has been developed for this range, leaving a technology gap between Categories A and B. While shallow water well drilling rigs (Category A) are small and portable enough to be transported and operated in high-density urban areas, they do not go very deep and are not very fast. The deep oil and gas rigs (Category B) are fast and powerful, but also

large and heavy, and are meant to operate in remote areas far from urban/suburban development. This leaves a gap for fast and strong, yet compact and portable drilling rigs that can operate in urban/suburban/rural areas.

**[0155]** In addition to the technology gap for mid-range length geothermal wells, several constraints exist when it comes to constructing geothermal boreholes in urban (e.g., high-density), suburban, and, or commercial areas. For example, the vehicle(s) transporting the construction equipment must satisfy the space and/or weight limitations imposed by the areas where the construction will be carried out. The drill rig and other construction equipment need to be transported to and from the construction site via paved concrete roads. Furthermore, any noise generated during well construction must be within the noise level imposed by these areas.

**[0156]** Coiled tubing has historically been an oil and gas supporting technology and is used primary for well interventions. Coiled tubing is versatile in terms of equipment, pipe size and configuration, fluids used, tools conveyance, and interaction with the reservoir. Its applicability and value for hydrocarbon production are vastly documented, from drilling and completions to production enhancement and well abandonment.

**[0157]** Despite its strengths, coiled tubing is not currently used in shallow (e.g., up to 1,000 feet) or mid-range (e.g., around 500 feet to 3,000 feet) drilling, nor has there been any attempt to utilize coiled tubing for geothermal loop insertion or grouting.

**[0158]** Some embodiments of the present disclosure are directed to an ultra-compact coiled tubing unit (UCCTU) for geothermal field construction. The UCCTU includes coiled tubing that can be employed for multiple phases of a geothermal field construction process, including a drilling phase, and/or a geothermal loop insertion phase, and/or a grouting phase. The UCCTU is designed and built considering inner-city weight, width, length, and height limitations, and can be deployed for geothermal drilling in dense urban areas, for decarbonizing buildings and reducing their dependance on the electrical grid.

**[0159]** According to some embodiments, the UCCTU disclosed herein fills the technology void for mid-range-length wells that are between ~500 feet and ~3,000 feet deep.

**[0160]** The present disclosure advantageously improves upon current geothermal well construction processes in several ways. First, the current practice of manually inserting geothermal loops into boreholes is difficult and physically demanding. The task becomes even more challenging when the well depths are ~800 feet and upwards. The challenge is exacerbated when groundwater is present, or when boreholes are not straight. The risk of geothermal loops getting stuck in the lateral borehole walls or pushed back by buoyancy is quite high, and limits the ability to install them to greater depths. Utilizing coiled tubing for geothermal loop insertion makes the process easier, faster, and more efficient. Coiled tubing can be driven into the ground using the injector. The driving force provided to the CT by the injector is large enough to push geothermal loops to depths unheard of by the current shallow geothermal drilling industry. Second, the industry currently uses Tremie lines for the grouting phase. Because grout is a viscous

paste, pumping grout through one-inch-diameter tremie lines at depths greater than ~500 to 600 feet is operationally impossible.

#### 1. Ultra-Compact Coiled Tubing Unit (UCCTU)

[0161] FIGS. 8A-8E, 10, and 20-30 illustrate components of an ultra-compact coiled tubing unit (UCCTU) 800 for geothermal field construction, in accordance with some embodiments. In some embodiments, the UCCTU 800 is also referred to as a coiled tubing unit (CTU); these terms are used interchangeably in this disclosure.

[0162] As used herein, a geothermal field refers to a set of (e.g., one or more) boreholes, a set of (e.g., one or more) geothermal loops connected to a building, and the rock volume surrounding the set of geothermal loops. In some embodiments, a geothermal field can refer to the entire set of boreholes and geothermal loops connected to a building, and the rock volume surrounding all the geothermal loops.

[0163] In this disclosure, the term “geothermal field” is used interchangeably with the term “geo-field.”

[0164] A geothermal loop (e.g., also described in FIG. 7) is constructed from a geothermal borehole. The geothermal loop can be a pipe or a long tube that is made of a material such as high-density polyethylene (HDPE), fiberglass or carbon steel. In some embodiments, the geothermal loop is installed as a U-loop into a borehole and connected to a heat pump of a building. Water or refrigerant (e.g., glycol) is flowing continuously through the geo-loop exchanging heat between the building and the subsurface rocks.

[0165] In this disclosure, the term “geothermal loop” is used interchangeably with the term “geo-loop.”

[0166] The UCCTU 800 can include multiple pieces of equipment that are modular in nature. In some embodiments, all the equipment of the UCCTU 800 can be installed and transported on one, two, three, or more chassis and skids (metal frames for supporting equipment) of a vehicle, such as one or more of: vehicle 810-1 in FIG. 8A, vehicle 810-2 in FIG. 8B, vehicle 810-3 in FIG. 20, vehicle 810-4 in FIG. 27, and vehicle 810-5 in FIG. 29. In some embodiments, the number of pieces of the equipment and/or their sizes and operating capacities can be scaled depending on a size, depth, and/or complexity of the geothermal field to be constructed.

[0167] In some embodiments, the UCCTU 800 includes a control cabin 812, which contains all of the necessary controls for operating the UCCTU.

[0168] In some embodiments, the UCCTU 800 includes a wet kit 814, which is a hydraulic component that uses a power take-off (PTO) to power functions.

[0169] In some embodiments, the UCCTU 800 includes a coiled tubing (CT) reel 816 for spooling and unspooling coiled tubing (CT) 818. The CT 818 is also referred to as a CT string (or a pipe). CT 818 is a continuous length of metal or composite tubing with no joints. The length of the CT 818 can be predefined according to a depth of geothermal borehole to be constructed (e.g., drilled). For example, the length of CT 818 can be 500 feet, 2000 feet, or 4000 feet, or 10,000 feet (e.g., within  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ). In some embodiments, CT 818 comprises a low-alloy carbon steel tubing. In some embodiments, the CT reel 816 is configured to be movable (e.g., with respect to its axle). For example, when the vehicle 810-1 or the vehicle 810-2 is parked at a site, the injector 820 and the CT reel 816 can be moved to

different locations, within the site, to facilitate drilling of multiple geothermal boreholes while the vehicle remains stationary.

[0170] An injector 820 (also referred to herein as a CT injector or an injector head) injects one end of the CT 818 from the surface into a subsurface to construct a geothermal borehole. For example, the injector 820 is hydraulically powered and provides the surface drive force to run CT 818 underground to drill/construct a geothermal borehole (e.g., a geothermal well), and retrieves CT 818 out of a geothermal borehole after the drilling is completed.

[0171] The injector 820 includes a gooseneck 819 (e.g., an arch) for guiding the CT 818 into a body of the injector 820. In some embodiments, the gooseneck 819 is coupled to (e.g., attached to) the injector 820 at a fixed position, meaning that the gooseneck 819 is not adjustable. In some embodiments, the gooseneck 819 is adjustably coupled to the injector 820. In some embodiments, the gooseneck 819 is detachable from the injector 820. In some embodiments, goosenecks of different sizes can be attached to the injector 820 (e.g., depending on the space available for drilling).

[0172] For borehole construction, a drilling motor and a drill bit (e.g., drill bit 502) can be installed on (or proximate to) the end of the CT 818 that is to be injected into the subsurface. In some embodiments, a drilling hammer, a steering tool and/or centralizers can be fitted the CT 818 for the borehole construction. The drilling hammer increases the push force applied on rocks during drilling, while the steering tool and/or centralizers keep the drilling as close to a straight line as possible.

[0173] In some embodiments, CT 818 includes downhole telemetry (e.g., wired or wireless), which is discussed in further detail in FIG. 11.

[0174] In some embodiments, one or more sensors (e.g., sensors 508 or sensors 1015), such as pressure sensors, temperature sensors, gamma ray sensors, inclination/azimuth sensors, vibration sensors, acceleration sensors, force sensors, torque sensors, pH sensors, salinity sensors, and/or thermal conductivity sensors, are also installed above or below the motor.

[0175] In some embodiments, the UCCTU 800 includes one or more cranes 824 for lifting and moving components of the UCCTU.

[0176] In some embodiments, the UCCTU 800 includes a stripper 822. The stripper 822 provides a pressure seal around a coiled tubing unit when it is being run into or pulled out of a live well. The sealing mechanism is activated by hydraulic pressure and is controlled by an operator.

[0177] In some embodiments, the UCCTU 800 includes a fluid pump 826, a solids separation management equipment or mud recycler 828, a geothermal loop reel 830 for spooling geothermal loop 1220 (see, e.g., FIG. 13), grouting equipment 832, and/or a mixing tank 834. The mud recycler 828 separates (e.g., filters) the cuttings from the returning drilling mud before re-pumping the mud into the well to bring new cuttings to surface. The mixing tank 834 is a tank used for mixing drilling mud and additives (e.g., clay stabilizers, salts, or anti-corrosion inhibitors) or for producing grout by mixing water and grout powder.

[0178] FIG. 8B shows that in some embodiments, the injector 820 is housed in a cradle 817 for transport by vehicle 810-2. In some embodiments, and as illustrated in FIG. 8C, the cradle 817 is removed and the injector is supported by forks 842 (e.g., extending from both sides of

the truck) at stand **838** (e.g., injector stand). The stand **838** is used for supporting the injector **820** while the injector **820** is in operation.

[0179] FIG. 8E illustrates an injector **820** in accordance with some embodiments. The injector **820** is mounted on a stand **838**. The stand **838** includes hydraulically actuated (e.g., telescopic) legs **840**. In some embodiments, the stand **838** and the stripper **822** are attached to the injector **820** during operation as well as transport (e.g., the entire time). The legs **840** can be configured (e.g., programmed) to control the drilling inclination of boreholes. For example, if the downhole telemetry data (e.g., obtained by sensors **1015**) indicates that a borehole is drilled vertically (i.e., not inclined), the legs **840** are hydraulically actuated so that the injector **820** positioned horizontally. If the downhole telemetry data indicates that the borehole is drilled at an angle that is offset from the vertical (e.g., the borehole is inclined/tilted/not vertical), the legs **840** can be programmed to tilt the injector at (e.g., to the angle) so that a subsequent trajectory of the borehole is vertical.

[0180] In some embodiments, at the start of borehole drilling, the legs **840** can be hydraulically actuated to tilt the injector **820** such that the initial drilling angle is tilted (e.g., inclined) with respect to a vertical axis (e.g., because less force is required to drill at an oblique angle than at 90 degrees). During drilling, the legs **840** can be hydraulically actuated to steer to the drilling angle to be parallel to the vertical axis, and reduce clearance between the injector and the ground.

[0181] In some embodiments, the UCCTU **800** includes downhole tools **836**, which can include any combination of one or more drilling motors, drill bits, fluid hammers, steering tools, centralizers, and/or washing nozzles.

[0182] In some embodiments, the UCCTU **800** includes a drill pipe **2910**, which is described with reference to FIG. 29.

[0183] In some embodiments, the UCCTU **800** includes a geothermal loop **1220** that is configured to be spooled onto and be unspooled from a geothermal loop reel **830**.

[0184] In some embodiments, the UCCTU **800** includes one or more geo-loop flanges **1200** and/or **2220**, which is described with reference to FIGS. 12, 21, 22, 23, and 24.

[0185] In some embodiments, the UCCTU **800** includes a CT bottom-hole assembly **2300**, which is described with reference to FIG. 23.

[0186] In some embodiments, the UCCTU **800** includes a borehole head stack **2500**, which is described with reference to FIG. 25.

[0187] FIG. 9 illustrates a coiled tubing reel **900** that is found on existing systems. As discussed above, the injector includes a gooseneck for guiding the coiled tubing into a body of the injector. In existing systems, the gooseneck is fixed (i.e., not movable) on the injector. Because the reel is large and the distance between the reel and the injector is small (e.g., about 5-25 feet), reels of existing systems, such as the one shown in FIG. 9, has a level-wind **902** that guides the coiled tubing onto the drum. The level-wind can move left and right, following the unspooling and spooling of the coiled tubing, and is prone to frequent mechanical failures.

[0188] In some embodiments disclosed herein, the CT reel **816** eliminates the use of a level-wind. Instead, the gooseneck **819** is modified so that it is capable of rotating to follow the coiled tubing misalignment. In other words, instead of using a level-wind to force the coiled tubing to align over a short distance between the level-wind and gooseneck, some

embodiments of the present disclosure employ a rotating gooseneck **819** with no level-wind for aligning the CT **818**. In some embodiments, the gooseneck **819** can be reinforced with a telescopic arm with a ball (e.g., a ball and socket joint) to ensure that the gooseneck **819** is only moving laterally.

[0189] In accordance with some embodiments of the present disclosure, the UCCTU is designed and built to be within size and weight limits for vehicles and/or loads moving within cities. The UCCTU **800** can be deployed for geothermal drilling in dense urban areas, for decarbonizing buildings and reducing their dependence on the electrical grid.

[0190] FIG. 10 illustrates a block diagram of a system **1000** for geothermal field construction, in accordance with some embodiments.

[0191] In some embodiments, the system **1000** can include at least a subset of the components of the UCCTU **800**.

[0192] In some embodiments, the system **1000** includes one or more processors (e.g., processing units, processing circuitry, electrical processing circuitry, hydraulic circuitry, signal processing circuitry, or CPUs) **1002**, one or more network or other communication interfaces **1004**, memory **1006**, and one or more communication buses **1008** for interconnecting these components. In some embodiments, the communication buses **1008** include circuitry (sometimes called a chipset) that interconnects and controls communications between system components.

[0193] In some embodiments, the system **1000** is communicatively connected to one or more sensors **1050**. The sensors **1050** can include one or more of: a force sensor, a torque sensor, a vibration sensor, an acceleration sensor, a gamma ray sensor, a temperature sensor, a pressure sensor, a pH sensor, and/or an azimuth/orientation sensor. There can be multiple sensors for each sensor type.

[0194] In some embodiments, the sensors are positioned on a drill bit of a CT and/or a joint pipe (e.g., as described in FIGS. 4 and 5). In some embodiments, drill bit and sensors **1050** are part of a telemetry system **1100** that is described with respect to FIG. 11.

[0195] In some embodiments, the system **1000** includes one or more input devices **1010** that facilitate user input, such as a keyboard, a mouse, a voice-command input unit or microphone, a touch screen display, a touch-sensitive input pad, a gesture capturing camera, or other input buttons or controls. In some embodiments, the system **1000** includes one or more cameras or scanners for capturing data. The system **1000** also includes one or more output devices **1012** that enable presentation of user interfaces and display content, including one or more speakers and/or one or more visual displays.

[0196] In some embodiments, the memory **1006** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and, optionally, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. In some embodiments, the memory **1006** includes one or more storage devices remotely located from one or more processing units **1002**. The memory **1006**, or alternatively the non-volatile memory device(s) within the memory **1006**, includes a non-transitory computer-readable storage medium. In some embodiments, the memory **1006** or the



computer-readable storage medium of the memory **1006** stores the following programs, modules, and data structures, or a subset or superset thereof:

- [0197] an operating system **1022**, including procedures for handling various basic system services and for performing hardware dependent tasks;
- [0198] a communications module **1024**, which is used for connecting the UCCTU **800** to other systems, computers and devices via the one or more communication interfaces **1004** (wired or wireless), such as the communication network(s), Internet, other wide area networks, local area networks, metropolitan area networks, and so on;
- [0199] a user interface module **1026**, which enables presentation of information (e.g., a graphical user interface for application(s) **1028**, widgets, websites and web pages thereof, and/or audio and/or video content, text, etc.) at the system **1000** via one or more output devices **1012** (e.g., displays, speakers, etc.);
- [0200] one or more user applications **1028**, which are executed by the system **1000** (e.g., web or non-web based applications for controlling another system, electronic device, or sensors, or for reviewing data captured by such devices);
- [0201] an optimization module **1030**, for determining operational parameters on the UCCTU **800** (e.g., the CT **818**) in accordance with downhole telemetry data obtained by the sensors **508** and/or **1050**;
- [0202] workflow module **124**, which is described with respect to FIGS. 2, 6A, 6B, and 6C;
- [0203] model **126**;
- [0204] data **1032**, including, for example:
  - [0205] input parameters **234**;
  - [0206] pump data **240**;
  - [0207] operational parameters **242**;
  - [0208] site information **246**; and/or
  - [0209] sensor data **1036** (e.g., sensor data **244**), which includes data obtained by sensors **508**, **1050**, and/or sensor data **244**. The sensor data can include pressure, temperature, depth correlation and azimuth, rock hardness, pumping rates, force, torque, pH, and/or vibration data.
- [0210] Although FIG. 10 shows a system **1000**, FIG. 10 is intended more as a functional description of the various features that may be present rather than as a structural schematic of the implementations described herein. In practice, and as recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated.
- [0211] Each of the above identified executable modules, applications, or sets of procedures may be stored in one or more of the memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various implementations. In some embodiments, the memory **1006** stores a subset of the modules and data structures identified above. Furthermore, the memory **1006** may store additional modules or data structures not described above (e.g., module(s) for machine learning and/or training models). In some embodiments, a subset of the

programs, modules, and/or data stored in the memory **1006** are stored on and/or executed by a server system (e.g., server system **120**).

[0212] In some embodiments, the system **1000** deploys a CT enabled drill bit and collects (e.g., in real time during the constructing) sensor data from sensors (e.g., sensors **1050**, FIG. 10) that are positioned on the drill bit and at the surface. In some embodiments, the sensors **1050** are part of a downhole telemetry assembly **1100**. In some embodiments, the drill bit is part of the downhole telemetry assembly **1100**. [0213] In some embodiments, the system **1000** uses the sensor data to determine (e.g., monitor) the drilling speed, the drilling force, and/or the drilling torque (e.g., the force and/or torque exerted on the drill bit). For example, the system **1000** can determine the force/torque on the CT **818** as it enters the injector, when it leaves the injector, and/or when it is injected into the subsurface. Additional details are described in U.S. Pat. No. 17/976,445, which is incorporated by reference herein in its entirety.

## 2. Downhole Telemetry System

[0214] FIG. 11 illustrates a downhole telemetry system (e.g., downhole telemetry assembly) **1100**, in accordance with some embodiments.

[0215] The telemetry system **1100** includes a wire **1102** (e.g., telemetry wire). In some embodiments, the wire **1102** is used for transferring power from the surface to the sensors. In some embodiments, underground data (e.g., downhole telemetry data) collected by the sensors **1050** (e.g., sensor sub-assembly **1106**) can be transferred to the surface automatically in real-time using the wire **1102**.

[0216] The telemetry system **1100** includes a sensor sub-assembly **1106**. In some embodiments, the sensor sub-assembly **1106** includes a directional control unit **1108** for controlling a drilling direction using data measured from an array of sensors **1104**. In some embodiments, the directional control unit **1108** is hydraulically activated. In some embodiments, the directional control unit **1108** is electrically activated.

[0217] The sensor sub-assembly **1106** includes sensors **1050**. The sensors **1050** can include one or more pressure sensors, one or more temperature sensors, one or more gamma ray sensors, one or more inclination sensors, one or more vibration sensors, one or more acceleration sensors, one or more force sensors, one or more torque sensors, one or more pH sensors, one or more salinity sensors, and/or one or more thermal conductivity sensors.

[0218] In some embodiments, the sensors **1050** are downhole sensors that measure subsurface sensor data. The sensors **1050** are positioned on both an interior surface and the exterior surface of the sensor sub-assembly **1106**.

[0219] In some embodiments, the system **1000** includes surface sensors that are located on above the ground, for collecting surface sensor data.

[0220] In some embodiments, the sensors **1050** include a first downhole temperature sensor that is positioned internal to the sensor sub-assembly **1106** and a second downhole temperature sensor that is positioned external to (e.g., facing the subsurface) the sensor sub-assembly **1106**. The combination of the first and second downhole temperature sensors are used to determine a geothermal gradient while drilling.

[0221] In some embodiments, the sensors **1050** include inclination and/or azimuth sensor(s) for determine whether lateral drift (e.g., a distance between an actual borehole

distance and a planned trajectory) while drilling. The lateral drift can be a displacement in a horizontal plane, in any direction, assuming that the borehole is vertical. In some embodiments, the lateral drift is a distance between an actual borehole distance and a planned trajectory, measured with respect to a plane perpendicular to the borehole axis.

[0222] In some embodiments, data collected from the downhole sensors **1050** is used for determining whether a geothermal loop is installed at target depth and/or grout is filling the borehole entirely.

[0223] In some embodiments, the sensors **1050** are arranged in an array. For example, in some embodiments, the sensors are arranged with a geometric pattern so as to be able to measure geological data (e.g., rock data) in three dimensions (e.g., directionally radial data).

[0224] In some embodiments, the sensors **1050** are positioned on the exterior surface of the drill bit or in/on the telemetry sub-assembly located between the drill bit and a CT connector. For example, the sensors that are positioned on the exterior surface of the drill bit can include: a force sensor, a torque sensor, a pressure sensor, a temperature sensor, a vibration sensor, a pH, a gamma ray sensor, and/or an azimuth/orientation sensor.

[0225] In some embodiments, the sensor sub-assembly (e.g., the telemetry system) uses force and/or vibration data obtained (e.g., measured) by the force and/or vibration sensors, to calibrate the data measured by the other sensors (e.g., pressure data, temperature data, etc.) based on the measured vibrations. Without this calibration, the data can be noisy.

[0226] In accordance with some embodiments, the telemetry system **1100** disclosed herein advantageously improves geothermal field construction. For example, a requirement for current drillers (e.g., in a city) is that they have to map/verify that the constructed boreholes are vertical. Because current geothermal borehole construction technologies do not incorporate built-in sensors, the industry practice is to drill about 100 feet every time, insert a wireline (e.g., a wire with sensors) to measure and confirm that the geothermal borehole is (e.g., still) in the direction the drillers believe the borehole should be, drill another approximately 100 feet, and repeat the measurement process. In contrast to the existing systems, the telemetry system **1100** system includes sensors positioned on (and/or within) the drill bit, thereby enabling the system to continuously monitor the drilling and intervene if needed.

### 3. Geothermal Field Construction

[0227] In some embodiments disclosed herein, a geothermal field construction process can include three phases: a drilling phase, a geothermal loop insertion phase, and a grouting phase.

[0228] According to some embodiments of the present disclosure, the system **1000** (e.g., UCCTU **800**) includes a coiled tubing that is used in at least two of the three phases (e.g., drilling and geothermal loop insertion; drilling and grouting; geothermal loop insertion and grouting; and/or drilling, geothermal loop insertion, and grouting) of the geothermal field construction.

[0229] In some embodiments, before the drilling phase, a rat hole of about 20-40 feet deep is drilled and a casing **1506** (e.g., a surface casing, a 6-7 inch diameter pipe) is installed in the rat hole. In some embodiments, coiled tubing is not used in the rat hole drilling process. The casing **1506**

protrudes about 1-3 feet above the ground and is capped with a casing flange **1504**, as illustrated in FIGS. **15A-1**, **15A-2**, and **15A-3**. After the rat hole is drilled, the injector **820**, the stripper **822**, and flanges are installed on top of the casing **1506** to commence the CT-enabled geothermal field construction. In some embodiments, the distance between the injector **820** and the ground can be adjusted via the legs **840** to account for the casing **1506**.

[0230] In some embodiments, the coiled tubing drilling technology disclosed herein can be combined with rotary drilling equipment. The CT and rotary drilling equipment (e.g., rotary drilling rig) can be mounted on the same chassis or on different chassis of vehicle(s) **810**. For example, in some embodiments, the vehicle **810-1** or the vehicle **810-2** can be fitted with a mast that is connected to a pump and a generator (e.g., control cabin **812**). The rotary drilling rig can drill the first 20- to 40 feet of borehole from the surface to facilitate installation of the casing **1506**. Then, coiled tubing drilling is used to construct the geothermal field.

[0231] The drilling phase includes drilling (e.g., constructing) one or more boreholes of the geothermal field. According to some embodiments of the present disclosure, the drilling is performed using a drilling motor and a drill bit installed at one end of CT **818**. The injector **820** pushes the CT **818** down into a borehole while water, drilling mud (e.g., drilling fluid) is pumped through CT **818** and the drilling motor (including a stator and rotor) rotates the drill bit. When the drill bit reaches the target depth, the CT **818** is pulled out of the hole. The drill bit and drilling motor are removed from the CT **818**, and a washing nozzle **1222** is attached to the CT **818** for use in the geothermal loop insertion phase.

[0232] During the geothermal loop insertion phase, geothermal loops are inserted into the boreholes that were constructed in the drilling phase. In some embodiments disclosed herein, during the geothermal loop insertion phase, a geothermal loop **1220** (e.g., illustrated in FIGS. **12B** and **13**) is physically coupled to the CT **818** and is injected (e.g., simultaneously or concurrently) with the CT **818** using the CT injector **820**.

[0233] In accordance with some embodiments, a geothermal loop flange **1200** (also referred to as a geo-loop flange) facilitates insertion of a geothermal loop **1220** into a borehole during the geothermal loop insertion phase. FIGS. **12A-1** and **12A-2** illustrate two views of the geo-loop flange **1200**, in accordance with some embodiments. FIG. **15A-1** and FIG. **17A** show that in some embodiments, the geo-loop flange **1200** is mounted between the casing flange **1504** (below) and the stripper **822** (above). The stripper **822** is located below the injector **820**, and a stripper flange **1502** is mounted below the stripper.

[0234] Referring to FIGS. **12A-1** and **12A-2**, the geo-loop flange **1200** includes a coiled tubing guide member **1202** (e.g., a flange) for guiding the CT **818**. The coiled tubing guide member **1202** includes a first end **1204** where the CT **818** enters. The coiled tubing guide member **1202** includes a second end **1206** where both the CT **818** and the geothermal loop **1220** exit. The coiled tubing guide member **1202** includes an opening **1208** on a side wall **1210** of the coiled tubing guide member **1202**.

[0235] In some embodiments, the coiled tubing guide member **1202** is cylindrical (e.g., annular) (or largely cylindrical) in shape. In some embodiments, an outer diameter of the geo-loop flange **1200**, as determined by the coiled tubing

guide member **1202**, is about 4-9 inches (e.g.,  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ), depending on the borehole size. In some embodiments, an inner diameter of the geo-loop flange **1200**, as determined by the coiled tubing guide member **1202**, is about 0.2-1 inch smaller than its outer diameter, depending on the material. In some embodiments, the geo-loop has a height that is about 1-3 feet (e.g.,  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ), depending on the borehole depth and an outer diameter of the geothermal loop.

[0236] The geo-loop flange **1200** includes a geothermal loop guide member **1212** for guiding the geothermal loop **1220** into the opening **1208** and out through the second end **1206** of the coiled tubing guide member **1202**.

[0237] In some embodiments, the geothermal loop guide member **1212** is arch-shaped (e.g., shaped like a gooseneck). The geothermal loop guide member **1212** can have a length that is about 1 to 3 feet long, with a radius of curvature between 6 inches and 18 inches, depending on a bending radius of the geothermal loop.

[0238] In some embodiments, the geothermal loop guide member **1212** has a rounded (e.g., beveled or chamfered) edge.

[0239] In some embodiments, the geothermal loop guide member **1212** includes rollers **1214** for guiding the geothermal loop **1220** into opening **1208**.

[0240] The coiled tubing guide member **1202** is physically and permanently coupled to a geothermal loop guide member **1212**. For example, in some embodiments, the coiled tubing guide member **1202** and the geothermal loop guide member **1212** are welded to each other.

[0241] The geo-loop flange **1200** can be made using a material such as carbon steel, fiberglass, high-density polyethylene (HDPE) or high-density polyethylene (HDPE). The geo-loop flange **1200** is installed after drilling phase commences and the drill bit is replaced by the washing nozzle tool. The geo-loop flange **1200** stays mounted between the casing flange **1504** and the stripper **822** during the geothermal loop insertion process, while the CT **818** travels to the bottom, and during the grouting phase, while the CT **818** travels back to the surface.

[0242] An exemplary usage scenario of the geo-loop flange **1200** during the geothermal loop insertion phase includes the steps of:

[0243] 1. Inserting the CT **818** through the geo-loop flange **1200**;

[0244] 2. Mounting the geo-loop flange **1200** below the stripper **822**;

[0245] 3. Inserting the geothermal loop **1220** through the geo-loop flange **1200**;

[0246] 4. Attaching the geothermal loop **1220** to the CT **818** when they have both exited from the second end **1206**. For example, FIG. 12B shows that the CT **818** and the geothermal loop **1220** can be physically coupled to each other using adhesive tape **1224**, a band, a bracket, a hook, or any other coupling mechanism. Regardless of the coupling mechanism, the geothermal loop **1220** should be positioned (and kept) above the CT **818** because the driving force of the CT **818** is necessary to push the geothermal loop **1220** into the borehole;

[0247] 5. After coupling the geothermal loop **1220** and the CT **818**, the geo-loop flange **1200** is mounted on top

of the casing flange **1504** (with the CT **818** and geothermal loop **1220** inside the geo-loop flange **1200** and the casing flange **1504**).

[0248] FIG. 13 illustrates geothermal loop insertion via the geo-loop flange **1200** and CT **818**, in accordance with some embodiments.

[0249] In some embodiments, rollers can be used to guide the geothermal loop **1220** during the insertion process. FIG. 14 illustrates rollers **1402**, positioned below the injector head **820**, for stabilizing the geothermal loop **1220** and guiding the geothermal loop **1220** into the borehole, in accordance with some embodiments.

[0250] In some embodiments, the rollers **1402** are used in conjunction with the geo-loop flange **1200**.

[0251] In some embodiments, the rollers **1402** are used without the geo-loop flange **1200**.

[0252] The geothermal loop **1220** is driven by the CT **818** into its desired depth. During this process, the washing nozzle **1222** that is mounted on the CT **818** is used for pumping high-pressure water to clear any obstacle/debris that is encountered in front of the geothermal loop **1220**, while pushing down the geothermal loop **1220**. When the geothermal loop **1220** has reached its target depth, it is detached from the CT **818**.

[0253] FIG. 15A illustrates a stripper flange in accordance with some embodiments. The stripper flange **1502** is coupled to the stripper **822** (e.g., via welding, screws, or bolts and nuts). The coupling between the stripper flange **1502** and the stripper **822** can be permanent or temporary. A casing flange **1504** is coupled to a casing **1506** (e.g., via welding, screws, or bolts and nuts). The holes of the stripper flange **1502** and the holes of the casing flange **1504** are aligned with one another, and screws (e.g., fasteners or bolts) are passed through respective sets of stripper flange hole and casing flange hole to secure the two flanges. The casing flange **1504** is not movable when the casing **1506** is fixed into borehole. When the injector **820** is moved with the crane **824** above the casing flange **1504**, significant lateral torque is applied to the injector **820** to align the holes of the two flanges. In some embodiments, the stripper flange **1502** has a rotating joint (e.g., illustrated at rotating flange **1508** in FIG. 15A), so only the stripper flange rim **1510** need to be rotated to align the holes of the two flanges.

[0254] In some embodiments, the stripper flange **1502** and the casing flange **1504** are secured using clamps. One advantage of using clamps is that only the flanges and not the holes-need to be aligned with each other. Deep oil and gas drilling use flanges that are secured to each other with thick bolts and nuts to ensure that the flanges do not explode in case of a pressure kick (e.g., ~5,000 to 10,000 psi). Because high pressures like these are typically not encountered in shallow-and mid-range-length well drilling, the flanges can be secured to each other using clamps without compromising safety.

[0255] In some embodiments, a cross-over flange **1520** (for diverting the return drilling mud) and/or a geo-loop flange can be installed between the stripper flange **1502** and the casing flange **1504**. This is illustrated in FIG. 15B.

[0256] FIG. 15B shows that when drilling mud (indicated by arrows **1522**) is being pumped down through CT **818**, the drilling mud picks up cuttings (e.g., small pieces of rock just drilled) and is diverted through the cross-over flange (indicated by arrows **1524**) through shakers and filers. The

cuttings are separated, and the cleaned drilling mud is pumped down again through the CT **818** (indicated by arrows **1522**).

**[0257]** The third phase of the geothermal field construction is grouting. Grouting is the process of injecting grout into spaces (e.g., gaps) between a borehole and a geothermal loop, to seal off fluid flow and reinforce well integrity. Grouting can also help maintain the desired temperature gradients necessary for efficient geothermal extraction.

**[0258]** In accordance with some embodiments of the present disclosure, after the geothermal loop **1220** detaches at the bottom and the CT **818** is pulled back to the surface, the washing nozzle **1222** is used to squeeze grout to fill spaces between the borehole and the geothermal loop **1220**.

**[0259]** The casing **1506** stays in the rat hole during the CT-enabled drilling and geothermal loop installation and grouting phases. Before the grout completely sets (i.e., when the grout is still soft), the casing **1506** is pulled out of the rat hole using a crane or forklift.

**[0260]** In some embodiments, drill cuttings (e.g., broken bits of rocks and other solid materials) removed from the borehole(s) during the drilling phase can be reduced to smaller pieces (e.g., by grinding the cuttings) to form gravel pack (e.g., a volume of ground cuttings), which are injected back to the subsurface during the grouting phase. Using drill cuttings in addition to or in place of grout has several benefits including reduced costs and improved logistics. Drill cuttings are generated during every drilling process and are available for use at no cost whereas grout needs to be sourced, paid for, and transported to the construction site. Furthermore, drill cuttings need to be disposed of after drilling, so re-introducing them back to the subsurface further reduces the time and costs associated with disposal.

**[0261]** FIGS. **18A** and **18B** illustrate respective cross sectional views of an inflatable packer **1802** in an uninflated state and in an inflated state respectively, in accordance with some embodiments.

**[0262]** In some embodiments, inflatable packers are used to seal the surface casing (e.g., casing **1506**) during drilling, geo-loop installation, and grouting.

**[0263]** In its simplest form, the packer **1802** is a cylindrical, elastic membrane that is sealed at the ends and that, when internally pressurized, inflates radially. The radial inflation is used to seal and anchor the packer **1802** in place in the hole or pipe in which it is inserted. In practice, the packer **1802** looks like a short length of hose with steel end fittings and, usually, a central through pipe. The packer **1802** includes an inflation medium that is introduced between the central pipe and the inside of the membrane. The membrane can be a simple rubber tube, or a fabric and wire-reinforced rubber element, or a thin metal sheath. The membrane material has a large bearing on the pressure rating for the complete packer and on the applications for which it is suitable. A simpler packer with an unreinforced rubber membrane will have a very low pressure rating, e.g., around 10-400 psi. The pressure rating of a packer with metal reinforcement can vary between around 500 to 5,000 psi.

**[0264]** As disclosed herein, one or more inflatable packers **1802** can be installed at the surface to seal the casing **1506**, instead of using grout. The advantage is that after completing the borehole grouting operation (e.g., filling the interior of the borehole with grout), the packer would be deflated and reused in another borehole. By contrast, if the casing **1506** is grouted on the outside, the casing and solidified grout will

form a solid structure and a crane will be needed remove the casing after the borehole installation.

**[0265]** The present disclosure differentiates from current geothermal well construction processes, which employ different pieces of equipment for each phase. At the present, the industry uses a drilling rig for drilling, some kind of manually operated reel to insert the geothermal loop into boreholes, and a tremie line to grout. To the best of the Applicant's knowledge, coiled tubing is not currently used for shallow geothermal drilling, nor has there been any attempt to utilize coiled tubing for geothermal loop insertion or grouting.

**[0266]** The present disclosure of employing coiled tubing for multiple phases (e.g., drilling, geothermal loop insertion, and/or grouting phases) of the geothermal field construction fills the technology void for mid-range-length wells that are between ~500 feet and ~3,000 feet deep.

**[0267]** The present disclosure also advantageously improves upon current geothermal well construction processes in several ways. First, the current practice of manually inserting geothermal loops into boreholes is difficult, physically demanding, and time-consuming when the well depths are ~800 feet and upwards, depending on the groundwater depth. The challenge is exacerbated when groundwater is present, or when boreholes are not straight. The risk of geothermal loops getting stuck in the lateral borehole walls or pushed back by buoyancy is quite high, and limits the ability to install them to greater depths. Using coiled tubing for geothermal loop insertion makes the process faster and more efficient. Coiled tubing, being carbon steel pipe, can push geothermal loops to depths unheard of by the current shallow geothermal drilling industry. Second, for the grouting process, the current industry uses Tremie lines which are hoses that are about an inch in diameter. Because grout is a viscous paste, pumping grout through one-inch-diameter tremie lines at depths greater than ~500 to 600 feet is operationally impossible. Larger-size tremie lines are not used because driller would encounter challenges to run them manually.

**[0268]** The advantages of using the UCCTU 800 for at least two of the three phases, or all three phases of the geothermal well construction, include: greater depths (e.g., 500 feet to 2,500 feet) can be achieved; less equipment is used; greater control, and faster operations. The improvements disclosed herein are novel and non-obvious not only because coiled tubing technology is not currently used for shallow geothermal drilling, but also because additional equipment (e.g., a geo-loop flange **1200**) and procedures need to be developed in order to attach the geothermal loop to the coiled tubing at the surface, push it through the geo-loop flange **1200**, detach the geothermal loop once it is at the bottom, and use washing nozzles **1222** coupled to coiled tubing for cleaning the borehole and for grouting.

**[0269]** FIG. **16** illustrates a geothermal loop insertion process **1600**, in accordance with some embodiments. In some embodiments, the process **1600** uses a removable arm **1602** that is configured to be inserted into a forklift pocket **1604** of the injector **820**. The arm **1602** includes designated axle points (e.g., defined by positions **1606** on the arm **1602**) that each corresponds to a respective geothermal loop minimum bend radius. The geothermal loop minimum bend radius can depend on the size (e.g., diameter) of the geothermal loop **1220**. For example, in FIG. **16**, the position **1606-1** can correspond to a minimum bend radius of 1 inch,

the position **1606-2** can correspond to a minimum bend radius of 1.5 inches, and the position **1606-3** can correspond to a minimum bend radius of 2 inches. Interchangeable spools **1608** for minimum radius can be positioned on the arm **1602**. A spool can have a diameter that is about 10-30 times the diameter of the geothermal loop **1220**. The geothermal loop **1220** passes through a window/opening **1610** and is physically coupled to the CT **818** (e.g., using the coupling methods discussed in FIG. **12B**).

[**0270**] FIG. **17A** illustrates a wellhead connection to an injector, in accordance with some embodiments of the present disclosure. FIG. **17B** illustrates a wellhead connection to an injector in existing systems.

[**0271**] In FIG. **17A**, the casing flange **1704** is connected to the stripper **822** via a flow cross flange **1706** (for drilling fluid mud returns) or a geo-loop flange **1708**. By contrast, FIG. **17B** shows that there is a region **1710** (e.g., gap) between the stripper **822** and the casing flange **1504**, where the CT **818** is exposed. The CT **818** in FIG. **17B** can rub against edges **1712** of the stripper **822** or edges **1714** of the casing flange **1504** when is moving, leading to wear and tear and eventual failure of the CT **818**. Thus, compared to existing systems, the improved wellhead design that is shown in FIG. **17A** reduces fatigue on the CT **818** and increases its useful lifetime.

[**0272**] FIG. **19** provides a flowchart illustrating an exemplary method **1900** for constructing geothermal fields, in accordance with some embodiments.

[**0273**] In some embodiments, the method **1900** includes, prior to drilling one or more boreholes of a geothermal field via a coiled tubing (e.g., CT **818**), drilling (**1902**) one or more ratholes. In some embodiments, the one or more ratholes are drilled (**1904**) via a rotary drilling rig that is different from the coiled tubing. In some embodiments, the method includes installing (**1906**), at each of the one or more ratholes, a respective casing. In some embodiments, the method includes sealing (**1908**) the respective casing via an inflatable packer (e.g., packer **1802**).

[**0274**] In some embodiments, the method **1900** includes drilling (**1910**), via the coiled tubing, the one or more boreholes of the geothermal field. In some embodiments, the method includes collecting (**1912**) drill cuttings while drilling the one or more boreholes. The method includes inserting (**1914**), via the coiled tubing, one or more geothermal loops into the boreholes. In some embodiments, the method includes grouting (**1916**), via the coiled tubing, the one or more geothermal loops. In some embodiments, the method includes injecting (**1918**) the drill cuttings into sub-surfaces around the one or more boreholes during the grouting.

#### Applications of Coiled Tubing Technologies for Geothermal Loop Installation

[**0275**] Some embodiments of the present disclosure are directed to the application of coiled tubing (CT) technologies for installing geothermal loops (e.g., geothermal loop **704** as shown in FIG. **7**, also referred to herein as a “geo-loop” or a “ground loop”) in boreholes, for geothermal heating and cooling applications. A geothermal loop can be a single-geo-loop that comprises two pipes that are connected at their ends with a U-bend, or a double geo-loop that comprises two pairs of pipes (i.e., four pipes) where a respective pair of pipes is connected at their ends with a respective U-bend. The geothermal loop can be made of one or more materials, such as plastic (e.g., high-density polyethylene (HDPE)),

fiberglass, and/or carbon steel. Typically, after a borehole has been drilled, a geothermal loop is manually installed by one or two workers, who manually push the geothermal loop from the surface into the subsurface of the borehole. In some instances, the geothermal loop may be spooled onto a carousel, with the workers are unspooling it and pushing it into the borehole.

[**0276**] Some embodiments disclosed herein simplify and/or at least partially automate the geothermal loop insertion process, by (i) physically coupling a geothermal loop (e.g., geothermal loop **1220**) to a coiled tubing string (e.g., CT **818**) and (ii) injecting the geothermal loop with the coiled tubing string using a CT injector (e.g., CT injector **820**) during the geothermal loop insertion process. Advantageously, this configuration reduces the equipment footprint on site and the personnel handling it. Please also refer to the discussion with reference to FIGS. **12B** and **13** above.

[**0277**] FIG. **20** illustrates a view of a vehicle **810-3** that includes a coiled tubing unit (CTU) (e.g., UCCTU **800**), in accordance with some embodiments. The CTU includes a geothermal loop reel **830**, a coiled tubing reel **816**, and a coiled tubing rig. A geothermal loop **1220** is configured to spool on and/or unspool from the geothermal loop reel **830**. A CT string **818** is configured to spool on and/or unspool from the coiled tubing reel **816**.

[**0278**] According to some embodiments of the present disclosure, the geothermal loop reel **830** and the coiled tubing reel **816** are both located on the CTU, and are both part of the CTU, to reduce the equipment footprint on site and the personnel handling it. In some embodiments, both the geothermal loop reel **830** and the coiled tubing reel **816** are mounted on the chassis of vehicle **810-3** that transports the CTU. In some embodiments, both the geothermal loop reel **830** and the coiled tubing reel **816** are mounted on a frame of the CTU, on the chassis of the vehicle. In some embodiments, the geothermal loop reel **830** is mounted on the same axle as the coiled tubing reel **816** (see FIG. **21A**). In some embodiments, the geothermal loop reel **830** and the coiled tubing reel **816** can be mounted on different axles, but both of these reels are positioned on the same vehicle chassis, and/or on a single frame (e.g., frame **2116**) that is positioned on a vehicle chassis.

[**0279**] In the example of FIG. **20**, the geothermal loop reel **830** and the coiled tubing reel **816** are both located on vehicle **810-3** (e.g., on the chassis of vehicle **810-3**). The geothermal loop **1220** is spooled on the geothermal loop reel **830**. The CT **818** is spooled on the coiled tubing reel **816**. In accordance with some embodiments, the CT **818** and the geothermal loop **1220** are configured to move at the same speed. In accordance with some embodiments, the CT **818** and the geothermal loop **1220** are configured to move at substantially similar speeds (e.g., a difference between the speed of the CT **818** and the speed of the geothermal loop **1220** does not exceed  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 3\%$ ,  $\pm 4\%$ ,  $\pm 5\%$ , or  $\pm 6\%$ ). Advantageously, controlling the CT **818** and the geothermal loop **1220** to move at substantially the same speed minimizes mechanical damage to the geothermal loop **1220** as the geothermal loop **1220** is inserted into a borehole using the CT **818**.

[**0280**] FIG. **21A** illustrates the CT reel **816** and the geothermal loop reel **830** from a front view of the vehicle **810-3**, in accordance with some embodiments. In this example, the CT reel **816** is mounted on axle **2102**, and CT **818** is spooled on the CT reel **816**. The geothermal loop reel

**830** is mounted on the same axle **2102** as the CT reel **816**, and the geothermal loop **1220** is spooled on the geothermal loop reel **830**. The CT reel **816** and the geothermal loop reel **830** are separated by a reel spacer **2104**. In some embodiments, the CT reel **816** has the same diameter as the geothermal loop reel **830**. In some embodiments, the CT reel **816** and the geothermal loop reel **830** have different diameters. As an example, in some embodiments, a diameter of the CT reel about 6-10 feet. In some embodiments, a diameter of the geothermal loop reel **830** is about 3-8 feet.

[0281] In some embodiments, both the CT reel **816** and the geothermal loop reel **830** are rigidly (e.g., fixedly) coupled to the axle **2102**. The CT reel **816** and the geothermal loop reel **830** are configured to rotate with the same angular speed (e.g., rotation speed or  $\omega$ ). In some embodiments, the CT reel **816** and the geothermal loop reel **830** have the same diameter. In some embodiments, the CT reel **816** and the geothermal loop reel **830** have different diameters. In some embodiments, the CT **818** and the geothermal loop **1220** are configured to move at the same speed. In some embodiments, the CT **818** and the geothermal loop **1220** are configured to move at substantially similar speeds (e.g., a difference between the speed of the CT **818** and the speed of the geothermal loop **1220** does not exceed  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 3\%$ ,  $\pm 4\%$ ,  $\pm 5\%$ , or  $\pm 6\%$ ). For example, the respective speed of the CT **818** or the geothermal loop **1220** is denoted by  $v$ , and can be determined using  $v=r\omega$ , where  $r$  is the radius of the respective reel.

[0282] In some embodiments, at least one of the CT reel **816** and the geothermal loop reel **830** is movably coupled (e.g., freely coupled) with the axle **2102**. Consequently, the CT reel **816** and the geothermal loop reel **830** may rotate with different angular speeds even though they are both mounted on the same axle **2102**.

[0283] In accordance with some embodiments, the CT reel **816** and the geothermal loop reel **830** may or may not rotate with the same angular speed, whereas the CT **818** and the geothermal loop **1220** are configured to move at the same speed or at similar speeds. For example, even though the CT reel **816** and the geothermal loop reel **830** share the same axle **2102**, in some embodiments the geothermal loop reel **830** can move freely on the axle **2102**. The CT injector **820** and the CT reel **816** move the coiled tubing string **818**, which in turn pulls the geothermal loop **1220**. In this example, the function of the geothermal loop reel **830** is to support to the geothermal loop **1220** and unspool the geothermal loop **1220** according to the driving force (e.g., pulling force) from the CT **818**. Because the movement of the geothermal loop **1220** and the coiled tubing string **818** are synchronized, the geothermal loop **1220** can be inserted into the borehole using the driving force of the CT **818** and the injector **820** while minimizing damage to the geothermal loop **1220**.

[0284] FIG. 21B illustrates another configuration of the CT reel **816** and the geothermal loop reel **830** from a front view of the vehicle **810-3**, in accordance with some embodiments. In this example, the CT reel **816** and the geothermal loop reel **830** are both mounted the vehicle **810-3**, such as on a frame **2116** of the vehicle. The CT reel **816** is mounted on an axle **2112**. The geothermal loop reel **830** is mounted on an axle **2114** that is different from the axle **2112**. The rotation speed ( $\omega$ ) of the CT reel **816** and the geothermal loop reel **830** are independently adjustable. The CT reel **816** and the geothermal loop reel **830** may or may not rotate with the

same angular speed, whereas the CT **818** and the geothermal loop **1220** are configured to move at the same speed or at similar speeds. The CT **818** and the geothermal loop **1220** are configured to move at the same speed or at similar speeds (e.g., within  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 3\%$ , or  $\pm 5\%$  of each other).

[0285] FIG. 21C shows a cross-sectional view of a single geothermal loop **1220A**, in accordance with some embodiments. The single geothermal loop **1220A** comprises a pair of pipes (i.e., two pipes) **2120-1** and **2120-2** that are connected at their ends with a U-bend **2122**.

[0286] FIG. 21C shows a cross-sectional view of a double geothermal loop **1220B**, in accordance with some embodiments. The double geothermal loop **1220B** comprises two pairs of pipes, namely a first pair of pipes **2130-1** and **2130-2** and a second pair of pipes **2132-1** and **2132-2**. A double U pipe fitting **2134** connects a respective pair of pipes at their ends.

[0287] FIG. 22 illustrates respective paths of the geothermal loop **1220** and the CT **818** during the geothermal loop insertion phase, in accordance with some embodiments.

[0288] FIG. 22 shows the CT string **818** is configured to unspool from the CT reel **816**. The CT string **818** passes through the gooseneck **819**, which guides the CT string **818** into a body of the injector **820**. The CT string **818** passes through the body of the CT injector **820**, the stripper **822**, and a geothermal loop flange (geo-loop flange) **2220**.

[0289] Referring to FIG. 22, the geothermal loop **1220** is configured to unspool from the geothermal loop reel **830**. The geothermal loop **1220** is configured to pass through a geothermal loop guide **2210**, and into the geo-loop flange **2220**, where it meets the CT string **818**. The geothermal loop **1220** is then coupled to the CT **818**. In some embodiments, the geothermal loop **1220** is coupled to the CT **818** via a coiled tubing CT bottom-hole assembly **2300** of the coiled tubing. In some embodiments, the geothermal loop **1220** is coupled to the CT **818** when they have both exited the geo-loop flange **2220**. The geothermal loop guide **2210** is positioned between the geothermal loop reel **830** and the geo-loop flange **2220**.

[0290] In accordance with some embodiments, the geothermal loop guide **2210** is configured to guide the geothermal loop **1220** it moves from the geothermal loop reel **830** to the geo-loop flange **2220**, and to reduce vibrations in the geothermal loop **1220** caused by its elasticity as it moves. For example, the geothermal loop guide **2210** can be a pipe that has a larger diameter than the geothermal loop **1220**. In some embodiments, the geothermal loop guide **2210** is a half-pipe. In some embodiments, the geothermal loop guide **2210** includes rollers for guiding and stabilizing the geothermal loop **1220**. In some embodiments, the rollers of the geothermal loop guide **2210** is configured to reduce a frictional force on the geothermal loop as the geothermal loop passes through the loop guide. The geothermal loop guide **2210** is configured to be positioned between the geothermal loop reel **830** and the geo-loop flange **2220**. For example, as illustrated in FIG. 22, the geothermal loop guide **2210** is positioned inside the frame of the CT injector **820** (e.g., by welding). In some embodiments, the geothermal loop guide **2210** is positioned on the coiled tubing unit, anywhere between the geothermal loop reel **830** and the geo-loop flange **2220**. In some embodiments, the system includes multiple geothermal loop guide **2210** that are each configured to guide the geothermal loop **1220** and reduce

vibrations of the geothermal loop **1220** as it moves from the geothermal loop reel **830** to the geo-loop flange **2220**.

[0291] FIG. 23 illustrates a CT bottom-hole assembly **2300**, in accordance with some embodiments. In some embodiments, the end of the CT string **818** that is unspooled from the CT reel **816** is coupled to the CT bottom-hole assembly **2300** via coiled tubing connector **2302**. The CT bottom-hole assembly **2300** can include any combination of sub-assemblies, such as a telemetry sub-assembly **2304** (e.g., the telemetry system **1100** described in FIG. 11), a drilling sub-assembly **2305** that includes a drilling motor and drill bit **502** (e.g., as described in FIGS. 5A to 5D), a geo-loop hook sub-assembly **2306** that includes a geo-loop hook **2312**, and a washing bottom-hole sub-assembly **2308** that includes washing nozzle **1222** and one or more jets **2310**. For example, the telemetry sub-assembly **2304** and the drilling sub-assembly **2305** are used for drilling one or more geothermal boreholes. The telemetry sub-assembly **2304**, the geo-loop hook sub-assembly **2306**, and the washing bottom-hole sub-assembly **2308** are used to for installing the geothermal loop **1220**. In some embodiments, the end of the geothermal loop **1220** (e.g., the u-bend) that is unspooled from the geothermal loop reel **830** is coupled to the coiled tubing bottom-hole assembly **2300** via the geo-loop hook **2312**. In some embodiments, the geo-loop hook **2312** comprises a first dimension  $\times 1$  and a second dimension  $\times 2$ , which can be varied (e.g., adjusted) according to a size (e.g., diameter) of the geothermal loop **1220**, a size (e.g., diameter and/or depth) of the borehole, a U-bend type of the geothermal loop **1220**, and/or a U-bend size of the geothermal loop **1220**.

[0292] FIGS. 24A and 24B illustrate various embodiments of the geo-loop flange **2220** (e.g., **2220-A** and **2220-B**) and their respective views, in accordance with some embodiments.

[0293] Referring to FIGS. 24A and 24B, in some embodiments, the geo-loop flange **2220** comprises a pipe and is largely cylindrical (e.g., annular) in shape. The geo-loop flange **2220** includes a first end **2402** and a second end **2404** opposite to the first end **2402**. A first opening **2406** is located at the first end **2402**. A second opening **2408** is located at the second end **2404**. The geo-loop flange **2220** includes a third opening **2410** that is positioned at a side wall **2414**. The CT **818** enters the geo-loop flange **2220** from the first opening **2406** and exits the geo-loop flange **2220** from the second opening **2408**. The geothermal loop **1220** enters the geo-loop flange **2220** from the third opening **2410** and exits the geo-loop flange **2220** from the second opening **2408**. In some embodiments, the geo-loop flange **2220** includes a guide member **2412** for guiding the geothermal loop **1220** into the third opening **2410** and out through the second opening **2408**.

[0294] In some embodiments, an outer diameter of the geo-loop flange **2220** is about 4-9 inches (e.g.,  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ), depending on the borehole size. In some embodiments, an inner diameter of the geo-loop flange **2220** is about 0.2-1 inch smaller than its outer diameter, depending on the material. In some embodiments, the geo-loop flange **2220** has a height of about 1-3 feet (e.g.,  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ), depending on the borehole depth and an outer diameter of the geothermal loop **1220**.

[0295] In some embodiments, the guide member **2412** is arch-shaped (e.g., shaped like a gooseneck). The guide member **2412** can have a length that is about 1 to 3 feet long,

with a radius of curvature between 6 inches and 18 inches, depending on a bending radius of the geothermal loop **1220**.

[0296] In some embodiments, the guide member **2412** has a rounded (e.g., beveled or chamfered) edge.

[0297] In some embodiments, the guide member **2412** includes side supports **2415** positioned along both sides of the guide member **2412**, for guiding the geothermal loop **1220** into opening **2410**.

[0298] In some embodiments, the guide member **2412** includes rollers **2416** positioned along one or both sides of the guide member **2412**, for guiding the geothermal loop **1220** into opening **2410**.

[0299] The coiled tubing guide member **1202** is physically and permanently coupled to a geothermal loop guide member **1212**. For example, in some embodiments, the coiled tubing guide member **1202** and the geothermal loop guide member **1212** are welded to each other.

[0300] The geo-loop flange **2220** can be made using a material such as carbon steel, fiberglass, or plastic (e.g., high-density polyethylene (HDPE)).

[0301] The geo-loop flange **2220-A** in FIG. 24A and the geo-loop flange **2220-B** in FIG. 24B differ in terms of the location of the third opening **2410** and the location of the guide member **2412**. In the case of geo-loop flange **2220-A**, the third opening **2410** and the guide member **2412** are positioned around midway between the first end **2402** and the second end **2404**, whereas the third opening **2410** and the guide member **2412** are positioned closer to the first end **2402** in the case of geo-loop flange **2220-B**. In some embodiments, the location of the third opening **2410** is based on a size and/or a flexibility of the geothermal loop **1220**, which can impact the shape of the geothermal loop **1220** as it enters the geo-loop flange **2220**.

[0302] FIGS. 25 and 26 illustrates a borehole head stack **2500** that is connected to an injector **820**, in accordance with some embodiments. As depicted in FIG. 25, the borehole head stack **2500** can include a stripper **822**, a geo-loop flange (e.g., geo-loop flange **1200** or geo-loop flange **2220**), and a flow cross flange **1706**. FIGS. 25, 26, and 27 depict that in some embodiments, the borehole head stack **2500** is configured to mounted on the injector **820** during mobilization, demobilization, and transportation of the injector **820**. In some embodiments, the injector **820** and the borehole head stack **2500** are collectively referred to as a coiled tubing rig.

[0303] FIG. 27 shows a view of a vehicle **810-4** for transporting the injector **820** with the connected borehole head stack **2500**, in accordance with some embodiments. The injector **820** is supported on a mast **2702**. In some embodiments, the mast **2702** is a telescopic mast that includes tubular sections that slide within one another in ordered to be extended or retracted pneumatically. In some embodiments, the mast **2702** is a rigid mast (i.e., not telescopic, cannot be extended or retracted). In some embodiments, the mast **2702** is vertically aligned (i.e., perpendicular with respect to as horizontal chassis of the vehicle **810-4**). In some embodiments, the mast **2702** is inclined at an angle. The minimum height of the injector undercarriage, denoted by H1 in FIGS. 27 and 29, should be large enough to allow the borehole head stack **2500** to stay mounted below the injector all the time, including when the mast and injector are in an angled position (e.g., at a certain angle from the vertical), as depicted in FIG. 29.

[0304] In accordance with some embodiments, the coiled tubing rig (that includes the injector **820** and the borehole

head stack **2500**) can be combined with a standard stick pipe derrick to form a hybrid coiled tubing-stick pipe rig.

[0305] FIG. 28 illustrates an existing stick pipe derrick **2800** in accordance with some embodiments. The derrick **2800** includes a mast **2802** that lifts and lowers a drill pipe **2804**, and rotates the drill pipe **2804** to drive it into the earth. The drill pipe **2804** is a straight pipe with a length of about 10- to 20-feet. The derrick **2800** also removes the drill pipe **2804** when the drilling operation is complete.

[0306] FIG. 29 shows a view of a vehicle **810-5** that includes a hybrid coiled tubing-stick pipe rig **2900**, in accordance with some embodiments. The hybrid coiled tubing-stick pipe rig **2900** is part of the UCCTU **800**. In some embodiments, the hybrid coiled tubing-stick pipe rig **2900** includes CT injector **820** and a drill pipe **2910** that is coupled to a rotary drilling drive **3010** (see FIG. 30). In some embodiments, the rotary drilling drive **3010** comprises a mast. In some embodiments, the hybrid coiled tubing-stick pipe rig **2900** includes the borehole head stack **2500**. In some embodiments, the hybrid coiled tubing-stick pipe rig **2900** includes the CT reel **816**, CT **818**, gooseneck **819**, geothermal loop reel **830**, and/or geothermal loop **1220**. The vehicle chassis **2902** is configured to switch between stick pipe drilling for drilling surface holes and coiled tubing drilling for drilling geothermal boreholes. The inclusion of both the stick pipe drilling mechanism and the coiled tubing drilling mechanism on the same chassis saves space, is more compact, and facilitates transportation of these drilling tools to and from the construction site via paved concrete roads.

[0307] In some embodiments, the stick pipe drilling (e.g., via the drill pipe **2910** and the rotary drilling drive **3010**) use the same power line and the same hydraulics as the coiled tubing drilling. In some embodiments, the rotary drilling drive **3010** with the drill pipe **2910** are used for drilling the first 20- to 40 feet of borehole from the surface. The CT bottom-hole assembly **2300** is then assembled sub-surface (e.g., at 20- to 40 feet deep), and the remaining borehole depth is drilled via coiled tubing drilling.

[0308] FIG. 30 illustrates a front view of the hybrid coiled tubing-stick pipe rig **2900** in the direction B-B, in accordance with some embodiments. The injector **820** is positioned next to the rotary drilling drive **3010** while the drill head **3020** for the stick pipe is on the external side of the rotary drilling drive **3010**. The rotary drilling drive **3010** should be kept at a minimal height to accept the smallest length of pipe commercially available for water well operations to avoid custom pipe manufacturing. The injector **820** and the drill head **3020** can travel on the same pulley system or operate independently of each other. In some embodiments, a telescopic hoist head can be added to handle pipes and/or casings that may be encountered during borehole construction.

[0309] Turning now to some example embodiments:

[0310] (A1) In accordance with some embodiments of the present disclosure, a system (e.g., system **1000**) for constructing geothermal fields comprises coiled tubing (e.g., a coiled tubing, such as CT **818**, or a CT string, or a continuous length of metal or composite tubing with no joints) that is configured for (e.g., performs operations comprising) (i) drilling (e.g., constructing) a borehole (e.g., one or more boreholes) of a geothermal field; (ii) inserting a geothermal loop into the borehole; and (iii) grouting the geothermal loop (e.g., using a Bentonite-based thermal grout and/or a sand mixture).

For example, in some embodiments, a geothermal field construction process uses the same coiled tubing for multiple operations including borehole drilling, geothermal loop insertion, and/or grouting.

[0311] (A2) In some embodiments of A1, the system includes a downhole telemetry assembly (e.g., downhole telemetry assembly **1100**) (e.g., wired or wireless) coupled to the coiled tubing and having a plurality of sensors (e.g., sensors **508**, **510**, or sensors **1050**). The system includes one or more processors **1002**, and memory **1006** storing instructions that are configured for execution by the processors **1002**. The memory stores instructions that, when executed by the one or more processors, cause the system to: (i) obtain (e.g., sense, measure, determine, collect), in real-time via the plurality of sensors, downhole telemetry data (subsurface data) during the drilling, the inserting, and the grouting.

[0312] (A3) In some embodiments of A2, the memory includes instructions that, when executed by the one or more processors, cause the system to: while drilling the one or more boreholes, minimize a lateral drift (e.g., a horizontal distance between an actual borehole and a planned trajectory) in accordance with the measured downhole telemetry data (that is measured by one or more downhole inclination and/or azimuth sensors).

[0313] (A4) In some embodiments of A2 or A3, the memory includes instructions that, when executed by the one or more processors, cause the system to determine, via the plurality of sensors, a borehole depth for a first borehole. The memory includes instructions that, when executed by the one or more processors, cause the system to, while inserting the one or more geothermal loops: (i) determine, via the real time telemetry data (via the sensors), an insertion depth for a first geothermal loop that is inserted into the first borehole; and (ii) in accordance with a determination that the insertion depth is within a predetermined margin of the borehole depth, cease the inserting process.

[0314] As an example, in some embodiments, the downhole telemetry data includes a borehole depth (X) that is obtained during drilling. During the loop insertion step, the geothermal loop is inserted to the borehole. The sensors measure a depth (Y) that is attained as the geothermal loop is inserted. In accordance with a determination that the depth X is within a predetermined margin of Y (e.g., when the values are within  $\pm 0.1\%$ ,  $\pm 0.3\%$ ,  $\pm 0.5\%$ ,  $\pm 1\%$ ,  $\pm 2\%$  or  $\pm 5\%$  of each other), the system (via the processors) determine that the desired depth has been reached.

[0315] (A5) In some embodiments of any of A2-A4, the memory includes instructions that, when executed by the one or more processors, cause the system to grout the one or more geothermal loops in accordance with the measured downhole telemetry data. For example, the grout flow rate and CT speed need to be correlated so that grout is fully filling the borehole. In some embodiments, the system (via the processors **1002**) monitor depth, downhole pressure (e.g., internal and external pressure), temperature (e.g., internal and external temperature), grout thermal conductivity (downhole), grout flow rate (from surface), and grout volume (from surface) to determine whether the borehole is fully filled with grout.



- [0316] (A6) In some embodiments of any of A2-A5, the plurality of sensors includes two sensors selected from the group consisting of: a pressure sensor, a temperature sensor, a gamma ray sensor, an inclination and/or azimuth sensor, a vibration sensor, an acceleration sensor, a force sensor, a torque sensor, a pH sensor, a salinity sensor, and/or a thermal conductivity sensor.
- [0317] (A7) In some embodiments of any of A2-A6, the plurality of sensors includes a first subset of (e.g., one or more) sensors that are positioned on an exterior surface of the downhole telemetry assembly; and a second subset of (e.g., one or more) sensors that are positioned on an interior surface of the downhole telemetry assembly. The exterior surface of the downhole telemetry assembly, or an outward-facing surface, is a surface that is directly exposed to a subsurface of the ground during drilling. The interior surface of the downhole telemetry assembly, or an inward-facing surface, is a surface that is not directly exposed to a subsurface of the ground during drilling.
- [0318] (A8) In some embodiments of any of A1-A7, the system includes a coiled tubing injector (e.g., injector 820) for injecting the coiled tubing from a surface into a subsurface during the drilling, the inserting, and the grouting.
- [0319] (A9) In some embodiments of A8, the system includes one or more roller mechanisms (e.g., rollers 1402) coupled to (e.g., mounted to) the injector, for guiding the one or more geothermal loops as they are inserted into the one or more boreholes.
- [0320] (A10) In some embodiments of A9, the one or more roller mechanisms include a depth gauge (e.g., level-wing depth counter) for measuring a length of a first geothermal loop installed in a first borehole.
- [0321] (A11) In some embodiments of any of A8-A10, the system includes a gooseneck (e.g., gooseneck 819) for guiding the coiled tubing into a body of the coiled tubing injector, wherein the gooseneck is configured to rotate to align the coiled tubing and the coiled tubing injector as the coiled tubing is unspooled (e.g., laterally). In some embodiments, the gooseneck is fitted with a telescopic arm with a ball to ensure that the gooseneck is only moving laterally.
- [0322] (A12) In some embodiments of any of A8-A11, the coiled tubing injector is mounted on a stand (e.g., stand 838) during the drilling, the inserting, and the grouting.
- [0323] (A13) In some embodiments of A12, the system is configured to, while drilling a borehole, in accordance with a determination that the borehole is tilted with respect to a vertical axis, adjust a respective height of one or more legs (e.g., legs 840) of the stand (e.g., to change a drilling angle).
- [0324] (A14) In some embodiments of any of A8-A13, the system includes a rotating flange (e.g., rotating flange 150) for aligning the coiled tubing injector on the casing flange (e.g., casing flange 1504).
- [0325] (A15) In some embodiments of any of A1-A14, a respective borehole has a depth in a range of about 500 feet to 3000 feet (e.g., within a threshold of  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ).
- [0326] (A16) In some embodiments of any of A1-A14, a respective borehole has a depth in a range of about 500 feet to 2500 feet (e.g., within a threshold of  $\pm 2\%$ ,  $\pm 5\%$ , or  $\pm 10\%$ ).
- [0327] (A17) In some embodiments of any of A1-A16, the coiled tubing is configured for performing the drilling, the inserting, and the grouting operations when the coiled tubing is unspooled from a reel (e.g., CT reel 816).
- [0328] (B1) In accordance with some embodiments, an apparatus (e.g., geo-loop flange 1200) for facilitating insertion of a geothermal loop into a borehole includes a coiled tubing guide member (1202) (e.g., a flange) having (i) a first end (1204), (ii) a second end (1206), and (iii) an opening (1208) on a side wall (1210) of the coiled tubing guide member. The coiled tubing guide member 1202 is configured for guiding a coiled tubing through the first end and the second end. The apparatus includes a geothermal loop guide member (1212) physically coupled to (e.g., attached to, connected to, by welding) the coiled tubing guide member at the opening. The geothermal loop guide member is configured for guiding the geothermal loop through the opening and through the second end.
- [0329] (B2) In some embodiments of B1, the geothermal loop guide member is arch-shaped (e.g., has a gooseneck shaped).
- [0330] (B3) In some embodiments of B1 or B2, the geothermal loop guide member comprises rollers (1214) for guiding the geothermal loop into the opening.
- [0331] (B4) In some embodiments of any of B1-B3, the geothermal loop guide member comprises a rounded (e.g., beveled or chamfered) edge.
- [0332] (B5) In some embodiments of any of B1-B4, the coiled tubing (e.g., CT 818) is part of a coiled tubing rig that includes a stripper (822) positioned below an injector (820) of the rig, and the apparatus is positioned below the stripper.
- [0333] (B6) In some embodiments of B5, the apparatus is positioned between the stripper (822) and a casing flange (1504).
- [0334] (B7) In some embodiments of any of B1-B6, the coiled tubing (e.g., CT 818) and the geothermal loop (1220) are physically coupled (abutted) together (e.g., via an adhesive tape, a bracket, a hook, or any other coupling mechanisms) after passing through the second end.
- [0335] (B8) In some embodiments of any of B1-B7, an outer diameter of the coiled tubing guide member is in the range of about 4 inches to 9 inches.
- [0336] (B9) In some embodiments of any of B1-B8, an inner diameter of the coiled tubing guide member is in the range of about 2 inches to 8.5 inches.
- [0337] (B10) In some embodiments of any of B1-B9, the coiled tubing guide member comprises a cylindrical flange.
- [0338] (C1) In accordance with some embodiments, a method for constructing geothermal fields includes (i) drilling, via a coiled tubing (e.g., CT 818), one or more boreholes (e.g., boreholes 402) of a geothermal field; (ii) inserting, via the coiled tubing, one or more geothermal loops (e.g., geothermal loop 1220) into the

constructed boreholes; (iii) and grouting, via the coiled tubing, the one or more geothermal loops.

[0339] (C2) In some embodiments of C1, the method further includes: prior to drilling the one or more boreholes, (i) drilling one or more ratholes (e.g., a borehole that is around 10-50 feet deep) via a rotary drilling rig that is different from the coiled tubing; and (ii) installing a respective casing at each of the one or more ratholes.

[0340] (C3) In some embodiments of C2, the method further includes sealing the respective casing via an inflatable packer (e.g., inflatable packer **1802**).

[0341] (C4) In some embodiments of any of C1-C3, the method further includes (i) collecting drill cuttings during the drilling; and (ii) during the grouting, injecting the drill cuttings into sub-surfaces around the one or more boreholes.

[0342] (C5) In some embodiments of any of C1-C4, the drilling, the inserting, and the grouting are performed when the coiled tubing is unspooled from a reel (e.g., CT reel **816**).

[0343] (C6) In some embodiments of C5, the unspooling is facilitated (e.g., utilizes) an injector (e.g., CT injector **820**).

[0344] (D1) In accordance with some embodiments, a system for installing geothermal loops comprises a coiled tubing unit (CTU) that includes a geothermal loop (e.g., geothermal loop **1220**) spooled on a geothermal loop reel (e.g., geothermal loop reel **830**) that is mounted on a first axle (e.g., axle **2102** or axle **2112**). The CTU includes a coiled tubing string (e.g., CT **818**) configured for inserting the geothermal loop into a borehole. The coiled tubing string is spooled on a coiled tubing reel that is mounted on a second axle (e.g., axle **2102** or axle **2114**). As the geothermal loop is inserted into the borehole, the geothermal loop is configured to unspool from the geothermal loop reel, the coiled tubing string is configured to unspool from the coiled tubing reel, and the geothermal loop and the coiled tubing string are configured to move at substantially similar speeds (e.g., at the same speed, or within  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 3\%$ ,  $\pm 4\%$ ,  $\pm 5\%$ , or  $\pm 6\%$  of each other). In some embodiments, the geothermal loop reel **830** and the coiled tubing reel **816** can move with the same speed, or at different speeds, to ensure that the coiled tubing string and geothermal loop move with the same speed. In some embodiments, the geothermal loop reel **830** and the coiled tubing reel **816** are mounted on rotate on different axles. In some embodiments, geothermal loop reel **830** and the coiled tubing reel **816** share the same axle and rotate on the same axle. In some embodiments, even though the geothermal loop reel **830** and the coiled tubing reel **816** share the same axle, the geothermal loop reel can move freely on the axle, independent of the coiled tubing reel (or vice versa). The injector **820** and coiled tubing reel **816** move the coiled tubing string, which in turn pulls the geothermal loop, so the loop reel just supports the loop and moves freely.

[0345] (D2) In some embodiments of D1, a first end of the geothermal loop is coupled to the coiled tubing string via a coiled tubing bottom-hole assembly.

[0346] (D3) In some embodiments of D1 or D2, the first axle and the second axle are the same axle.

[0347] (D4) In some embodiments of any of D1-D3, the system further comprises: a gooseneck (e.g., gooseneck **819**) configured to guide the coiled tubing string into a body of an injector (e.g., CT injector **820**); the injector configured to inject the coiled tubing string from a surface into a subsurface during the inserting; and a geothermal loop flange (e.g., geo-loop flange **2220**) configured to be mounted below the injector, the geothermal loop flange configured to facilitate insertion of the geothermal loop into the borehole.

[0348] (D5) In some embodiments of D4, as the geothermal loop is inserted into the borehole (e.g., borehole **2230**); the coiled tubing string is configured to unspool from the coiled tubing reel, over the gooseneck, through the injector, through the geothermal loop flange, and into the borehole; and the geothermal loop is configured to unspool from the geothermal loop reel, through the geothermal loop flange, and into the borehole.

[0349] (D6) In some embodiments of D5, the system further comprises one or more geothermal loop guides (geothermal loop guide **2210**), configured to be positioned between the geothermal loop reel and the geothermal loop flange (e.g., geo-loop flange **2220**), wherein the geothermal loop is configured to unspool from the geothermal loop reel and pass through (e.g., move through) the one or more geothermal loop guides and the geothermal loop flange, into the borehole; and each of the one or more geothermal loop guides includes a respective set of rollers configured for reducing a frictional force on the geothermal loop as the geothermal loop passes through the one or more geothermal loop guides.

[0350] (D7) In some embodiments of D6, at least one of the geothermal loop guides is positioned on a frame of the injector.

[0351] (D8) In some embodiments of any of D4-D7, the geothermal loop flange includes a first opening (e.g., first opening **2406**) at a first end (e.g., first end **2402**); a second opening (e.g., second opening **2408**) at a second end (e.g., second end **2404**) opposite to the first end; and a third opening (e.g., third opening **2410**) at a side wall (e.g., side wall **2414**) of the geothermal loop flange.

[0352] (D9) In some embodiments of D8, the coiled tubing string is configured to enter the geothermal loop flange at the first opening and exit the geothermal loop flange at the second opening as the geothermal loop is inserted into the borehole.

[0353] (D10) In some embodiments of D8 or D9, the geothermal loop is configured to enter the geothermal loop flange at the third opening and exit the geothermal loop flange at the second opening as the geothermal loop is inserted into the borehole.

[0354] (D11) In some embodiments of any of D8-D10, the geothermal loop flange further includes a guide member (e.g., guide member **2412**) configured to guide the geothermal loop into the third opening.

[0355] (D12) In some embodiments of any of D1-D11, the geothermal loop reel and the coiled tubing reel have different diameters.

[0356] (D13) In some embodiments of any of D1-D12, the geothermal loop is made of one or more of: steel, carbon fiber, plastic, or a composite material.

[0357] (D14) In some embodiments of any of D1-D13, the geothermal loop comprises a plastic material.

[0358] (D15) In some embodiments of any of D1-D14, the geothermal loop spooled on the geothermal loop reel and the coiled tubing string spooled on the coiled tubing reel are part of a coiled tubing unit (CTU); and the CTU is configured to be mounted on a vehicle chassis. This is illustrated in, e.g., FIGS. 20, 27, and 29.

[0359] (D16) In some embodiments of D15, during the inserting of the geothermal loop into the borehole, the geothermal loop is configured to unspool from the geothermal loop reel and the coiled tubing string is configured to unspool from the coiled tubing reel while the geothermal loop reel and the coiled tubing reel are both positioned on the vehicle chassis.

[0360] (D17) In some embodiments of any of D1-D16, the system further comprises electrical processing circuitry and hydraulic circuitry, wherein the coiled tubing string is configured to be operated via the electrical processing circuitry and hydraulic circuitry.

[0361] (D18) In some embodiments of D17, the system further comprises a rotary drilling rig configured to operate via the same electrical processing circuitry and hydraulic circuitry as the coiled tubing string.

[0362] (D19) In some embodiments of D18, wherein the coiled tubing string and the rotary drilling rig are configured to be mounted on the same vehicle chassis. This is illustrated in FIG. 29.

[0363] The terminology used in the description of the invention herein is for the purpose of describing particular implementations only and is not intended to be limiting of the invention. As used in the description of the invention and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0364] As used herein, the phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on.”

[0365] As used herein, the term “exemplary” means “serving as an example, instance, or illustration,” and does not necessarily indicate any preference or superiority of the example over any other configurations or implementations.

[0366] As used herein, the term “and/or” encompasses any combination of listed elements. For example, “A, B, and/or C” includes the following sets of elements: A only, B only, C only, A and B without C, A and C without B, B and C without A, and a combination of all three elements, A, B, and C.

[0367] The foregoing description, for purpose of explanation, has been described with reference to specific implementations. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The implemen-

tations were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to best utilize the invention and various implementations with various modifications as are suited to the particular use contemplated.

What is claimed is:

1. A system for installing geothermal loops, comprising: a coiled tubing unit (CTU) that includes:
  - a geothermal loop spooled on a geothermal loop reel that is mounted on a first axle; and
  - a coiled tubing string configured for inserting the geothermal loop into a borehole, wherein the coiled tubing string is spooled on a coiled tubing reel that is mounted on a second axle;
 wherein, as the geothermal loop is inserted into the borehole:
  - the geothermal loop is configured to unspool from the geothermal loop reel;
  - the coiled tubing string is configured to unspool from the coiled tubing reel; and
  - the geothermal loop and the coiled tubing string are configured to move at substantially similar speeds.
2. The system of claim 1, wherein a first end of the geothermal loop is coupled to the coiled tubing string via a coiled tubing bottom-hole assembly.
3. The system of claim 1, wherein the first axle and the second axle are the same axle.
4. The system of claim 1, further comprising:
  - a gooseneck configured to guide the coiled tubing string into a body of an injector; and
  - the injector configured to inject the coiled tubing string from a surface into a subsurface during the inserting; and
  - a geothermal loop flange configured to be mounted below the injector, the geothermal loop flange configured to facilitate insertion of the geothermal loop into the borehole.
5. The system of claim 4, wherein:
  - as the geothermal loop is inserted into the borehole:
    - the coiled tubing string is configured to unspool from the coiled tubing reel, over the gooseneck, through the injector, through the geothermal loop flange, and into the borehole; and
    - the geothermal loop is configured to unspool from the geothermal loop reel, through the geothermal loop flange, and into the borehole.
6. The system of claim 5, further comprising:
  - one or more geothermal loop guides, configured to be positioned between the geothermal loop reel and the geothermal loop flange, wherein:
    - the geothermal loop is configured to unspool from the geothermal loop reel and pass through the one or more geothermal loop guides and the geothermal loop flange, into the borehole; and
    - each of the one or more geothermal loop guides includes a respective set of rollers configured for reducing a frictional force on the geothermal loop as the geothermal loop passes through the one or more geothermal loop guides.
7. The system of claim 6, wherein at least one of the geothermal loop guides is positioned on a frame of the injector.
8. The system of claim 4, wherein the geothermal loop flange includes:

- a first opening at a first end;
  - a second opening at a second end opposite to the first end;
  - and
  - a third opening at a side wall of the geothermal loop flange.
9. The system of claim 8, wherein the coiled tubing string is configured to enter the geothermal loop flange at the first opening and exit the geothermal loop flange at the second opening as the geothermal loop is inserted into the borehole.
10. The system of claim 8, wherein the geothermal loop is configured to enter the geothermal loop flange at the third opening and exit the geothermal loop flange at the second opening as the geothermal loop is inserted into the borehole.
11. The system of claim 8, wherein the geothermal loop flange further includes a guide member configured to guide the geothermal loop into the third opening.
12. The system of claim 1, wherein the geothermal loop reel and the coiled tubing reel have different diameters.
13. The system of claim 1, wherein the geothermal loop is made of one or more of: steel, carbon fiber, plastic, or a composite material.
14. The system of claim 13, wherein the geothermal loop comprises a plastic material.
15. The system of claim 1, wherein the geothermal loop reel and the coiled tubing reel are configured to be mounted on the same vehicle chassis.

16. The system of claim 15, wherein during the inserting of the geothermal loop into the borehole:
- the geothermal loop is configured to unspool from the geothermal loop reel and the coiled tubing string is configured to unspool from the coiled tubing reel while the geothermal loop reel and the coiled tubing reel are both positioned on the vehicle chassis.
17. The system of claim 1, further comprising:
- electrical processing circuitry and hydraulic circuitry, wherein the coiled tubing string is configured to be operated via the electrical processing circuitry and hydraulic circuitry.
18. The system of claim 17, further comprising:
- a rotary drilling rig configured to operate via the same electrical processing circuitry and hydraulic circuitry as the coiled tubing string.
19. The system of claim 18, wherein the coiled tubing string and the rotary drilling rig are configured to be mounted on the same vehicle chassis.
20. The system of claim 1, wherein the geothermal loop comprises a single geothermal loop or a double geothermal loop.

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